

THE CAMPAIGN FOR NORTH AFRICA THE DESERT WAR 1940-43

AIR & LOGISTICS GAMES RULES OF PLAY AND SCENARIOS

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THE AIR GAME

The following rules Sections cover the air war in CNA. In essence, all planes are represented individually, and fighters have pilots. Each type of plane has differing characteristics and mission capabilities. All of these are listed in the Aircraft Characteristics Charts for each nationality. Yet, in keeping with the tone of the land game, the feel of the game is operational. While there is plane-to-plane combat, it is not resolved tactically, and all strengths have been considered operationally.

Ideally, the air portion of the game requires a separate Player for each side. There is a large amount of paperwork involved, and, even though there were relatively few planes in Africa, a Player may have to juggle operations for 500 to 1000 aircraft during a given OpStage. The rules for the Air Game are a lot simpler than those for the land game. Air Commanders should simply keep their Characteristics Charts handy and be prepared to do a lot of planning.

[33.0] SEQUENCE OF PLAY: Air Game

Note: It is possible to use the Land Game and Air Game Rules while omitting the Logistics Game rules.

LAND/AIR GAME-TURN SEQUENCE OF PLAY

Play of the game should proceed in accordance with the following outline.

I. INITIATIVE DETERMINATION STAGE

The Players determine who will have the initiative for the coming Game-Turn (see Section 7.0). The Player going first within the Operations Stage (see below) is known as "Player A"; the other is "Player B".

II. STRATEGIC AIR PLANNING STAGE

A. Designation Phase:

The Players assign their airplanes to fly Land Support or Strategic missions.

B. Axis Malta Availability Determination Phase

The Axis Player determines the amount of support he will receive from his abstracted N. African Theatre airforce for Raids on Malta.

C. Strategic Mission Assignment Phase

Planes designated as flying Strategic missions are assigned. The Axis Player assigns his planes to Raids on Malta or naval convoy protection. The Commonwealth Player assigns his planes to naval Recon missions or Bombing Reserve.

D. Malta Raid Phase

The Axis Player resolves flak suppression, anti-aircraft fire and bombing missions against Maltese air facilities (Note that Commonwealth warships stationed at Valetta may only be attacked by Land Support Missions).

III. NAVAL CONVOY STAGE

A. Naval Convoy Schedule Phase

The Axis Player refers to the Axis Naval Convoy Level Chart and then to the Axis Convoy Capacity Table and rolls one die to determine the total tonnage available for the next Game-Turn. He then plans what specific cargoes the convoys will carry and their routes.

B. Convoy Resolution Phase

1. *Naval Convoy Reconnaissance Segment:* The Allied Player resolves Strategic Convoy Reconnaissance missions.

2. *Convoy Lane Assignment Segment:* Axis convoy protection aircraft are assigned CAP missions in specific Convoy Lanes. Commonwealth Bombing Reserve aircraft are assigned CAP, Flak Suppression or Convoy Bombing missions in specific convoy lanes.

3. *Convoy Bombing Segment:* All air-to-air combat, Flak Suppression, anti-aircraft, and convoy bombing is carried out.

IV. FIRST OPERATIONS STAGE

A. Initiative Declaration Phase

The Player who gained the initiative in Stage I now states whether he will be "Player A" or "Player B" for this particular Operations Stage.

B. Weather Determination Phase

The Player with initiative rolls for weather (see Section 29.0). All adjustments for unusual weather are made at this time, including any additional evaporation of Fuel/Water from "Hot Weather."

C. Organization Phase

(The following Segments are undertaken in any order desired).

1. *Reorganization Segment:* Attachment and/or assignment of reinforcements, replacements, or any other non-assigned units including Trucks may be undertaken. Detachment of attached units may also be done.

2. Construction Segment:

a. *Construction Completion Step:* Any work scheduled for completion is finished, markers being removed or added as required. Units that have completed construction may freely be moved.

b. *Construction Initiation/Continuation Step:* Any continuing (or ensuing) work on projects is noted. Any units involved in such work may not be moved (voluntarily) during the remainder of the current Operations Stage.

3. Training Segment:

a. *Training Completion Step:* Units and Replacement Points completing a level of training are noted and the effects in Morale Applied.

b. *Training Initiation/Continuation Step:* Any units beginning or continuing training are noted. They may not be moved (voluntarily) in the remainder of the Operations Stage (exception: Reaction).

4. *Supply Distribution Segment:* Supplies in the same hex as land units may be redistributed at this time, subject to the maximum innate capacity of the unit(s) and the hex. Trucks may be loaded/unloaded:



5. *Tactical Shipping Strength:* Both Players execute the transport of cargo between African ports. **Note:** Axis coastal ships are represented in the game by counters. Allied coastal shipping is not represented by counters, and is limited only by the port capacities.

D. Naval Convoy Arrival Phase

All land and air reinforcements, Replacement Points, and Supplies scheduled for arrival (and actually arriving) appear in their designated ports of arrival or entrance hexes. Check Reinforcement Tables and Convoy Rules. The Axis Player plans for the arrival of future Replacement Points from the Axis Replacement Points Pool, if he is the phasing Player. In his first Naval Convoy Arrival

Phase of each month, the Commonwealth Player consults the Commonwealth Production Table and determines how many Replacement Points he will receive one month hence and plans for their arrival.

E. Commonwealth Fleet Phase

1. *Fleet Assignment Segment:* The Commonwealth Player assigns his Ships to any Sea or Coastal hexes for Bombardment purposes.

2. *Fleet Repair Segment:* Any Repair work on ships is undertaken at this time.

F. Land Support Air Phase

(Only those planes assigned to Land Support Missions may be used in this Phase.)

1. *Mission Assignment Segment:* All planes that are fueled may be assigned Missions by the Player.

2. *Mission Deployment Segment:* Counters representing each type of plane assigned a Mission are placed in their hex of assignment.

3. *Air-to-Air Combat Resolution Segment:* Combat between opposing aircraft, including any caused by Scramble and Interception, is resolved. Mission aborts may take place before combat resolution.

4. *Flak Resolution Segment:* Aircraft in the same hex as enemy anti-aircraft units undergo Flak fire.

5. *Mission Completion Segment:* All missions that can be completed (e.g., bombing, strafing, etc.) are completed. Abortions may take place here, too.

6. *Return to Base Segment:* All surviving aircraft are returned to their base of origin (excepting those on transfer missions and some other rare occasions).

7. *Tactical Maintenance Segment:* Both Players may attempt to ready all planes eligible to fly land support missions for the next OpStage.

G. Reserve Designation Phase

Player "A" designates which of his units he is placing in Reserve Status, indicating that with a Reserve marker (see Section 18.0).

H. Movement and Combat Phase

The following four Segments comprise the Movement and Combat Phase. Player "A" may, if he so wishes, repeat Segments 1 through 4 as many times as he desires within the restrictions of the Continual Movement rules (8.2). Each repetition must include all four Segments.

1. *Movement Segment:* All units except unattached trucks and tank recovery squadrons not in reserve and eligible to be moved may be moved. Non-Phasing Player ("B") may move units in accordance with the Reaction movement rules (see Case 8.6).

2. *Breakdown Determination Segment:* All vehicles and Motorized units of both sides are checked for Breakdown (Section 21.0). Broken down vehicles are indicated with an appropriate marker.

3. Combat Segment:

a. *Position Determination Step:* Both Players determine the "position" (Front or Back) of all Gun and Armor class units.

b. *Barrage Step:* Players secretly plot and then execute any barrages.

c. *Retreat Before Assault Step:* Player "B" may retreat before assault any of his units eligible to be so moved (see Section 13.0).

d. *Force Assignment Step:* Both Players secretly assign TOE Strength Points to anti-armor or close assault. Player "A" determines which assaults will be probes (although he need not reveal this decision) and which TOE Strength Points will be withheld from assault.

e. *Anti-Armor Step:* Both of the Players simul-

taneously resolve any anti-armor fire, extracting any casualties. Destroyed tank markers are deployed as required.

f. Close Assault Step: Close Assaults are resolved in any order as determined by Player "A." He announces which of them are actually probes after each is resolved.

4. **Reserve Release Segment:** Player "A" may release any of his reserves that he wishes to.

J. Truck Convoy Movement Phase

Player "A" may move his Second and Third Line (unattached) Trucks, replacement points and any POW's and Guards.

K. Commonwealth Rail Movement Phase

If Player "A" is the Commonwealth Player, he may utilize rail movement (see Case 8.7) to move land units or supplies. Otherwise this Phase is omitted.

L. Repair Phase

1. **Towing Segment:** Player "A" may tow broken down and recovered vehicles.

2. **Maintenance Segment:** Player "A" may attempt to repair broken down, or destroyed vehicles which were not towed in this Phase.

M. Patrol Phase

If the Phasing Player has not engaged in Assault, he may use patrols for reconnaissance purposes.

At this point, Phases G, H, J, K, L, and M are repeated, except that for "Player 'A'" read "Player 'B'".

V. SECOND OPERATIONS STAGE

Both Players repeat all facets of the First Operations Stage.

VI. THIRD OPERATIONS STAGE

Both Players repeat all facets of the First Operations Stage.

VII. STRATEGIC AIR RECOVERY STAGE

A. Return to Base Phase

All surviving planes from missions flown in Stage II are returned to their base or origin if possible.

B. Aircraft Maintenance Phase

Both Players may attempt to ready planes that have flown missions during the Strategic Air Stage.

VIII. END OF GAME-TURN

You have now completed one full Game-Turn, replete with airplanes and sandstorms and the lot! Congratulations. You may put in for a transfer, but before doing so, keep in mind that the Russian Front is the only other active Theatre.

[34.0] THE AIRCRAFT

GENERAL RULE:

Players will be handling individual aircraft in CNA, each type of plane having its own characteristics. There are, basically, seven different types of planes: fighters, fighter-bombers, bombers (also classified as to night bombing capabilities), dive-bombers, transports, pure reconnaissance (some planes with additional capabilities may also fly recon), and flying boats. These planes can perform a variety of missions, all of which fit into one of the following roles: patrol (support of other planes), combat (bombing or strafing), recon, or transport.

Each plane is assigned to a specific squadron, whose Squadrons Ground Support Unit (base) is placed on the game-map — usually in an air facility (airfield, air landing strip, flying boat basin or flying boat alighting area). The Squadron Ground Support Unit (SGSU) services the planes for that squadron. The planes themselves are rated as to

range, tactical air combat, bomb load and/or transport capacity, maneuver capability, fuel consumption and mission capability. Fighters are also assigned individual Pilots. Planes are useful only if they have been fueled and refitted after a mission.

CASES:

[34.1] AIRCRAFT CAPABILITIES

[34.11] Range

A Plane's range is the maximum distance, in hexes, that it may be flown to a hex to perform a mission. It is also the maximum distance from which it may return to a base from a mission. There is one exception to this: planes flying a transfer mission (simply moving from one air facility to the next) may be flown a distance equal to twice that of their listed range. **Example:** An Me109E based on a Landing Strip in hex C3101 may fly to hex C3133 for a strafing mission — a distance of 32 hexes, within its Range of 41 — and then fly back to its squadron base at C3101. The Me109E could not be flown from the mission hex to an Axis airfield in A0915 for two reasons: it is too far away and, generally, all planes must return to their Base of origin. Planes may not save hexes not flown (within a plane's Range) going out and use them coming back (e.g., the Me109E flies only 10 hexes to a Mission; it still has a Range of 41 on its Return, not 41 + 31 — the hexes not used flying to the Mission hex).

[34.12] Certain aircraft have more than one listed Range, usually as a result of an increased fuel capacity or bombload. Players are free to choose whichever range load they wish.

[34.13] Tactical Air Combat (TacAir)

This is the aircraft's combat rating *vis a vis* other aircraft, based on the number and type of armament carried by the plane. A unit with a parenthesized TacAir Rating (3) may not initiate air-to-air combat.

[34.14] Bombload Capacity

This is the number of Bomb Points, called tonnage (but not actually in tons), that a Plane can carry. If a Plane can carry torpedoes that fact is listed along with the bombload capacity.

[34.15] Transport Capacity

This is the capacity of the plane, given in TOE Strength Points (infantry) and/or tons of supplies (see Case 54.5 for tonnage equivalents).

[34.16] Maneuver

This is a somewhat abstract rating that simulates the ability of a plane to fly faster, higher, and with greater agility than other planes. Maneuver ratings are always considered in relation to other planes. Maneuver ratings are used to modify Basic TacAir Differentials.

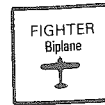
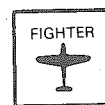
[34.17] Fuel Consumption

This is the number of Fuel Points a plane requires to perform any mission or emergency flight (see Case 37.3). All Fuel Points are consumed during a mission, regardless of the type or distance of the mission.

[34.18] Mission Capability

These are the types of missions that the plane can undertake. Planes may only undertake missions for which they have capability. For a more complete description, consult the Section on missions, 39.0.

[34.19] All aircraft Ratings are listed on the Aircraft Characteristics Chart, 34.6. No aircraft counter contains any numbers; players keep track of all ratings on their various sheets.



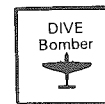
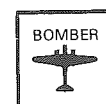
[34.2] FIGHTERS

[34.21] Fighters, as a class of plane, include fighters (F) as well as fighter-bombers (FB). Fighter-bombers may fly either as a fighter or bomber, but not both at the same time. Thus a fighter-bomber on a bombing mission may not break off to *initiate* air-to-air combat, although it may defend against such normally.

[34.22] Fighters and fighter-bombers usually fly Combat Air Patrol (CAP) missions; i.e., support and protection for bombers or a ground facility. Certain fighters may also *strafe* Enemy units/facilities. See the individual missions for details.

[34.23] Every fighter and fighter-bomber is assigned a pilot (see Case 40.1). Pilots affect the plane's air-to-air combat capability.

[34.24] Certain fighters/fighter-bombers may fly reconnaissance missions. Fighters and fighter-bombers flying recon may not initiate air-to-air and may not bomb.



[34.3] BOMBERS

[34.31] Bombers, as a class of plane, includes bombers, dive-bombers and fighter-bombers.

[34.32] Most Bombing Missions are concerned with dropping bombs (air bombardment) on targets in order to destroy, reduce or otherwise incapacitate them. Different bombers have varied capabilities as regards this.

[34.33] Bombers, in and of themselves, are usually considered as one or sometimes more of the following: Bombers, Night Bombers (NB), Dive Bombers (DB), Torpedo Bombers (TB).

[34.34] Dive-bombers are the same as regular bombers, with one difference: they can perform two Missions at once. Dive-bombers may both bomb and strafe the *same* target.

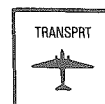
[34.35] Bombers may never initiate air-to-air combat; they may only defend if attacked. However, bombers may increase their TacAir Ratings if they fly in large formations (see Case 45.36).

[34.36] Certain bombers are capable of transporting (TT) personnel or supplies.



[34.4] FLYING-BOATS (SEAPLANES)

Several planes in the game are flying boats, or seaplanes (FlyBt). These aircraft have all the normal range of capabilities that other planes have, with one exception: they may not be based in, take off from, or land in airfields or air landing strips. They must use flying boat basins or flying boat alighting areas.



[34.5] TRANSPORTS

[34.51] Both sides are given planes of minimal combat value that are used for transporting (TT) personnel and/or supplies.

[34.52] Transports are given a Transport Capacity, expressed in TOE strength points or Supply tons they can carry. Thus, a Ju52 can carry up to 2½ tons (or 20 Points) of Fuel or 1 TOE point of infantry. Transports can never transport vehicles or motorized units of any type (except motorcycle units).

[34.53] Transports may never initiate air-to-air combat. They receive no benefit from formation flying, as do bombers. Transports are relatively helpless in the air and must have heavy CAP to insure their arrival.

[34.6] AIRCRAFT CHARACTERISTICS CHARTS (see Charts and Tables)

[34.7] AIRCRAFT COUNTERS

[34.71] The aircraft counters do *not* represent individual planes in the game; they are used to note the location of planes and facilitate air-to-air combat.

[34.72] Planes on the ground are not represented by the Squadron Ground Support Unit counter (see Section 35.0). The actual types and numbers of plane in that squadron are kept on the Squadron Composition Sheet. No aircraft counters are needed.

[34.73] When placing a squadron or squadrons (or part of a squadron) of planes in a target hex (the hex in which the mission will occur) the Player simply places one aircraft counter for each class of planes (fighter, bomber, transport) in the target hex to denote that planes of that class are in the hex. The counters are lettered so that the Players can note which planes/squadrons are represented by what counters.

[34.74] If any planes are eliminated, the Player simply reflects those losses by adjusting that total on his sheet, not by changing counters. The counter representing the planes remains in place.

[34.75] When resolving air-to-air combat, Players may want to use aircraft counters to represent individual fighters, bombers, etc., so that they may better visualize what is happening. Players may therefore take any aircraft counters not on the game-map and use them as they see fit in representing the individual planes.

[34.8] AIRFORCE REINFORCEMENTS AND WITHDRAWALS

The Players receive airforce reinforcements in the forms of individual pilots, individual planes and Squadron Ground Support Units (SGSU's). Low ranked pilots arrive as needed while higher ranked pilots arrive randomly. SGSU's arrive as needed. Airplane reinforcements follow a fixed schedule: in addition, the Commonwealth Player is required to remove various squadrons during the course of the game.

[34.81] The Commonwealth Player's airforce reinforcements may be placed in any air facility in Malta, Cairo, Alexandria, or Off-map except Ethiopia. The reinforcements may be divided amongst the receiving air facilities in any distribution the Commonwealth Player chooses except as follows:

A/No more than 10% of a month's airplane reinforcements may be sent to Malta.

B/No airplane reinforcements may be sent to a Malta/N African Off-map air facility in excess of the facility's *current* squadron capacity.

The Axis Player's airforce reinforcements may be divided amongst the bases in the Tripoli/Tunisia boxes and Italy/Sicily/Crete as he chooses within the requirements concerning basing German bombers in the Mediterranean (see Case 43.1).

[34.82] SGSU's arrive in the Naval Convoy Arrival Phase of the Operations Stage. Axis SGSU's may arrive in any Tripoli/Tunisia box. Commonwealth SGSU's may arrive in any Alexandria/Cairo hex. SGSU reinforcements may be brought in as the Players desire subject to the following restrictions:

Both Players: An SGSU lost through enemy action may not be replaced for one complete Game-Turn. This is in addition to any restrictions below.

Commonwealth Player: A/U.S. Army Airforce SGSU's (i.e., those with "USAAF" historical designations and CPA's of 30) may not be brought in until August, 1942 (1/91 Game-Turn). B/The Commonwealth Player may never have more SGSU's in play than the total number of his planes at *on-map* facilities divided by twelve, plus two. E.g., if the Commonwealth Player has 438 planes at on-map facilities he may have 39 GSU's $438 \div 12 = 36\frac{1}{2} + 2 = 39$ (rounded up).

C/ The Commonwealth Player may, because of a counter shortage, find that he is somewhat short of SGSU's (there are enough commonwealth counters to hold approximately 1,200 planes). The Commonwealth Player may create additional SGSU counters only if he has the maximum number of planes possibly based at Commonwealth N. Africa off map air facilities (i.e., calculate the number from the base's printed rather than current capacity). In such a case, he may create SGSU counters but is still bound by "B" above. These counters may not be U.S. SGSU's.

Axis Player:

A/ The Axis Player may never have more German SGSU's in play than the total number of his German planes based at on-map facilities divided by ten. E.g., if the Axis player has 286 planes based on-map he may have 29 German SGSU's.

B/ The restrictions on Italian SGSU's in play are the same as those for German SGSU's.

C/ Captured Commonwealth planes may count as part of the on-map planes for the Axis nation to whose SGSU they are assigned.

D/ The Axis Player may run short of Italian and/or German SGSU counters. He may only create more SGSU's for a nation if at least 30% of that nation's total airforce is based at off-map (N. African and/or Mediterranean) air facilities. This percentage is of total planes, not by types.

[34.83] One-to Six-Rated Pilot reinforcements arrive during the Naval Convoy Arrival Phase of the First Op Stage of each month. The Players always possess as many zero-rated pilots as needed for any situation (including emergency flight).

The Players determine the number and rating of pilot reinforcements once a month, at the instant of their arrival. The owning Player consults the appropriate Pilot Arrival Table (Case 34.88 or 34.89) and if pilots are available, makes a single roll of two dice (although separate rolls for the German and Italian pilots). The Player adds the dice together and cross indexes the result under all of the columns to determine the number of pilots of that nationality of each rating that he receives for that month (note that Hauptmann Hans-Joachim Marseille arrives automatically.)

[34.84] Airplane reinforcements arrive in the Naval Convoy Arrival Phase. The reinforcements are listed as to the total of each type of plane arriving during that month (the beginning of the campaign has arrivals during certain specific game-turns instead of months). The planes must be divided amongst the weeks as evenly as possible and all of the planes arriving during a game-turn may arrive in the First Op Stage of that game-turn at the owning Player's choice (i.e., the Players need only divide the total by four, not twelve). The

individual planes arriving are determined freely by the owning Player. E.g., in January of 1941 the Commonwealth Player receives a total of 59 planes (44 Hurricanes, 12 Blenheims, 2 Wellingtons and 1 Maryland). The Commonwealth Player could have 15 Hurricanes arrive the first game-turn; 10 Hurricanes, 2 Blenheims, the Maryland and both Wellingtons arrive the second game-turn; 15 Hurricanes arrive the third game-turn and 10 Blenheims and 4 Hurricanes arrive the fourth game-turn.

[34.85] The Commonwealth Player is required to withdraw certain squadrons during the course of the game. The dates, squadron types and plane requirements are listed in the Commonwealth Airplane Reinforcement and Squadron Withdrawal Schedule (Case 34.86). Squadron withdrawals are performed at the beginning of the month indicated. Squadrons may be withdrawn from any air facility on Malta or game map section "E". The Commonwealth Player simply removes from play all of the planes and pilots assigned to that squadron.

SGSU's are not removed unless they now exceed the limit determined in Case 34.82.

[34.86] Commonwealth Airplane Reinforcement and Squadron Withdrawal Schedule (see Charts and Tables)

[34.87] Axis Airplane Reinforcement Schedule (see Charts and Tables)

[34.88] Commonwealth Pilot Arrival Table. (see Charts and Tables)

[34.89] Axis Pilot Arrival Tables. (see Charts and Tables)

[35.0] SQUADRON GROUND SUPPORT UNITS

COMMENTARY:

All planes are attached to a specific squadron, and each squadron is represented by a Squadron Ground Support Unit (SGSU), which is the base for that squadron. Each squadron has its own Squadron Composition Sheet, and the planes and pilots — as well as supplies — attached and available to that squadron are listed on it. The squadron is the basic unit for the Air Game, and finds its counterpart in the German "Staffel" and Italian "Squadriglia." Actually, the Squadron was next-to-lowest in air hierarchy (above the Flight); squadrons were part of wings, which were, in turn, part of groups (and their Axis counterparts). However, most of this was for purely administrative purposes. Therefore, the only organizational level *CNA* covers is the Squadron.

GENERAL RULE:

Squadrons — specifically the SGSU's — are used to maintain and supply the aircraft assigned to them. Each SGSU may contain only one class of airplane, and it is limited as to the number of planes it may handle effectively. The SGSU counters are considered vehicles (medium trucks) and have their own Capability Point Allowance printed on the counter. They have no combat values.

[35.1] SGSU CHARACTERISTICS

[35.11] An SGSU counter acts as base for all planes assigned to that squadron. Each SGSU counter is placed on the game-map to indicate where that squadron is located. They are usually placed at air facilities. SGSU counters do not include the airplanes attached to the squadron;

SGSU's are the mechanics, trucks, equipment, etc., that are part of that squadron. They are not "air" units.

[35.12] SGSU's have their Capability Point Allowance printed on the counter; they are vehicles. SGSU's move in the Truck Convoy Phase. They have *no* Stacking Point Value and may stack freely with other units. (Their capability to handle the planes in their squadron is limited by the Capacity Level of the air facility they are based at.) SGSU's have no combat strength of *any* type; if an Enemy combat unit moves adjacent to an SGSU that has no Friendly combat unit stacked with it, that SGSU may react (if possible). If it cannot react, it is eliminated.

[35.13] SGSU's arrive according to the orders of the player (See Case 34.8). Each SGSU has a unit designation (squadron ID number). These numbers are historical, but they are not *used* historically. If an SGSU counter is eliminated, it may be returned to play, as per 34.8, one Game-Turn later.

[35.14] SGSU's require a certain amount of supplies themselves to function. Each SGSU must expend one Stores Point per Game-Turn. In addition, each SGSU requires one Fuel Point and one Water Point per Operations Stage. SGSUs without the required supplies (for themselves) may not repair their planes. (Maintaining the planes will, of course, require additional supplies of Fuel and Ammunition).

[35.15] Truck units may be attached to an SGSU as First Line Transport. They are used to carry the supplies that the SGSU needs to keep its planes fit and readied.

[35.16] SGSUs may be used to construct air landing facilities or flying boat alighting areas (see Case 24.7).

[35.17] SGSUs are necessary to refuel and refit airplanes; only SGSUs can perform such tasks. Aircraft may *refuel* at *any* SGSU: their own squadron or another squadron base. However, *refitting* must be performed at their assigned squadron base (SGSU) for maximum efficiency. If a plane attempts to refit at an SGSU other than its assigned squadron, the Player adds *one* to his die-roll for refit. In addition, Players should remember that planes may not be refueled or refitted beyond the capacity of the air facility (see Case 36.12, 36.2, 36.3 and 36.4).

[35.18] The United States Army Air Force (USAF on the counters) SGSU's may not be used prior to Aug. 1, 1942.

[35.2] SQUADRON COMPOSITION

[35.21] Each Squadron must contain the same class of aircraft: i.e., bomber, fighter or transport (recon planes may be assigned anywhere). Fighter-bombers may go into either. Furthermore, it is recommended (but not a rule) that only one type of plane (e.g., Hurricane IIC or Ju87D, etc.) be assigned to a squadron. The latter is for ease in handling and supervision. The composition of each squadron is noted on the squadron's Composition Sheet.

[35.22] Planes may not change squadron assignments, with two exceptions:

- The squadron's SGSU has been eliminated. In such case the aircraft are free to transfer to any other squadron, or they may be placed in reserve.
- A squadron falls below 50% ready Capacity. Planes may then be transferred from other squadrons to bring that squadron back to a more than 50% capacity.

Planes reassigned to a new squadron are given a transfer mission (to fly to their new base), unless the new SGSU is in the same hex as the plane.

[35.23] The number of planes that may be assigned to a squadron is limited and dependent on the nationality of the squadron. Each squadron's limit is the maximum number of *ready* planes it can handle; a squadron can always keep $\frac{1}{3}$ more in reserve (whether ready or not). Thus, a squadron with a capacity of 12 may contain 16 planes: 12 ready to fly, and 4 ($\frac{1}{3}$ of 12) in reserve (and incapable of performing a mission). The capacities are as follows:

Nationality	Ready Capacity	Reserve Capacity
Italian Squadriglia	9	3 (12 total)
German Staffel	12	4 (16 total)
Commonwealth Squadron		
(1940-June '41)	15	5 (20 total)
Commonwealth Squadron (July 41-43)	18	6 (24 total)

Players will note that starting with July 1941 Commonwealth Squadrons increase their capacity.

[35.24] Pilots (see Case 40.1) for fighter and fighter bombers are assigned to *squadrons*, *not* to specific planes. A record of pilots in a squadron is kept on the Squadron Composition Sheet, and pilots are given specific planes when they fly missions.

[35.25] Assignment and transfer of planes, where flying is unnecessary, is done in the Organization Phase. Otherwise, assignment and transfer are done in the Mission Deployment Segment (as a transfer mission).

[35.26] As noted, each squadron may keep a certain number of planes in reserve. These planes may be readied and armed (or not, as the Player sees fit), but they cannot fly if the ready capacity of that squadron is flying. Reserves are used to supplement aircraft that were not readied. Reserves may *not* scramble. They may use emergency flight (see Case 37.3).

[35.27] If an SGSU is eliminated (see Case 35.12) and there are planes on the ground with the SGSU, the planes are captured. They may be used by the opposing Player one Game-Turn later or destroyed as he sees fit.

[35.28] Italian and German planes may *not* coexist in the *same* squadron. They require separate maintenance. However, Italian squadrons may consist *entirely* of German planes (but *not* vice versa).

[35.29] A historical note: Although the Commonwealth and Italian SGSU's are indistinguishable as to type of aircraft handled (at least on the face of the counter), the German SGSU's are not. Each SGSU represents a specific type of plane. The Players need not follow this representation, as the SGSU's do not represent strict squadron OB. However, for those so inclined, JG Squadrons are Fighters, St.G. Squadrons are for Dive Bombers, KG for Bombers, ZG for Fighter-Bombers, and H and F Squadrons for Recon. (All, of course, with a few exceptions.)

[36.0] AIR FACILITIES

GENERAL RULE:

Air facilities include airfields, air landing strips, flying boat basins and flying boat alighting areas. These facilities are differentiated by the type and number of planes they can handle in terms of flight and maintenance. Air facilities are constructed (see Case 24.7), and counters are provided for that purpose for each type of air facility. Some air facilities exist at the start of each Scenario; these

are noted in the instructions for that Scenario. Air facilities may be damaged and/or destroyed by Enemy bombardment. The number of SGSU's and airplanes which can use an air facility is dependent upon its current capacity level. Thus take-off, recovery, and maintenance are limited.



CASES:

[36.1] AIRFIELDS

[36.11] Airfields are fully-equipped and engineered air facilities, with repair shops, full maintenance facilities and runway capacity for a large number of squadrons. Airfield construction is covered in Case 24.7. Each airfield is noted by an airfield counter in the hex in which the airfield has been constructed.

[36.12] Each Airfield has a *capacity level* of six (maximum). This means that it may handle a maximum of six squadrons (regardless of squadron size) at any one time. There may never be more than six SGSU's in a given airfield hex. However, see Case 36.5.

[36.13] If an airfield is landing six squadrons planes in a given Operations Stage, it may not receive any more airplanes, even in an emergency. Such excess planes must seek other fields or strips at which to land. Likewise, no more than six squadron's worth of planes may be readied at a given airfield (see Aircraft Maintenance Section 38.0).

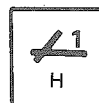
[36.14] The capacity of an airfield (its levels) may be reduced by enemy bombardment (air or artillery). Players should keep track of the capacity level of an airfield. Thus, if bombing has reduced the capacity of an airfield from six to three, that airfield may handle only three squadrons at a time, until it is built back up to six. Note that, regardless of the capacity of an airfield, each airfield may still have six SGSU's with it, even though some may not be able to function because of a reduced capacity level. If an airfield is reduced to zero capacity, it is considered destroyed for all purposes.

[36.15] Airfields may also be damaged by desert raiders. However, normal land units never damage airfields (except by barrage; see Cases 36.14 and 12.5). Land combat units may capture or destroy (entirely) an airfield by occupying its hex. Airfields are, in essence, non-denominational; they may be used by anyone.

[36.16] Players should keep in mind that, while it is the airfield that has the *capacity* for maintenance, it is the SGSU that actually performs that maintenance (see Case 35.17).

[36.17] An airfield is a supply dump for supplies to be used by the SGSU's on that airfield. Fuel, ammunition, stores, etc., may be stored at an airfield as if it were a dump. Land units may *not* use airfield supply dumps unless it is an emergency. (Exactly what constitutes an emergency is left to the Player.) Any SGSU at an airfield may make use of the supplies there to maintain and ready its planes.

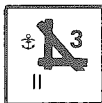
[36.18] Airfields have an intrinsic anti-aircraft Strength Point of one (in addition to any other anti-aircraft units present). The Strength Point may only be used against planes on strafing and/or dive-bombing combat missions.



[36.2] AIR LANDING STRIPS

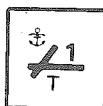
Air landing strips are simply locations cleared and leveled for take-offs and landings, with only minimal maintenance facilities. They are

represented by air landing strip counters, and they are treated exactly as airfields except that each air landing strip has a maximum squadron capacity of *one*. If that capacity level is destroyed, the strip is eliminated and removed from the game-map (see Case 24.7 for construction details, and Case 36.1 for capacity details).



[36.3] FLYING BOAT BASINS

Flying boat basins are "airfields" for flying boat seaplanes. All flying boats *must* use a flying boat basin (or alighting area) for landing and maintenance. They may not use airfields or landing strips. Basins have the same features as an airfield, with the exception that their Capacity is *three* Squadrons. Construction details are in Case 24.7. Players should note that they can bring supplies to a flying boat basin simply by bringing trucks into the hex; they need no other transport. Flying boat basins are placed in any coastal hex. As with airfields, flying boat basins have one intrinsic Strength Point of anti-air (see Case 36.18). Flying boat basins may not be used for normal aircraft.



[36.4] FLYING BOAT ALIGHTING AREAS

Alighting Areas are the same as flying boat basins, except that they have a capacity of *one* Squadron and they are immune to artillery barrage. They are still susceptible to Air Bombardment (see Cases 24.7 and 41.36).

[36.5] OFF-MAP AIR FACILITIES

There are several off-map air facilities: e.g., the Tunisia boxes, the Commonwealth facilities such as Deversoir, on map E, et cetera. These are treated the same as other, on-map facilities with three exceptions:

- There are unlimited supplies for airplane maintenance and repair;
- They may exceed the SGSU limit up to the limit listed on the field;
- They may never be destroyed by bombing (although they may be reduced to zero), and may be rebuilt using the Malta AirFacility Construction Table (one roll gives total levels for all CW air facilities each game-turn).

Axis Mediterranean bases are also different in one other way: they may not be bombed.

[37.0] FLIGHT

GENERAL RULE:

Flight is movement of aircraft. Flight is handled operationally: planes are simply assigned to fly to a given hex from a given hex. The path of flight in hexes may not exceed the plane's range. When the plane is to be moved, it is simply placed in its target hex, noting the path of flight (hexes it flies through) for purposes of CAP interception. Planes must be refueled to fly, and all planes except those flying transfer missions must be refitted to fly again (after they have flown any other mission). Flight is never affected by the presence of ground units, with the exception of flak fire.

CASES:

[37.1] THE MECHANICS OF FLIGHT

[37.11] All planes have a range (see Case 34.11). This determines how far a plane may fly in any one direction (unless the plane is flying a transfer mission, in which case it may double its range for the one-way flight).

[37.12] To move a plane, the Player notes the location of the target hex for that mission and counts the distance in hexes, using any path he wishes, from the base/air facility to the target hex. If the number of hexes falls within the range of the plane, the plane may be flown to that hex (see Cases 37.4 and 56.18).

[37.13] Having determined the path of flight, the Player takes an aircraft counter to represent each class of plane that is assigned to a mission in that hex and places that counter (or counters) in the target hex. The Player will have a record of what missions each squadron is undertaking. Flight takes place in the Mission Deployment Segment of the Land Support Air Phase, or in the Malta and Convoy Bombing Segments of the Strategic Air Stage.

[37.14] There is a form of flight called scramble, wherein planes on the ground may take off to counterattack a nearby air threat (see Case 40.3).

[37.15] No plane may fly unless it has been fueled. Furthermore, other than units flying transfer missions, no plane may fly unless it has been refitted (see Case 37.32).

[37.16] If a plane that has been fueled flies any distance it uses *all* its Fuel and must be refueled to fly again.

[37.17] Planes may be flown without Ammunition, although this could be a dangerous practice.

[37.2] RESTRICTIONS ON FLIGHT

[37.21] Planes may never exceed their range in any one direction, unless they are flying transfer missions. Planes flying transfer may double their range for the one-way transfer trip.

[37.22] No plane may fly into or out of a sandstorm or rainstorm hex.

[37.23] Planes must return to their base of origin; i.e., where the Flight started from. If, for some reason, that air facility had been so reduced that it cannot hold the returning planes then — and only then — may planes attempt to land elsewhere. If such a plane cannot land at any air facility it must crash land as close to an air facility as it can get. The Player then rolls one die. If he rolls a 1, 2, or 3 the plane — and pilot — survive the landing. The plane and the pilot are recovered during the next OpStage, and both may be ready to fly (pending refitting, etc.) in the following OpStage. If the plane does not survive (a roll of 4, 5, or 6), roll the die again. If the Player rolls a 1 or 2 the second time, the pilot survives the crash and escapes serious injury and may return to play *one* Game-Turn following his crash.

[37.24] No planes may fly in excess of the air facility's capability level. Moreover, no planes may fly in excess of an SGSU's ready capacity (see Case 33.23). Thus, if there are five SGSU's on an airfield, but the capacity level of that airfield has been reduced to two, only two of those SGSU's may refit and ready their planes (thus enabling them to fly). The other three squadrons are forced to remain inactive because of the reduced field capacity. Likewise, an Italian squadron (for example) could send no more than nine planes on a mission, regardless of how many planes it has ready (as reserves).

[37.3] EMERGENCY FLIGHT

[37.31] If an Enemy land combat unit moves adjacent to a Friendly air facility that has readied planes on the ground, and those planes are in danger of being destroyed or captured by that Enemy unit, the readied planes may attempt an emergency flight to escape such danger. Emergency flight is a transfer mission

[37.32] Only fueled planes — including reserves — may make an emergency flight.

[37.33] The Player assigns pilots and then rolls once for *each* plane attempting to escape. Using one die, if the plane is a fighter or fighter bomber and he rolls a 1, 2, or 3, or if the plane is a bomber, recon, or transport and the Player rolls a 1 or 2, then the plane escapes. On any other die roll, the plane must remain on the ground.

[37.34] Emergency flight follows all normal rules of flight, except that it occurs out of sequence. Planes performing emergency flight may be flown up to double their range.

[37.35] The Player must exercise this option immediately. If an Enemy unit moves adjacent and the Friendly Player does not roll for emergency flight prior to the movement of another Enemy unit, all planes are stuck on the ground.

[37.4] DISTANCE CHARTS

(see Charts and Tables)

The Air Distance Table is provided as a handy reference to determining distances in hexes between major points. Considering the size of the game-map, counting hexes can be a chore.

[38.0] AIRCRAFT MAINTENANCE

GENERAL RULE:

In order to fly, planes must be *ready*. A *readied* plane is one that is refitted and fueled (being armed is optional, although it is a major contributor to plane survival). SGSU's have the capability of *readying* (i.e., performing maintenance on) planes. Aircraft maintenance is limited to each air facility's current squadron capacity. Planes may be refueled and rearmed by an SGSU without penalty, although refitting is performed with maximum efficiency by the squadron's own SGSU.

CASES:

[38.1] READYING AIRCRAFT

An airplane that is readied is one capable of performing a mission. Readying aircraft is performing maintenance so that the planes may be capable of flight and combat. There are three different types of maintenance. The different types of maintenance are:

- Fuel.** A plane must have fuel to fly; no plane may fly without being refueled up to its capacity.
- Refit.** Refitting is repairing wear and tear from the previous mission. Desert flying proved extremely tough on planes, and keeping them flying was even tougher.
- Arming.** In order to engage in air-to-air a plane must have ammunition. In order to engage in bombing missions, a bomber must have bombs.

[38.2] REFUELING AIRCRAFT

[38.21] Planes must have fuel to fly. The amount of fuel needed to refuel a plane is its Fuel Consumption Rating, listed on the Aircraft Characteristics Chart. That is the number of Fuel Points required to enable one plane of that type to fly any one mission.

[38.22] Planes are refueled *at* air facilities *by* SGSU's. The SGSU does the actual refueling; the airfield or air strip simply limits the number of SGSU's that may operate there.

[38.23] An SGSU may refuel planes from any squadron. However, it may never refuel more planes than the maximum capacity of that SGSU

(readied *plus* reserve). Therefore, an Italian SGSU can handle refueling chores for a total of 12 planes, regardless of squadron assignment, in a given Operations Stage or 12 planes in a given Strategic Maintenance Phase.

[38.24] To refuel a plane the necessary fuel must be in the hex containing the air facility (which usually acts as a supply dump for its planes). The fuel is subtracted from the total supply in the air facility, and the plane is marked as refueled on the Squadron Composition Sheet. This is done for each plane that a Player wishes to refuel. Refueling takes place in the Tactical Maintenance Segment or the Strategic Maintenance Phase, depending on the class of mission assigned.

[38.25] Planes do not have to be refueled (if the Player does not wish to use them), nor do refueled planes have to fly. Planes in reserve for a squadron may be refueled. Planes may sit refueled without flying for the entire game; fuel never gets stale.

[38.26] Players will note that certain aircraft have the capability of adding to their range by carrying extra fuel (or, to be exact, extra fuel tanks). This decreases their maneuverability but adds greatly to their range. Such tanks may be jettisoned if the planes become involved in air-to-air combat, in order to increase their maneuverability. In such case the Maneuver Rating reverts to its highest (normal) value, but the range of the plane also reverts to its lower range. This could cause problems in terms of getting back to base.

[38.3] REFITTING AIRCRAFT

Refitting aircraft is repairing and maintaining them so that they can fly another mission. In order to fly any mission other than a transfer, a plane must be refitted.

[38.31] At the start of a Scenario, all planes are considered refitted, unless otherwise indicated. As soon as a plane flies any mission other than transfer, it must be refitted again. A plane that is not refitted may fly no mission other than transfer, even if it is refueled.

[38.32] Refitting takes place in the Tactical Maintenance Segment, for planes assigned tactical missions, and in the Strategic Maintenance Phase, for planes assigned strategic missions.

[38.33] Planes are refitted at air facilities by their SGSUs. It is the capacity of the SGSU that counts; the air facility only limits the number of SGSUs allowed. Each SGSU can refit up to the maximum planes the SGSU can contain (Ready plus Reserve). An SGSU can Refit planes from other squadrons; however, when doing so, the Player must add *one* to his refit die roll if the plane being refitted is not assigned to the SGSU doing the work.

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[38.34] Refitting is not a guaranteed process, like refueling. Players must roll for each squadron undergoing refit. If planes not belonging to that squadron undergo a refit attempt, they are rolled for separately from the rest of the squadron. The Player simply determines the number of aircraft in the squadron undergoing refit and throws one die, making any adjustments (see Case 38.35) and consulting the Aircraft Refit Table (38.38). The table gives him the percentage of planes successfully refitted. (Round all fractions up.)

[38.35] When rolling for refit, the Player adds one to his die roll if the planes do not belong to the squadron attempting the refit. If the planes at-

tempting refit are Italian, the Player (be he Axis or Commonwealth) adds *two* to the die roll. If the planes are German, add *one* to the die roll. The affects may be cumulative.

[38.36] For every squadron undergoing an attempted refit — whether successful or not — the Player must have present and actually expend one Stores Point. A Player is not required to try to refit any plane; the choice to refit or not is up to him.

[38.37] Twenty percent (rounded up) of all refitted planes in a hex on the ground in which there is a sandstorm must be refitted again before they can be flown. Players should distribute this effect as equally as possible.

[38.38] Aircraft Refit Table (see Charts and Tables)

[38.39] Players wishing to use a more painstaking (if slightly) more accurate, method of refitting planes may roll for each plane *individually*. The Player, in this case, rolls two dice. The Commonwealth Player refits a plane on a roll of 2 through 8; the Axis Player refits German planes on a roll of 2-7, the Italian planes on 2-6. The effects of Case 38.35 are increased to adding *two* to the die roll to account for the “wrong” squadron. (This was, with some minor variations, the method used by the Playtesters. I can’t truthfully say they loved it. . .).

[38.4] ARMING AIRCRAFT

[38.41] Aircraft are armed to provide them with ammunition for combat and bombing. Arming is not necessary for flight; it is necessary to fly a bombing mission — and for safety.

[38.42] Planes are rearmed by SGSUs with the same limits and restrictions as applied to refueling. Ammunition comes from the air facility (supply dump).

[38.43] Each *squadron* requires *one* Ammunition Point to allow its planes to use their TacAir Rating. If that point is not supplied, each plane has a TacAir Rating of zero and may not fire at any Enemy planes (see Case 46.21).

[38.44] For bomb tonnage, one Ammunition Point is required for all bombs loaded onto a plane. Thus a bomber with a tonnage capacity of 15 would need and expend only one Ammunition Point to carry its maximum bomb load (see Case 41.22).

[38.45] A plane’s bombload, whether used or not, is expended during a mission including Emergency Flight. (It was dangerous to land with bombs, so if unused after an abort they were often dropped in the sea or into the desert).

[38.46] The armament status of a plane may be noted on the Squadron Composition Sheet.

[38.47] The cost in Ammo Points to load a bomber with mines (see Case 41.39) or torpedoes (see Case 41.7) is one Point.

[39.0] MISSIONS

GENERAL RULE:

No plane may fly without being given a specific mission. A mission is a one or two word description of the undertaking, the reason the plane is flying and what it is supposed to do. Generally, no plane may perform more than one mission at any one time. There are a large variety of missions, and they are all dependent on the class of plane and the target. The various missions and their descriptions are covered in a series of Cases following 39.1, which is general in nature. Missions may be aborted, or discontinued before completion.

Note: Often players may not know precisely what is in a hex, but will have to assign missions “blindly,” and only find out what target are present when the planes arrive.

PROCEDURE:

Most missions involve different planes going after different types of targets. However, virtually all missions (except transfer) follow the same pattern. The sequence below should be followed for all missions, with exceptions taken as specifically noted in a mission description.

- All Planes are placed in their mission target hex.
- Resolve all interception and scramble.
- Any air-to-air combat between fighters is resolved.
- Fighters on CAP that survive air-to-air may, under certain circumstances, attack Enemy non-fighter planes.
- Any planes remaining in hex after air-to-air undergo flak fire if there are Enemy AA/Flak units in that hex.
- Each mission is run in any order the Player wishes, the opposing Player noting and taking any damage inflicted.
- All surviving planes return to base.

CASES:

[39.1] MISSIONS IN GENERAL

[39.11] Unless a plane is attempting emergency flight (see Case 37.3), no plane may fly unless it has been assigned a specific mission. Nor may any plane be flown if an Enemy land combat unit is adjacent (see 37.3).

[39.12] Each Player may use any shorthand he wishes to note which mission is which. Each Squadron Composition Sheet has a section for listing the mission that a specific plane is undertaking in a given Operations Stage or Strategic Air Phase. The Player simply writes the mission next to the plane, and that plane will attempt that mission.

[39.13] There are two major areas of missions: strategic air missions and Tactical Land Support. The former refers to any Commonwealth Missions flown against Axis Naval Convoys as well as any Axis Missions flown *against* Malta. Both of these Missions are flown *only* in the Strategic Air Phase of the Sequence of Play; moreover, they are flown only once per Game-Turn.

[39.14] Tactical land support missions are missions flown during an Operations Stage of the Game-Turn, affecting the movements and usage of land (or naval) units. Tactical land support missions are numerous in type and they are flown a possible three times a Game-Turn (or once an Operations Stage).

[39.15] There is no limit to the number of planes that may be assigned to a given mission to a given hex. There *is* a restriction to the type and class of plane that can undertake a given mission, as no plane may perform a mission for which it does not have the Capability. Plane Capabilities are listed on the Aircraft Characteristics Chart (34.6).

[39.16] Planes from different squadrons may be assigned to the same mission (or even target), and planes from the same squadron may be assigned to different missions. However planes from the same squadron may not be divided between strategic and land support missions.

[39.17] There is no limit to the number of the type of missions that may occur in (or rather over) a given hex (limited, of course, by what is in that hex

worth going after). Players should keep the planes assigned to each specific mission separate, so as to avoid confusion.

[39.18] There are basically five "types" of missions: fighter, bomber, transport recon and transfers. The later mission is the only mission that may be undertaken by *any* plane (see Case 42.1)

[39.19] Generally, a plane may fly only one mission per Operations Stage or Strategic Phase (with the exception of certain fighters and dive bombers; Case 39.2). A plane flying a mission in an Operations Stage may not fly in the Strategic Phase of that Game-Turn and vice versa.

[39.2] COMBINED MISSIONS

Planes can generally fly only one mission per flight. However, certain fighter aircraft had the ability to carry small bombloads, dropping them as they strafed a target. The Stukas, as dive bombers, also had this capability, although their armament placement prohibited them from performing in a fighter capacity. Thus, any plane with a capability notation of D("dual") may strafe and bomb the same target as a combined mission.

[39.3] VOLUNTARILY ABORTED MISSIONS

[39.31] An aborted mission is one that is cancelled *before* any action is taken to complete that mission. No mission may be voluntarily aborted once some action, be it bombing, strafing, or even air-to-air combat, has taken place.

[39.32] To voluntarily abort a mission a Player simply states that he wishes to abort; he then sends his planes back to base — if they qualify (see Cases 39.36 and 39.37).

[39.33] Aborted missions consume fuel, and bombers must expend their bombload, even though they do not use it.

[39.34] A Player is not required to abort *all* his planes on a given mission. He may choose to continue with some and send some back. The choice, when available, is up to him, within the restrictions below.

[39.35] Any fighter (any plane with a pilot, including fighter-bombers) may voluntarily abort if its *Maneuver Rating* is no lower than 10 Points the highest maneuver-rated Enemy Fighter on CAP in the Target hex. To determine this, subtract the "aborting" fighter's Maneuver Rating from that of the Enemy CAP fighter. If the result is -10 or better (-8, -2, +3, etc.) the plane may abort. Of course, if there is no Enemy CAP, the fighters may always abort (exception: see Case 40.26).

[39.36] Non-fighters flying with a sufficient fighter Screen (Friendly fighters flying CAP for that hex; see Case 39.37) may abort if they wish. If there is not sufficient fighter Screen, they may *not* abort, unless the planes' Maneuver Rating is higher than the best Enemy fighter (a highly unlikely occurrence).

[39.37] A sufficient fighter screen is defined as a number of Friendly fighters (simply counting numbers of planes, not ratings) equivalent to at least 25% of the number of Enemy Fighters flying CAP in that hex. Thus a bomber raid with 12 fighters as escort encountering 40 Enemy fighters on CAP could be aborted; if there were 50 Enemy planes flying CAP, the bomber mission could *not* be aborted. Fighters acting as such a screen may *not* abort if they act as a screen. Fighters cannot screen fighters.

[39.38] Aircraft may also abort *involuntarily*; e.g., as a result of AA/flak fire (see Case 46.3 and 45.14).

[39.4] NIGHT MISSION

[39.41] Night Missions are essentially a Phase within the Land Support/Air Phase. All Night Scramble, air-to-air combat, anti-aircraft fire and land support bombing are conducted separately from other Land Support Missions.

[39.42] Only certain missions may be flown at night and only planes with night missions capability may participate in any night missions actions (exceptions see Case 39.43).

[39.43] Planes, with the exception of planes assigned Offensive CAP or Scramble missions, may only perform a mission at night if specifically assigned a night mission. In all other situations a plane, even if it possesses night mission capability, is considered to be performing a regular Land Support Mission.

[39.44] Planes flying at night benefit from the decrease in the ability of enemy aircraft to find them (e.g., planes on night missions may not be Intercepted), may not engage enemy aircraft in air-to-air combat, and benefit from a decrease in the effectiveness of anti-aircraft fire. Conversely, non-fighters may not benefit from Formation Flying while performing a night mission.

[39.5] AIR MISSION SUMMARY (see Charts and Tables)

[40.0] FIGHTER COMBAT

GENERAL RULE:

Fighters may fly a variety of missions. The most common fighter mission is combat air patrol (or CAP). Fighters may also fly strafing missions and flak suppression missions. Fighters are also capable of scramble, a form of interception CAP. A number of fighters are capable of carrying bombloads (especially fighter-bombers); these fighters are so noted and they may participate in bombing missions (see Section 41.0). Each fighter is flown by a pilot, whose Pilot Rating will affect the plane's air-to-air combat ability. Fighters may be assigned to only one mission in an Operations Stage (three in one Game-Turn) or Strategic Air Stage (one in one entire Game-Turn; see Case 39.2).

PROCEDURE:

Players assign each fighter/fighter-bomber or squadron a mission in a specific hex. Once the planes reach the target hex, the Player must then assign each plane or squadron to a particular *target* of the type previously selected (e.g., fighters assigned to offensive CAP would be assigned particular Enemy aircraft to attack only after they were in their target hex; fighters assigned to a strafing mission against "first-line Trucks" would be assigned trucks attached to a specific parent formation only after the planes had reached the hex (and any air-to-air combat and flak fire had been resolved)).

[40.1] PILOTS

[40.11] Pilots are not represented by counters. Rather, pilots enter the game through the Reinforcement Track as "numbers," the number referring to the pilot's Rating. Each Player notes the pilots he is receiving and assigns them to his squadrons as they arrive. Consult the various Pilot Arrival Tables to see how this works.

[40.12] Pilots are assigned to squadrons. They are assigned to a particular squadron — and become available for use by that squadron — in the first Operations Stage of the turn of their Arrival. As each squadron usually contains only one type of plane, *that* is the type of plane the pilot is trained to fly. Thus a pilot assigned to a squadron of Hurricanes is trained to fly Hurricanes.

[40.13] Pilots may be reassigned to different squadrons at any time. A pilot is reassigned simply by switching him from one Squadron's Composition List to another. A pilot that is being reassigned may not fly in the Operations Stage (or Strategic Air Phase) of his reassignment.

[40.14] If a pilot is reassigned to a squadron that contains planes of a different type than for which he was trained, the pilot must be trained to fly those new planes. It requires one full Game-Turn (or three Operations Stages) to train a pilot. Thus a pilot switching from a squadron of Hurricanes to a squadron of Beaufighters may not fly for three Operations Stages.

[40.15] The Pilot Rating is his capability. Pilot Rating's are either 1,2,3,4, or 6. (The 6 pilot represents the infamous Hans-Joachim Marseille, the German Desert Ace). The Pilot Ratings are added to the plane's TacAir Rating when that plane is involved in *any* air-to-air combat. Thus an ME Bf109F with Marseilles would have a TacAir Rating of 14.

[40.16] As Pilots are assigned to squadrons, the Player may have them fly any plane within that squadron (remember Case 40.14). Any plane that does not have a pilot assigned for a mission is considered to have a zero-rated pilot. There is an unlimited supply of zero-rated pilots.

[40.17] If a fighter is shot down, there is a chance that the pilot bailed out and can be recovered (see 45.65). If the pilot has been recovered, he may fly again three Operations Stages after his Recovery. He is returned to his squadron of assignment.

[40.18] There is an Optional Rule available for Players who are attempting a full campaign game and wish to create their own Aces. In such a case, all pilots start the game with a zero Rating; there is no need to bring in Reinforcement Pilots (ignore any Reinforcement Pilots). However, for each five "kills" a pilot gets, he increases his Rating by one. A kill is any Enemy plane destroyed in air-to-air combat. There is no limit how high a rating a pilot may get. This should not be used for the shorter Scenarios.

[40.2] COMBAT AIR PATROL (CAP)

[40.21] Any plane that has F (fighter) capability (see the individual capabilities on the Aircraft Characteristics Charts) may fly Combat Air Patrol. Usually, fighters fly CAP. To denote that a fighter is flying a CAP mission, write CAP and the target hex in his mission column on the Squadron Composition Sheet. He should also denote whether this is offensive CAP (CAP-O) or defensive CAP (CAP-D).

[40.22] Offensive CAP concerns flying to a hex with the intention to shoot down Enemy planes. Defensive CAP is undertaken by fighters wishing to protect Friendly bombers or transport (flying as escort/screen for them) or to protect an installation or facility. The distinctions can sometimes become a bit blurred.

[40.23] If three or more fighters are assigned to *offensive* CAP in a given hex, the fighters possess the equivalent of an air Zone of Control (i.e., they are flying CAP over the hex they are in as well as all six hexes surrounding that hex). They may thus engage any Enemy planes that enter any one of those seven hexes.

[40.24] Fighters (or any plane with F capability) may fly CAP over any hex on the game-map that is within range. However, no plane based in Africa (on the game-map) may fly CAP of any type for or against Axis Naval Convoys. Commonwealth planes based on Malta may fly CAP

against Axis Naval Convoys only. Axis planes based in Italy, Sicily, and Crete may fly CAP for Axis convoys. Remember, a plane flying CAP may not fly any other mission in that Stage.

[40.25] Example of CAP Missions

The Axis Player wishes to bomb Tobruk. He sends two squadrons of Ju88's to accomplish this. He also sends to the same target hex (Tobruk) as escort/screen two squadrons of ME Bf109E's and a squadron of Italian CR. 42's. (They could have come from any air facility within range.) The Commonwealth Player has placed on defensive CAP above Tobruk two squadrons of Hurricanes. The Commonwealth Hurricanes and Axis ME's and CR 42's engage in air-to-air to see whether the bombers can get through to complete their mission.

[40.26] Using the example in Case 40.25 we can best illustrate the differences between offensive and defensive CAP. If the Commonwealth planes were flying defensive CAP they would, first off, have *no* air Zone of Control (40.23). They would also not automatically initiate air-to-air combat. However, the Axis Squadrons would have to get through the defensive CAP to perform their Mission. If the Hurricanes were flying offensive CAP, they would have an air Zone of Control and they could initiate combat against the Axis planes. There is one additional — and important — difference. If fighters are flying offensive CAP, they would *have to* attack any Enemy planes entering their hex or any hex in their Zone of Control to perform a mission in that hex; they could not decide to not have combat. Exception: only Enemy planes in one such hex must be attacked. Defensive CAP planes may decline combat and abort; offensive CAP planes may not.

[40.27] Fighters flying Offensive CAP *may intercept* Enemy planes whose paths of flight enter the hex they are in or their air ZOC. After all plane counters have been placed to indicate their mission target hex, but before revealing the numbers of planes involved, a Player with fighters on offensive CAP may ask to see a path of flight. If that path coincides with his fighters, the Player with the offensive CAP *may* choose to intercept the Enemy mission right there. If so, air-to-air combat will take place, and any surviving planes may continue their mission. Fighters on offensive CAP may intercept only one mission per Operations Stage.

[40.3] SCRAMBLE

There may be a time when a Player wishes to either conserve fuel or, perhaps, retain some flexibility with his fighter force. *Scramble* gives him the ability to do that. Planes assigned to scramble missions are planes that stay on the ground until Enemy aircraft come within range. They may then, under certain circumstances, take-off and engage the Enemy in air-to-air combat.

[40.31] Only planes with scramble capability may be assigned a scramble mission. Planes assigned a scramble mission are noted with an SC in the Mission column. No hex is assigned. Scramble mission may be assigned only in the Land Support Air Phases.

[40.32] Scramble occurs at the end of the Mission Deployment Segment. After all planes have been placed in their assigned mission hexes, any fighters assigned to scramble may take off and engage Enemy planes within scramble interception range. The farther the range, the lesser the chance of successful interception. A Player is not required to scramble his planes even if they are assigned to a scramble mission, but only planes assigned a scramble mission may scramble.

[40.33] If a Player decides to scramble his fighters so assigned he notes the hex which he wants them to fly to and consults the Scramble Table (40.4), determining the distance in hexes from the base of the scrambling planes to the target hex. The path of flight may not include a hex (other than the target hex) containing other enemy fighters. The Player then rolls one die and checks the table to see if the scramble has succeeded. If it has, the planes have made contact with the Enemy. They are considered to be on offensive CAP now and will attack those enemy planes. If the Scramble is unsuccessful, the planes have flown and they still must refuel and refit. The choice of whether to Scramble is up to the Player, but all decisions are made in secret and revealed at the same time (at the end of the Mission Deployment Segment).

[40.34] The Player wishing to scramble rolls for each *squadron* separately, even if more than one squadron is heading for the same target. Squadrons may not be split between different hexes for scramble purposes.

[40.4] SCRAMBLE TABLE (see Charts and Tables)

[40.5] STRAFING

[40.51] Any plane with strafing capability (STR) may strafe ground forces and other selected targets. Mostly, fighters have strafing capabilities. Dive-bombers have the capability to both Strafe and Bomb a given target. Strafing is a mission, and precludes the undertaking of any other mission (with the exception of planes with STR/B capability).

[40.52] Planes on a strafing mission are assigned a target hex and a specific type of target (Case 40.54). This would be listed as, say, STR/C1234/dump. Strafing missions follow the general course of all missions (39.0).

[40.53] Each individual plane may strafe only one target in a hex, regardless of how many different types or number of target of a single type there are in that hex. Furthermore, strafing is resolved as a single attack against a target; it is not a repeated series of attacks against, say, each truck in a convoy. **Note:** a group of fighters assigned to strafe, for example, "infantry-type units" may attack each of two battalions attached to a Brigade in a target hex, so long as no one plane attacks more than one battalion.

[40.54] The following types of targets may be Strafed:

- a. Infantry-type types of units
- b. Trucks in Convoy
- c. First Line Trucks (attached)
- d. Supply Dumps
- e. Grounded Aircraft
- f. Tanks (only Hurricane II D's)
- g. Ports
- h. Water Pipeline

[40.55] Strafing may not occur in major city hexes.

[40.6] STRAFING PROCEDURES

[40.61] *Infantry Units.* The strafing Player totals his TacAir Points (Remember, do *not* add Pilot Ratings in here; they apply only to Air-to-Air) and consults the Strafing Table. The Player may strafe as many infantry-type battalions or companies as he wishes, but each plane may strafe only one target (battalion) per mission. The Player then rolls two dice and checks that diceroll under the column for the number of TacAir points applied to that battalion/company. The result is the number of TOE Strength Points eliminated. If such infantry toe Strength Points were actually being transported by trucks, such trucks are also eliminated.

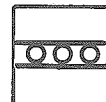
[40.62] *Truck Convoys.* This is treated in the same fashion as Case 40.61, with losses being in Truck Points. The Player being strafed determines which trucks are eliminated, except that the losses *must* be distributed as evenly as possible between the various types of "cargo" (i.e., water, fuel, Replacement Points, etc.) A strafing Player may choose to go after first line attached trucks in the same fashion. In such a case he would strafe the parent unit, designating its attached trucks as the target. Again, losses are distributed as evenly as possible.

[40.63] *Grounded Aircraft* Treated similarly to Case 40.61, with the target being planes on the ground. The result on the Strafing Table however, is not the number of planes but rather the number of *dice* that the strafing Player must roll to see how many planes are shot up. Thus, if he gets a 1 on the Strafing Table, that Player then rolls one die; the result of that dieroll is the number of planes strafed and destroyed on the ground. Strafed Planes may be Recovered (45.6).

[40.64] *Supply Dumps:* The mechanics are the same as 40.62, the results are in the percentage of all supplies in that dump destroyed. Thus a result of 1 equals 10%, 2 equals 20%, etc.

[40.65] *Tanks:* Strafing of enemy tanks by planes was done only towards the end of the campaign, and only by certain planes equipped with heavy-enough cannon to pierce the tank's armour. The commonwealth Hurricane Mark IID's were the only planes so equipped (with Vickers 40mm cannon). Strafing Tanks is accomplished much the same as strafing planes, with the second die or dice-roll producing the minimum number of Armour Protection Points that must be removed. Anything with an armour protection rating may be strafed. Any dummy tank counter strafed is automatically removed from play.

[40.66] *Ports.* Strafing of ports is combined — and is considered part of — a bombing raid on the port. The Player adds his Strafing TacAir Points then halves them, rounding down. He adds this to the total bombload delivered on the target (the port), and consults the Air Bombardment Table (See 41.5)



[40.67] *Water Pipeline.* Planes may strafe an Enemy pipeline in order to render it useless. The Player assigns the fighters a mission to strafe the pipeline in a given hex, uses the Strafing Table, and rolls the dice. If he obtains any result of 2 or more, the pipeline is destroyed. (This is not a "cost efficient" way of destroying a pipeline.)

[40.7] FIGHTER FLAK SUPPRESSION

[40.71] Fighters may use their TacAir Strength to "suppress" enemy anti-aircraft fire *before* it is used against Friendly planes.

[40.72] Fighter flak suppression does not produce any losses on either side. Rather the fighters divert the flak fire and cover for the other planes. However, any unsuppressed flak may fire at any planes in the hex, including any fighters on a flak suppression mission. Flak suppression is a mission.

[40.73] Fighter flak suppression is flown after air-to-air is resolved but *before* any flak fire against planes is undertaken.

[40.74] For every three fighters (individual planes) that fly flak suppression, *one* point of anti-aircraft/flak fire is neutralized and may not be used that Operations Stage. Only light AA points

and ship AA points may be suppressed by fighters. Heavy AA points may not be suppressed by fighters.

[40.75] Fighters flying flak suppression expend ammunition. The flak units neutralized also expend ammunition, as they are firing at the fighters!

[40.76] Anti-aircraft units may only be destroyed by bombing; see Case 41.3

[40.77] Planes may strafe ships to suppress flak, as a regular flak suppression run. They may not damage a ship by strafing. Ships may be damaged by bombing and torpedoes (41.34). Remember 39.2 (Combined missions).

[40.8] STRAFING TABLE (see Charts and Tables)

[40.9] NIGHT FIGHTERS

[40.91] Night fighters (so indicated on the Aircraft Characteristics Chart) have all the characteristics or regular fighters, with one exception: night fighters *may* scramble against a night bombing mission, but when they do, the Player rolling for scramble *adds* two to the die roll.

[40.92] The following night fighters have their Maneuver Ratings lowered by 4 points when used during the day: Blenheim IVF, Bf 110, and Ju88c.

[40.93] Planes on offensive CAP have no 20C at night.

[40.94] Any CAP must be designated as being flown at night, and such CAP/Escort has no effect on other missions.

[41.0] BOMBING MISSIONS

GENERAL RULE:

Any plane with a Bombload Capacity greater than "zero" may engage in a bombing mission. There are many varieties of bombing missions, delineated by the target type. In addition, certain planes have the capability of engaging in night bombing missions, which decrease the chances of getting attacked by enemy CAP, while lessening the likelihood of damage to a target. Certain planes can engage in bombing missions while strafing. See 39.2. To determine the effect of a bombing mission the Player simply totals the Bombload Points (Tons) delivered to the target and consults the Air Bombardment CRT corresponding to the target type to see what damage, if any, has resulted.

PROCEDURE:

During the Strategic Air Planning Stage of each Game-Turn, the Players must assign any bombers that they wish to use in that Game-Turn either to land support missions, (not specifying what missions each plane or squadron will perform, but just assigning them to general land support missions for that Game-Turn), or to strategic convoy reconnaissance and/or bombing missions (commonwealth player only). All such assignments should be in writing. Any Malta bombing missions are resolved immediately, in the Strategic Air Planning Stage, and any bombers assigned to such a mission may not be reassigned or used again until the next Game-Turn. Any convoy missions are resolved during the appropriate Segments of the Convoy Resolution Phase of the Naval Convoy Stage of each Game-Turn; any bombers assigned a convoy mission may not be reassigned or used until the following Game-Turn. During *each* Land Support Air Phase (there are three in each Game-Turn, one in each Operations Stage), both Players may assign any eligible bombers assigned to "land

support missions" to specific missions and after air-to-air combat and flak fire are resolved, the missions themselves are resolved.

[41.1] RESTRICTIONS ON BOMBERS

Note: The term "bombers" generally refers to planes that do not require pilot points.

[41.11] Bombers may not scramble, nor do bombers have an air zone of control.

[41.12] Bombers may bomb only one target per mission, and that target must be selected prior to actual flight. (Sorry, you can't drop your excess bombs on Tobruk on the way back.)

[41.13] Bombers may abort before dropping their bombs on their selected target. (See Case 41.3.) However, whether the bombers abort or not, they are considered to have dropped their bombs somewhere, and thus expend their ammunition.

[41.14] In order to fly any mission (excepting Transfer), a bomber must be refueled and refitted (as well as armed).

[41.15] Players will note that bombers have parenthesized TacAir Ratings. This means that they may not be used except to fire back at Enemy fighters that attack them. See the Section on Air-to-Air combat. (45.0).

[41.16] Certain planes may undertake two missions simultaneously (see Case 39.2) dive bombers may strafe and bomb the same target and fighters with bombload capacities may both strafe and bomb a target, if they have a "D" capability. *However*, fighter-bombers may *either* bomb *or* undertake a fighter mission; they do not have the capacity to undertake both types of missions at once. If a fighter-bomber is flying a bombing mission, it is considered to have a parenthesized TacAir Rating (unless it jettisons bombload).

[41.17] Torpedoes may be used only against ships and ports.

[41.2] LAND SUPPORT MISSION PROCEDURE

[41.21] Land support missions are concerned with bombing land installations, ports, personnel and equipment in furtherance of operations on land. Bombers assigned to land support missions for a Game-Turn are assigned a specific mission in a specific hex during the Mission Assignment Segment of a Land Support Air Phase. Thus, to assign a squadron of Wellingtons to bomb the port facilities at Tobruk, the Commonwealth Player would write "B-PF/Tobruk" (or any other reasonable representation of that intent). A counter representing all of the bombers of each type assigned to a particular mission in a hex is placed in that target hex (players will have to keep track on paper of which bombers are represented by each counter), unless they are intercepted en route by Enemy CAP (see Case 40.27). (That is why it is important for a Player to be aware of the path of flight; if it passes through an enemy offensive CAP Air ZOC, the Enemy fighters may attack the bombers). Bombers may also be intercepted by Enemy fighters assigned to scramble missions.

[41.22] Once a bomber has reached the target hex, any air-to-air combat is resolved. Then any flak fire is resolved. Then, after any losses are subtracted, any remaining bombers may drop their bombload, and the opposing Player determines the effects of same. He does this by totaling the number of bomb tons delivered against the particular target and referencing the proper portion of the Air Bombardment and Secondary Barrage Target Combat Results Table (41.5). He then rolls two dice reading sequentially and applies any results. Any surviving planes are then returned to base (subject to Enemy CAP interception on the way).

[41.3] LAND SUPPORT BOMBING MISSIONS

All Land Support bombing missions except "mining harbors" are resolved using the Air Bombardment and Secondary Barrage Target Combat Results Table (41.5). Mining harbor attempts are resolved using the Mining Harbors by Plane Table (41.9).

[41.31] **Bombing Personnel (B-CU):** The effect of the anti-personnel bombing mission is to *Pin* a number of units (battalion-equivalent) in the target hex; see Case 41.9 for the explanation of Air bombardment pins. The Player totals his bomb total and consults the Air Bombardment CRT. He rolls two dice, reads the results sequentially, and cross-references that diceroll with the bombload tonnage. The result is the number of battalion-equivalent units Pinned. However, for this purpose, each tank or gun (artillery/anti-tank) unit is worth *two* units (AA/Flak units may be pinned in the way). The choice of units to be Pinned is determined the same as an artillery barrage (See 12.2). There is one exception to this section: Units in *Major Cities* may not be bombed unless and until the fortification level of the city is reduced to one (1) or less.

[41.32] **Bombing Truck Convoys (B-TC):** The results are given in the number of truck units destroyed. The defending Player must divide the losses as evenly as possible amongst type of trucks and types of cargo. Trucks in major cities may not be bombed by air until the city is reduced to zero ('O'). In addition, first line trucks (attached) may be bombed in similar fashion, except that first line trucks may be bombed only by planes with a "D" capability. The method and results are the same, however.

[41.33] **Flak Destruction (B-FS):** The results are in terms of the number of TOE strength points with an Anti-Air rating that are destroyed. Flak/AA may also be pinned; see B-CU above. Flak destruction bombing takes place *after* flak fire, unlike fighter flak suppression.

[41.34] **Bombing Ships (B-CF):** The result is the number of Damage Points inflicted on the ships in the hex (See 30.34). Bombing of the Commonwealth Fleet may not be undertaken at night. Torpedoes may be used against ships (See 41.7). See also 41.6. Axis coastal shipping may not be bombed.

[41.35] **Bombing Supply Dumps (B-SD):** The result is the percentage of each type of supply in that dump that is destroyed. In addition, if there are any unattached trucks in the hex, for every 10% of supplies destroyed, one Truck Point is lost, choice of defender, dividing losses as evenly as possible.

[41.36] **Bombing Air Facilities (B-AF):** The result is the number of capacity levels. That that facility is reduced. In addition, for every level destroyed, remove 10% of the planes on the ground (e.g., 2 levels, 20% planes), rounded down.

[41.37] **Bombing Major Cities/Fortifications (B-F/C):** Any major city or hex with fortifications may be bombed. If the target is a major city, the mission may be flown at night (41.4). Only one level of fortification may be destroyed in any Operations Stage. See also 22.35. Players may use this mission to neutralize a temporary repair facility that is not in a major city hex. If the Player obtains a result that would reduce the fortification level by one, the temporary facility is neutralized (and may not be used) for one Operations Stage.

[41.38] **Bombing Roads or Railroads (B-R or B-RR):** This is done in the same fashion as Bombing Cities, using the same table (but a different row, according to the target). Roads may be bombed at night if the Road is in a Coastal Hex.

The results are simply having the road or railroad hex destroyed. (Railroads are very easy to repair; whether they are worth the effort to destroy is the player's problem.)

[41.39] A. Mining Harbors (B-MH): Players use the Harbor Mining Table, with the Player rolling for each six bombers attempting to successfully lay the mines. Rolling one die, if the Player succeeds, place one mine counter in the port hex. The result of a mine is the same as blocking harbors: **Note:** The capacity column refers to the average bombload for all six planes (Not this total of six). **B. Bombing Ports (B-P):** The Purpose of this mission is to reduce the efficiency level (55.1) of the port. Players total bombload and strafing delivered to the hex and consult the Air Bombardment Table. The number result is the loss in efficiency rating for that port. (See Case 41.8 for bombing Tripoli, etc.) Torpedoes may be used against ports.

[41.4] NIGHT BOMBING MISSIONS

[41.41] Certain bombers may fly night missions. Bombers capable of flying and bombing at night have a capability code of "N" or "O" in their capability column.

[41.42] In night bombing, accuracy is sacrificed for the sake of safety. Only fighters with night fighting capacity may escort bombers on night missions.

[41.43] The Player must designate that a given bombing mission is a night mission as such: B-P(N). See the Air Mission Summary (Case 39.5) for a list of which missions may be flown at night.

[41.44] All regular missions are resolved before night missions, and the "defending" Player must be informed if a given mission is a night mission prior to all resolving of missions, scramble, etc.

[41.45] Bombers flying night missions have their bombload tonnage halved, rounded down. Torpedoes may not be used at night.

[41.46] Only Night Fighters may Scramble against a Night Bombing Mission (see Case 40.9). Fighters without NF capability may be given an offensive CAP versus night raid mission, but have no ZOC, and may only search for (and thus attack) Enemy planes with a mission in the same hex (see Case 41.96). Night Fighters also go through the search procedure in 41.96

[41.47] Fighters of CAP as well as Enemy night fighters scrambling against night bombers must go through a search procedure in order to find the bombers. The Player rolls one die for each Squadron; if he rolls a '1' the fighters have found the bombers (and any NF escort). Subtract '2' from any search die roll involving squadrons of night fighters (at night). If the die roll is not good, search is unsuccessful and the fighters go home. Any air-to-air combat proceeds normally.

[41.48] Flak/AA fire against night missions is adjusted one column to the left. If already in the lowest column, the flak fire is considered ineffective.

[41.49] Night bombers do *not* get any benefits from formation flying (45.36).

[41.5] AIR BOMBARDMENT TABLES (see Charts and Tables)

[41.51] Air Bombardment & Secondary Barrage Table

[41.52] Mining Harbors by Plane Table

[41.6] STRATEGIC PHASE BOMBING MISSIONS: AXIS NAVAL CONVOYS

[41.61] The Axis Player uses the convoy shipping lanes to place his arriving convoys of supplies and

personnel (see 55.1). Each of these convoy shipping lanes has distances in hexes from certain key cities listed in the lane. This is the airflight distance from that city (see 56.18).

[41.62] Commonwealth bombers assigned to the Strategic Air Phase may fly bombing missions against Axis naval convoys. This includes Africa-based bombers as well as those based on Malta. The only requirement is that the distance flown is within that plane's range. In tracing the range from Africa-based air facilities, the planes must fly to one of the cities listed in the lane and then fly to the lane from that point. The two distances are added to obtain the full distance.

[41.63] Axis fighters based in North Africa may not fly CAP for Axis Naval Convoys. Axis fighters based in the Mediterranean may fly CAP for any Convoy Lane within range. Commonwealth fighters may fly CAP or Flak Suppression missions to any Convoy Lane within range. The distances to the Convoy Lanes are given in the Axis Naval Convoy Air Distance Chart (Case 56.18). Note that planes flying from North Africa must be able to fly to one of the cities listed plus the additional distance from that city to the Convoy Lane.

[41.64] The Commonwealth Player may only attack Axis Convoys that he has located. Axis Convoys may only be located as a result of a Strategic Convoy Reconnaissance Mission.

[41.65] The procedure for resolving attacks against Axis Naval Convoys is as follows:

A. The Players divide their planes between those assigned to Land Support and those assigned to Strategic Missions. The Axis Player subdivides his planes to those performing Air Raids on Malta and those flying convoy protection.

B. The Commonwealth Player performs his naval recon missions. Note that the planes may not be attracted by air-to-air combat or anti-aircraft fire.

C. The Axis Player assigns his convoy protection aircraft CAP missions in specific Convoy Lanes. The Commonwealth Player assigns his planes to a CAP, flak suppression, or bombing mission in any convoy lane in which a convoy has been located (remember range restrictions). If no convoys have been located, the planes never take off and therefore do not have to undergo readying.

D. Air-to-air combat is resolved in the individual convoy lanes.

E. Flak Suppression of the convoy is performed.

F. Anti-aircraft fire attacks from the convoy are resolved.

G. Commonwealth bombing of the convoy is resolved. The Axis Naval Convoy Missions are now completed. All of the surviving planes will return in base during the Strategic Air Recovery Stage.

[41.66] Convoy bombing is resolved on the Air Bombardment and Secondary Barrage Target Results Table. The Commonwealth Player totals his attacking bomb points, modifying for torpedoes if necessary (see Case 41.7), and rolls two dice. The cross-indexed result is in tens of percent of the cargo in that convoy lost.

[41.67] All losses are evenly divided percentage-wise among the various cargo types, rounding fractions up. All truck replacement points are considered to be one cargo type. All Italian tanks are considered to be one cargo type. All German tanks are considered to be one cargo type. All Italian guns are considered to be one cargo type. All German guns are considered to be one cargo type.

[41.68] Bombing of Axis Naval convoys never affects the arrival of listed reinforcements.

[41.7] TORPEDOES

[41.71] Certain planes have the capability of carrying torpedoes, which are effective against ships and port facilities (See Case 38.47 for arming Bombers with Torpedoes.)

[41.72] When Axis planes are carrying torpedoes against Enemy ships (this refers to Axis plane vs. Commonwealth Fleet) the Axis Player uses the torpedo line, not the normal bombload line. Thus, if he has a mixture of planes (some carrying bombs and some carrying torpedoes) he rolls *twice* for that mission: once for the bomb-carrying planes and once for the torpedo-carrying planes.

[41.73] When Commonwealth planes fly against Axis convoys and at least 50% of the planes are carrying torpedoes, in determining the level (Case 41.66), increase the bombing tonnage by 25%, rounding upward. Thus, if 160 tons of bombs were delivered to convoy route Nr. 3 via 30 planes, 17 of which were armed with torpedoes, the tonnage would be increased to 200 tons (effective).

[41.74] Torpedoes may be used against port facilities, but as such count as normal bombs.

[41.75] Only planes with "T" capability can carry torpedoes.

[41.8] BOMBING TRIPOLI

[41.81] Commonwealth bombers may bomb the Tripoli/Tunisia boxes in order to reduce the port capacity level and destroy any planes on the ground. These are the only two types of missions that may be flown against Tripoli, etc.

[41.82] Distances from game-map hexes to Tripoli, etc., are printed on the Air Distance Chart. Add that distance to any actually flown on the Game-Map to get the actual flight distance for the raid.

[41.83] Bombing missions against Tripoli (the port) or Bizerta (Tunis) are conducted as any other similar mission, and the results are the same as any such mission. (See Case 41.39)

[41.84] Bombing missions against grounded Axis planes in Tripoli/Tunisia air facilities are conducted as a Bombing Air Facilities Mission, although the results against the facility are ignored (see Case 41.36).

[41.85] The AA/Flak capacity for Tripoli, et al, is constant, non-destructable, and printed in the respective boxes. CAP coverage depends on the Axis Player, but any CAP coverage is for all targets in *that* box.

[41.86] Night missions may be flown against the port of Tripoli.

[41.9] PINNED BY BOMBING

The effect of a *Pinned* result inflicted on a unit by Air Bombardment is different from an artillery barrage pinned. (This rules Section supercedes the somewhat incorrect definition of Pinned given in Case 3.1; in essence this case is an early errata.) The Pinned by Bombing result depends on whether the "pinned" unit belongs to the "A" Player or the "B" Player.

[41.91] Pinned Unit: "A" Player

The following results occur from a Pin to a unit belonging to the A Player for that OpStage:

1. The unit may voluntarily use only ½ (rounded-down) of its CPA; and
2. Such unit may not voluntarily enter an Enemy ZOC; and
3. If such unit is in an enemy ZOC when Pinned such unit may not leave that ZOC; and
4. Such unit may not voluntarily engage in combat; i.e., it may not barrage or use Offensive Close Assault or Anti-Armour.

As soon as the OpStage switches to the B Player these results are no longer in effect.

[41.92] Pinned Unit: "B" Player

The following results occur from a Pin to a unit belonging to the B Player for that OpStage: such unit may not move voluntarily or engage in combat voluntarily during A OpStage. As soon as it is B's turn, the effects of Section 41.91 apply. The unit becomes unpinned at the end of that OpStage.

[42.0] NON-COMBAT MISSIONS

[42.1] TRANSFER MISSIONS

[42.11] A transfer mission is flying a plane from one air facility to another.

[42.12] Transfer missions should be flown before undertaking any other missions, so as not to forget them in the heat of battle. Unless they run into Enemy CAP (see 40.27) they should not run into any problems.

[42.13] Planes flying transfer missions may double their range. Transfer is a one-way flight; transfer planes do not (cannot) return to their base of origin.

[42.14] Planes flying transfer need not be refitted to fly. However, a plane completing a transfer mission *must* be refitted to fly another mission in the next Stage or Turn. Transfer missions consume fuel; whether planes conducting them are armed is a question of choice.

[42.15] To assign a plane a transfer mission, write "Transfer to" and the hex of destination in the mission box of the Squadron Control Sheet. The plane is then placed in the new hex. Transfers are flown only in Tactical Land Support Phases of the Operations Stage.

[42.2] RECONNAISSANCE MISSIONS

[42.21] Reconnaissance (Recon) missions may be flown by any plane with a *Recon Capability* ("R").

[42.22] The object of a Recon mission is to find out what units—in general terms—the Enemy has in a given hex. Therefore, Recon missions are flown over a specific hex with an Enemy unit in it. Recon missions may be flown over any hex except for major cities.

[42.23] The Player assigns a number of planes to fly Recon to a specific hex. After all air-to-air and flak, the Player totals the number of planes on Recon in that hex and consults the Air Reconnaissance Table (42.27), rolling one die. The result is the number of units in hex the Player has learned information about (42.24).

[42.24] For each battalion-sized unit the player flying Recon can learn about, the opposing Player must reveal the type of unit and its TOE strength under the following conditions:

- All tank units (including dummy tank units, which are considered to have a TOE Strength of "three" for this purpose) revealed first, then infantry, then artillery.
- The opposing player must give the TOE strength, but may "lie" in either direction to a deviation of '2'. Thus, for a unit with a TOE of '3' he could say it had '1' or '5' or any number in between.

[42.25] No Recon mission may be flown in a hex that is undergoing air bombardment.

[42.26] Land Recon missions are flown in the Tactical Land Support Air Phase of the OpStage. Planes flying Recon require refit. Recon missions

against Axis Naval convoys are flown during the Strategic Air Phase.

[42.27] Air Reconnaissance of Land Units Table (see Charts and Tables)

[42.3] TRANSPORT MISSIONS

[42.31] Certain planes with transport capability may transport personnel or supplies by air. The cargo capacity of each transport is listed on the Aircraft Characteristics Chart in terms of TOE Strength Points or tons of supplies.

[42.32] For a Transport to carry a TOE Strength Point or supplies to a given hex, that "cargo" must start the Operations Stage in the hex with the transport. The transport may then fly one type of transport mission:

- The Transport flies to the target hex, lands the cargo, and flies back; or
- The transport flies to the target hex and lands, remaining in that hex. In this case, the transport may use its transfer range (double its normal range).

[42.33] Transport missions must be flown to Friendly air facilities, as the planes have to land. Supplies and personnel may be *airdropped* under special circumstances (See 42.4)

[42.34] Personnel transported by air may not voluntarily move in the Operations Stage they are transported in, having used their allotted CPA.

Personnel may be flown into a hex in an Enemy Zone of Control; however, such units may not be used for any attacks. That is, they may not voluntarily expend CP's on combat. They may defend. Personnel air-transported may never voluntarily exceed their basic CPA, through reaction or retreat before assault.

[42.35] Supplies air-transported may be used as soon as they can be assigned to a dump or specific units. They cannot be used directly "off the plane." They must be distributed in the Organization Phase first.

[42.36] Transport missions may be flown only in Operations Stages. Planes require refitting, even when flying Transport Missions.

[42.37] Only infantry (without trucks), motorcycle infantry/recce units, and airborne units may be transported by air. **Exception:** *dead* camels may be transported by air. *Live* camels may *not* be transported by air.

[42.4] AIRDROP MISSIONS (PARATROOPS)

Although there were several airborne units in the desert campaign (The Ramcke Brigade and the Folgore Division), there were no major airborne missions. The following rules are, in essence, hypothetical.

[42.41] Units capable of being airdropped are so noted on the OA Sheets. Such units may be airdropped by transport planes during a tactical land support Air Phase, under certain circumstances.

[42.42] An airborne mission is the same as a transport mission (42.32a). The only difference is that the units are *dropped* into the hex, not landed, so an air facility is not needed for the transports. However, they must start from a Friendly air facility.

[42.43] Units may be airdropped into any clear, gravel or desert hex. They may not be airdropped into major cities and they may not be airdropped into hexes occupied by Enemy units or hexes in an Enemy Zone of Control. Units may be airdropped into a Friendly occupied hex, within stacking restrictions, as long as such hex is not in an Enemy ZOC. In addition, the drop-zone hex may not be more than two hexes from a recognizable land feature (42.44).

[42.44] A recognizable land feature is a major city, a road (unfinished, destroyed or not) or railroad hex, a village, air facility, or oasis.

[42.45] Airdrops may not occur in sandstorms or rainstorms. However, there is no drift; units always drop in the target hex. There are no casualties in an airdrop.

[42.46] Airdropped units require and use supplies the same as any unit: they are *not* commandos. Thus, only certain types of supplies may be airdropped: ammunition and stores may be airdropped (using the precepts of 42.42 and 42.43). Supplies may be airdropped into a *Friendly-occupied hex* even if it is in an Enemy Zone of Control. In the latter case, there is no need to use the recognizable land feature provision of the airdrop rule. Airdropped supplies may be used as soon as they arrive, unlike transported supplies (42.35).

[42.47] Airdropped combat units may move and fight normally, with one exception: they may not voluntarily exceed their CPA in the Operations Stage in which they are dropped.

[42.48] There is no limit to the number of TOE Strength Points that may be dropped (within the limits of how many transport planes you can muster). However, no unit may be airdropped more than once a month (every 12 Operations Stages).

[42.5] NAVAL CONVOY RECONNAISSANCE MISSIONS

[42.51] The Commonwealth Player may fly reconnaissance missions in Convoy Lanes to determine the existence and size of the Axis Naval Convoy assigned to that Lane.

[42.52] The Commonwealth Player must assign each reconnoitering plane to a specific Convoy Lane. A plane may only be assigned to a naval recon mission if it possesses Recon capability (R). A plane may be assigned to any Lane within its range. The distances to the convoy lanes are listed in the Axis Naval Convoy Air Distance Chart (Case 56.18). A plane flying a recon mission from North Africa must possess a range at least equal to the distance from its air facility to one of the key cities listed on the chart *plus* the distance from that city to the Lane.

[42.53] Naval recon missions are resolved on the Air Reconnaissance of Axis Naval Convoys Table. If a successful reconnaissance of the convoy lane has been made the Axis Player must indicate whether or not a convoy is assigned to that lane this game-turn; and if so, its approximate tonnage.

[42.54] Planes flying naval recon missions may not be attacked in air-to-air combat or by anti-aircraft fire.

[42.55] Naval recon missions are Strategic missions that are flown in the Naval Convoy Stage

[42.56] Naval convoy recon missions are assigned in the Strategic Mission Assignment Phase of the Strategic Air Planning Stage. An unlimited number of planes may be assigned recon missions in the same convoy lane. Naval recon missions are resolved in the Naval Convoy Reconnaissance Segment of the Convoy Resolution Phase.

[43.0] AXIS ITALIAN-AEGEAN AIR BASES

GENERAL RULE:

The Northern "shore" of the Mediterranean was almost exclusively controlled by the Axis, especially after the fall of Greece and Crete in early 1941. The Axis often used Italy, Sicily, Greece and Crete to base its bombers, so that they could be used for multi-purpose duty. The German Player is,

therefore, *required* to base a certain percentage of his Bombers in either Italy, Sicily or Crete. Boxes for these are provided on the game-maps, as are their flight distances. There are also certain restrictions pertaining to missions based in the Mediterranean, in terms of fighter support.

[43.1] AXIS MEDITERRANEAN BOMBER BASE REQUIREMENTS

[43.11] At least 75% of all German He 111's, Ju88D's and FW220's *must* be based in Mediterranean Bases.

[43.12] Until 1/35 Game-Turn 1941, 75% of all German bombers must be based in Italy/Sicily.

[43.13] From June 1/35 Game-Turn 1941 to the end of the game at least 50% of all He111's, Ju88's, and FW220's must be based in Crete; the remaining 25% may be based in Sicily/Italy or in Crete.

[43.14] It takes one OpStage to transfer from Italy to Sicily and/or to Crete, and vice-versa.

[43.2] MEDITERRANEAN BASE RESTRICTIONS AND CAPABILITIES

[43.21] German Bombers based in Italy/Sicily and Crete do not need SGSU's. All their requirements are met by the specific "Box" they are in, including fuel and ammunition. The Axis Player does not expend fuel or ammo for these planes. However, all planes must be refitted (38.3)

[43.22] When refitting planes as per 38.37 (per Squadron), the Axis Player should divide all bombers in a given area into groups of 6 to 12, considering each such group a squadron.

[43.23] Bombers based in Crete must be used to attempt to raid the Suez Canal area (not on the game-maps) each month. This is handled abstractly in that during *four* OpStages within a given month there may be *no* bombing missions flown from Crete. There is no effect on the game of such missions (as they had little effect operationally on the Commonwealth, other than harassment and extended work hours). There are also no losses.

[43.24] Bombers flying from Crete may *not* receive any protective CAP from planes in Africa. They fly "naked".

[43.25] Bombers based in Italy/Sicily *may* coordinate with African-based fighters for CAP. They may also be used in raids on Malta (see 44.0).

[44.0] MALTA

COMMENTARY:

The importance of the tiny island of Malta to the campaign in North Africa cannot be understated. Lying, as it does, astride the Axis Shipping Lanes, it provided the Commonwealth with an excellent position from which to hinder the transport of Axis supplies and reinforcement.

GENERAL RULE:

The Commonwealth has several air facilities on Malta on which he may base planes for strikes at Axis convoys and bases. In retaliation, the Axis may attempt to reduce the effectiveness of Malta by bombing the island. The Axis Malta reduction effort is controlled by external policies which have been abstracted into several charts proscribing the extent of such effort. Likewise, the Axis may not invade Malta — at least not in this game. Planes performing missions over Malta must be assigned a specific hex.

[44.1] THE MALTESE AIR BASES

[44.11] The map of Malta represented on Game-Map "A" is not in the same scale as the African portions of the game-maps, nor is it in scale in terms of geographic location vis-a-vis the African

coast. The air distances to key points in Africa are listed in the Malta Box. The map-of Malta itself shows different terrain and cities, mostly for historical purposes. The only items of importance on the Malta map are the air facilities and the port of Valetta.

[44.12] Maltese air facilities are printed directly on the game-map (unlike most others). They are permanent and may never be moved. No further air facilities may be built on Malta. Although they may not be destroyed, Maltese air facilities may be reduced (by Axis bombing) to zero effectiveness.

[44.13] The initial capacity of Maltese air bases is given in each scenario. It may be increased—up to the standard levels—by using the Maltese Air Facility Construction Table (44.5) The Commonwealth Player may use the table once each Game-Turn for each air facility on Malta, the table informing the Commonwealth Player how many "levels" he may construct for each facility. No supplies need be expended.

[44.14] The Commonwealth Player does not need—nor does he use—SGSU's on Malta. Each "level" of air facility can handle up to 18 planes of any type. Thus an airfield operating at maximum capacity (6 levels) could handle 108 planes. However, for the sake of play ease, the Commonwealth Player should group his planes as if they were squadrons.

[44.15] Planes on Malta may be moved from one field to another without flying. Such a movement is not considered a mission and happens "outside" the play sequence. In essence, the Commonwealth Player simply moves his squadrons or planes from one field to the other at the beginning of the OpStage, noting such movement on his squadron sheets, prior to any missions being flown. The Axis Player need not be informed of such moves.

[44.16] Planes based on Malta do not need fuel or ammo; they are automatically refueled and rearm-ed each Stage. However, they must be refit like all other planes, using the same method as all other planes.

[44.17] Commonwealth planes may transfer to and from Malta and the African fields, provided they have the range.

[44.18] The Commonwealth Player may divert AA/Flak points (Replacements) to Malta. For each light AA Flak point he diverts to Malta he receives one AA Point on Malta; for each heavy AA point he diverts he receives four AA points on Malta. These points come from the Commonwealth Replacement Pool for AA points. No more than one AA Replacement Point (Light or Heavy) may be diverted to Malta per Month, and the maximum AA point capacity of Malta is 48 AA points.

No flak counters are placed on the Malta map; the Commonwealth Player simply notes which bases have how many Flak Points. Any losses from bombing are noted as they occur. Flak points may move between airfields/strips as planes do in 44.15 but flak/AA units may not be moved between Malta and the mainland.

[44.2] AXIS AIR RAIDS ON MALTA

[44.21] Axis bombers based in Italy/Sicily as well as any African-based Axis planes that can reach the island can raid Malta, bombing the Maltese air facilities to reduce their capability to refit planes.

[44.22] In addition, to 44.21 the Axis may draw on Italian and German aircraft based *permanently* in Italy and Sicily (and otherwise not available to the Axis Player). These planes are reflected by the Axis Malta Availability Levels. (See 44.41.)

[44.23] There are four Axis Malta Availability Levels, representing the levels of Axis commitment to Malta. Each level gives the number of Game-Turns during the Campaign Game that that portion may be used. Each shorter scenario has separate listings. No Availability Level may ever be used more Game-Turns than it is listed as available. It is up to the Axis Player to plan his raids according to what he wants to accomplish and when he wants to do it.

[44.24] Axis Raids on Malta take place in the Strategic Air Phase of the Game-Turn, and thus occur only once per Game-Turn. (This is, admittedly, somewhat abstract—and the various Malta tables reflect this abstraction.) The Axis Player chooses the planes in the game he wishes to bomb Malta with by consulting the portion of the Malta Availability Table of his choice (44.23). He assigns each squadron to a specific airfield in Malta and conducts each bombing and/or strafing raid normally—including air-to-air and AA/flak.

[44.25] To compute how many — and what types — of planes he may use, the Axis Player uses the Axis Malta Table (44.42). In addition to the planes he receives from this Table he may then add in — up to the maximums he gets from the Table — any planes he wishes from Africa. In essence, the Table shows the Axis Player how many of his planes in the game, but in Italy/Sicily he may use plus how many planes not in the game (for normal use) but available nonetheless that may be added to this total. The two are combined to provide the number of Planes Available, to which are added any Africa-based planes. If, at any time, the Axis Player does not wish to bomb Malta he is considered to have used Level I (which is unlimited).

[44.26] In using the Axis Malta Availability Tables, the Axis Player always assigns his map-based planes *after* determining how many planes he will get from the tables.

[44.27] In no case may the Axis Player assign more planes of any one type from map bases than are assigned from the Availability Tables. That is, if the table says 6 S.M. 81's may be used, the Axis Player may add in no more than 6 map-based SM 81's. Furthermore, if a plane is *not* assigned from the table it may not be assigned from the map.

Example: The Axis Player decides to launch a Level III raid on Malta. He has, based in Italy and Sicily, 12 SM 79's, 6 BR 20's, 16 CR 42's, and 4 CR 32's. He consults the table and rolls a "6". As per the number on the left of the slash he may assign 50% of these available planes to the mission. He may add to this, as per the percentage to the right of the slash, a number of planes *equal to* (100%) what he has based in Italy/Sicily. These planes are not actually in the game but have been provided by the Axis in Europe. In essence, the Axis player now has available 150% of the total planes he has based in Italy/Sicily. He may also add to this the same number of planes (or less) — by type — as he has available from the above from bases in Africa.

[44.28] To determine what losses are absorbed by planes actually in play — as opposed to those taken by planes that are used only for Malta bombing as per 44.4 — the player must obtain a ratio between the planes he uses that come from (1) Africa, plus (2) the percentage to the left of the slash on the Table, and compare that total to the planes he uses from the percentage to the right of the slash on the table (i.e., those planes not really in the game but available for bombing Malta). This ratio is used to obtain an Actual Loss Percentage, which is then applied to any total losses (from the CRT's), rounding *up*. The Axis Player applies this revised (and usually lower) loss

number to those planes he actually has in the game. Planes not actually in play never take losses; they are an applied constant.

Example: The Axis Player has one squadron of Ju88A's from Benina and two squadrons of Ju88A's from the Malta Availability Table. All squadrons have 12 planes, for a total of 33% of the total comes from Benina. The three squadrons attempt to bomb Luqa airfield. The squadrons lose 6 planes to air-to-air and 2 to flak. In addition, 2 are forced to abort, a factor not considered—simply noted—when determining who takes losses. 33%, rounded up, of 8 means 3 bombers from Benina have been shot down. The number of bombers lost in Italy/Sicily from the Availability Table is not considered.

[44.29] The Axis Player may, *after* rolling on the Table, decide not to raid. The raid is cancelled, but he still has used the table he rolled for once—regardless of whether he cancels or not (see Case 44.23.) In such a case, the Axis Player may not, of course, use map-based planes.

[44.3] INVASION OF MALTA

Realizing the importance of Malta to the Axis supply effort, the German High Command contemplated invading Malta on several occasions, even going so far as to set a definite date. However, that plan was aborted when Rommel borrowed the air cover assigned to such invasion for his thrust to take Tobruk—and then never returned them. In any case, the German High Command had deep reservations about an invasion of the strongly-defended island, and given the fearsome losses taken by German paratroopers on Crete it is more than likely that the Germans never would have carried their planned invasion through to the end. They certainly would have received little valuable support from the Italians who, despite several professed "airborne" divisions (like the Folgore) had little practice in this sort of thing. It is mainly for these reasons that—at least in this game—the Axis Player may not invade Malta. Sorry.

[44.4] STRATEGIC AIR RAIDS ON MALTA

(see Charts & Tables)

[44.41] Axis Strategic Airforce Commitment Chart

(see Charts & Tables)

[44.42] Axis Malta Availability Table

(see Charts & Tables)

[44.5] MALTA AIR FACILITY CONSTRUCTION TABLE

(see Charts & Tables)

[45.0] AIR-TO-AIR COMBAT

GENERAL RULE:

Although CNA uses individual planes, the resolution of air-to-air combat is done on an operational level, with much abstracted (in terms of plane ability, etc.). Air-to-air combat occurs before flak/AA fire and before completing any missions. Essentially, only fighters with non-parenthesized tacair ratings may *initiate* air-to-air combat, and fighters on offensive CAP must initiate Air-to-air combat. Air-to-air combat is resolved by comparing tacair ratings, augmented by the pilot ratings, and further adjusted by any difference in maneuver ratings. Bombers' tacair ratings may be augmented if they are formation flying. Planes resolve air-to-air in pairs, one plane against another, and all combat takes place within the hex.

PROCEDURE:

The Players first determine which planes in the hex will be engaging in air-to-air. This means all scrambles must be performed first. Each Player then declares whether or not he wishes to initiate air-to-air, with any player having planes on offensive CAP forced to initiate. Fighter vs fighter (or screen) combat takes place before fighters may attack any Enemy bombers/ transports, etc. (unless they are unescorted).

Having determined the planes involved, the Players now decide who will be the Attacker (see Case 45.16). (Remember, all Missions are flown simultaneous). The defender places all his planes being attacked down on a flat surface, using available aircraft counters (see 34.75). The attacker now allocates his attacking fighters against the defending planes. Each individual air-to-air combat is now resolved, using the Air-to-Air Combat Tables. After attacking fighters have attacked escorts/screens they may, if possible, now attack the bombers/ transports. After all air-to-air is resolved, players complete their missions (after, of course, undergoing flak/AA fire.)

All opposing planes involved in air-to-air combat are individually "matched up", (see Case 45.2). The Players now determine each airplane's tacair differential. In order to do this, the Players first adjust the tacair rating of their aircraft in one of two possible ways:

- For each *fighter*, the owning Player adds the rating of the pilot to that of his plane.
- For each *bomber*, add to the tacair rating any applicable formation flying bonus: if 6-17 bombers are on the mission, add one to each bomber's tacair rating; if 18 or more bombers are on the mission, add two to each plane's tacair rating. (Note: Dive bombers are not eligible for formation flying bonuses.)

Each Player now subtracts the adjusted tacair rating of the Enemy plane from his own plane's adjusted rating. This is the *tacair differential*. The tacair differential is now adjusted for each plane to account for the relative difference in maneuverability of the planes. The maneuverability rating of the less maneuverable plane (the lower rating) is subtracted from that of the higher, and the Maneuver Adjustment Chart is consulted to obtain the appropriate maneuver adjustment. This Maneuver Adjustment number is *added* to the tacair differential of the plane with the higher maneuverability rating and subtracted from the tacair rating of the plane with the lower maneuverability rating. (Note: If the maneuver ratings are equal, no such adjustment is possible.) Thus, if prior to the adjustment for maneuver ratings one plane had a tacair differential of -3 and the other +3, but the plane with the lower tacair differential had a maneuver rating 16 points higher than the second plane, then the tacair differentials would be adjusted, respectively, from -3 to +1 and from +3 to -1, since the Maneuver Adjustment Chart calls for an adjustment of "4" in favor of the plane with the higher maneuver rating.

In fighter vs. fighter match-ups, the fire of the plane with the "minus" tacair differential is resolved first. (For "0" differentials, the plane with the highest rated pilot is considered to fire first, or if that is a tie, the attacking Player's plane.) If there is more than one fighter attacking one Enemy Fighter, the lone fighter fires first. In fighter vs. non-fighter, the non-fighter plane's attack is always resolved first.

Using the final adjusted tacair differentials, the Players refer to the tacair Kill Table (45.5), and the Player firing first rolls two dice and combines the results sequentially to obtain a result (e.g., "2", "4" equals "24"). If the result falls within the kill

range given on the table corresponding to the appropriate tac air differential, the opposing plane has been shot down. Otherwise, the opposing Player with the aircraft with the positive tacair differential gets to fire. If his dice roll result falls within the kill range as delineated on the TacAir Kill Table, the first plane is shot down; if not, then there is no result. (Note: Non-fighters fire individually at each Enemy fighter that is matched against them; fighters may *never* fire at more than one plane, even if attacked by more than one.)

Figure A:

PLAYER A			
Plane Nr./Type	Pilot	TacAir	Manuever
1 Gladiator MkII	1	2	27
2 Gladiator MkII	0	2	27
3 Tomahawk	0	2	32
4 Tomahawk	0	2	32
5 Hurricane MkI	0	4	32
6 Hurricane MkI	3	4	32
7 Hurricane MkI	1	4	32
8 Hurricane MkI	0	4	32
9 Beaufighter MkIF	0	11	30

Figure B:

PLAYER B			
Plane Nr./Type	Pilot	TacAir	Manuever
A Fiat C. R. 42 Falco	1	3	28
B Fiat C. R. 42 Falco	0	3	28
C Messerschmitt Bf. 100	2	5	32
D Me Bf 110	1	5	32
E MeBf 110	0	5	32
J Me Bf 109 E	2	6	34
K Me Bf 109 E	1	6	34
L Me Bf 109 F	0	6	39

Figure C:

PLAYER A'S PLANES								
1	2	3	4	5	6	7	8	9
D	E	G	H	I	J	K	L	B
A	C					M		
	F							

PLAYER B'S PLANES

The preceding is an illustration of determining the plane match-ups in the following example of planes about to engage in air-to-air combat. Player A (Commonwealth) possesses nine planes while Player B (Axis) possesses thirteen planes, making the former the defender and the latter the attacker for this particular group of air-air combats.

Figure: The Commonwealth Player has two Gladiator MkII's, one piloted by a 1 pilot; two Tomahawks; four Hurricane MkI's, one piloted by a 3 pilot and one by a 1 pilot; and one Beaufighter Mk IF. The Commonwealth Player numbers his planes (for convenience) in ascending order by combat worth. Within the numbers assigned to each type of plane he randomly determines the rank of the pilot handling the aircraft.

Figure B: While the Commonwealth Player is listing his planes the Axis Player is performing the same actions. His job is slightly easier as he is not troubled by having to jumble his pilot ratings.

Note that if either or both Players possess identical aircraft with identically rated pilots that arrived in this particular hex from different squadrons the owning Player would have to indicate the parent squadron for each of those planes.

Figure C: The Commonwealth Player announces the type number and ID numbers for each of his planes *without* stating the pilot's rank. The Axis Player then matches up his planes in any pattern and order against the Commonwealth Player's aircraft, making sure that each Commonwealth aircraft is opposed by at least one Axis aircraft. In the case of a plane being attacked by a number of planes and only able to shoot back at a single one of the attackers (this particular type of example), he names the order in which the multiple attacks on the opposing aircraft are to be performed (as the defender may only shoot at one of the attacking planes). The Players then proceed to resolve the individual air-to-air combats.

Note that in this example the Axis Player has apparently decided to take the long view, with the worst plane (Italian C. R. 42) being sacrificed to the Beaufighter and ganging up on the two worst (pilot distribution being equal) Commonwealth planes.

CASES:

[45.1] HOW AND WHEN AIR-TO-AIR COMBAT OCCURS

[45.11] Air-to-air combat takes place within a hex. Only planes in a given hex may engage in air-to-air combat with each other, regardless of air Zones of Control. (Remember, fighters with air ZOC's *may* move into an adjacent hex to engage in air-to-air; if they are on offensive CAP in a hex without opposing aircraft they *must* move into an adjacent controlled hex to engage.)

[45.12] All planes in a given hex are subject to air-to-air combat. However, there are priorities as to who attacks whom, and when. The primary rule is that only fighters or fighter-bombers on a fighter combat mission may initiate air to air combat.

[45.13] Air-to-air combat occurs before flak/AA fire and before any missions in that hex may be completed. Remember, fighters may scramble or intercept into a hex, and that must be done before attempting to resolve air-to-air.

[45.14] If there are fighters in a hex on offensive CAP, those fighters must attack all Enemy planes in that hex. Thus, fighters on offensive CAP automatically initiate air-to-air combat. If there are no fighters on offensive CAP, both Players must decide (secretly; to be revealed simultaneously) whether they want to initiate combat. If *either* Player decides to initiate combat, combat is joined.

However, if *neither* Player desires air-to-air, *both* Players send their aircraft home, and all missions in that hex are cancelled. (This is sort of a free abort; planes may always try to abort even if the other Player desires combat, but this is not automatic. See 39.3.)

[45.15] Once air-to-air has been initiated, the Players determine which planes, if any, are being *screened* by Friendly fighter escort (planes flying defensive CAP for the bombers, etc.). Only planes flying a mission *other than* CAP may be screened. Furthermore, planes flying a mission other than CAP are automatically screened, unless they have no escort. Planes without Friendly defensive CAP are *not* screened. The Players now set the screened units aside, and engage in air-to-air with their fighters on CAP.

[45.16] The Players must now decide who is the attacker. The Player with the most fighters (pure numbers) is the Attacker in all situations. In the instance where both Players have the same number of fighters, the Player with initiative for that Game-Turn (7.0) is the attacker. At this point, Players attempt to abort any missions they desire to abort. See 39.3.

[45.17] A plane without ammo (bombs aside) may not fire in Air-to-Air combat. However, its normal ratings are still used to resolve the effects of opposing fire.

[45.18] Players now resolve the first wave of air-to-air combat (45.2). Once this is completed, the screened planes may be attacked. (See 45.3). This is resolved in the same way, with certain restrictions and changes.

[45.19] A plane assigned to a particular match-up may never be switched to another match-up, even if the plane it was matched-up against is shot down before it has a chance to fire.

[45.2] FIGHTER VS FIGHTER

[45.21] In resolving air-to-air combat between fighters (planes with nonparenthesized tacair ratings), the Player follows the general outline of the Procedure augmented as below.

[45.22] The Defender places his fighters down on a flat surface, using the aircraft counters. He then states what *type* of plane (e.g. Me109F, CR42, etc.) each fighter counter represents. He does *not* reveal the rating of his pilot at this time.

[45.23] The attacker must now assign at least one fighter to *every* fighter the defender has placed down. Once every defending fighter is "covered", the attacker may assign more than one fighter to a defending fighter. There is no limit to the number of attacking fighters that may engage a solitary defending fighter, as long as all the defending fighters are attacked.

[45.24] Each individual air-to-air match-up is now resolved (45.5). A *defending fighter* may fire at only (any) *one* attacking fighter, while, if there is more than one fighter *attacking* a defending fighter they all may fire at the defending plane. However, all air-to-air combat is resolved individually. No planes ever add their strengths together. The Players may resolve the individual match-ups in any order they wish.

[45.25] After fighter vs fighter has been resolved, there is a possibility that some of the fighters may fire, again, against enemy bombers and other screened planes (See 45.3).

[45.3] FIGHTER VS NON-FIGHTER (SCREENED PLANES)

[45.31] When all fighter vs fighter combat has been finished, Friendly fighters remaining may, if possible (see Case 45.32) attack any Enemy screened planes. They do not have to do so, even if they are on offensive CAP—unless *only* non-fighter planes were in the hex to start with (and so there was no fighter vs. fighter combat). In such case, offensive CAP fighters must initiate air-to-air with such unscreened planes.

[45.32] If the Player wishing to attack screened planes has *more* fighters remaining (after fighter vs. fighter combat) in the hex than does the Enemy Player, he may attack the screened planes with the *excess* number of fighters; e.g., the Axis Player sends a squadron of He111's escorted by 6 Me Bf109F's to a Target. The Commonwealth Player scrambles two squadrons of Hurricane IIC's. The Hurricanes attack the Messerschmitts, and, after fighter vs fighter combat, there are 19 Hurricanes remaining along with 5 Messerschmitts. The Commonwealth Player may now take any of his 14 excess Hurricanes (19-5=14) and attack the Heinkels. The 14 Hurricanes are now engaging in a second round of air-to-air.

[45.33] The Defender is automatically the Player with the screened planes. He places his planes down as in 45.22. The attacker may now assign his fighters in *any* way he sees fit; he does not have to cover all the defender's planes, as in 45.23. He

may take, using the above example, all 14 Hurricanes and attack one of the Henkels, or he may divide them amongst the 12 Henkels. The choice is the attacker's.

[45.34] Air-to-air fighter vs non-fighter is resolved in the same way as fighter vs fighter (see 45.2) with one exception: defending, screening planes may shoot at *each and every* fighter attacking that plane — not just one, as in 45.24. (A dive bomber may shoot at *one* attacking fighter.)

[45.35] Any screened planes that survive may now proceed to the flak/AA fire portion of the mission run.

[45.36] Bombers flying in formation receive a bonus: if 6-17 bombers are on the mission, add one to each plane's tacair rating; if 18 or more bombers are on the mission add two to each plane's tacair rating.

Examples of Air-to-Air Combat Resolution:

- A CR 42 and a Hurricane I engage in air-to-air. The CR 42 has a '3' pilot for a total tacair of '6' (3 + 3 = 6), while the Hurricane has a '2' pilot for a total of '6' (4 + 2 = 6). But the Maneuver Rating of the Hurricane is four better than that of the CR 42 (32 to 28), so this gives an adjustment of 2 tacair points in favor of the Hurricane. The final tacair differential is +2 for the Hurricane, -2 for the CR 42. The CR 42 fires first at -2, needing an 11 or 12 for a kill. He rolls a 25 and misses. (Had he rolled a '12' the combat would have been over.) Now the Hurricane rolls, needing an 11-22 to get a kill. If both miss, there is no more rolling.
- Three Hurricane IIB's attack a Junker Ju88D, flying in a formation of 12. Each Hurricane has a '2' pilot. The Hurricane has a tacair of '6', plus '2' for the pilot. The Ju88 has a tacair of '5', plus one more for formation flying. That gives the Hurricane a differential of +2. However, the Hurricanes have an 11 point maneuver advantage, for a further adjustment of 3 points, making the final differential for each Hurricane +5. The Junker 88 fires first at each Hurricane. If he rolls an 11 he scores a kill. Then each Hurricane fires at the Ju88, needing an 11-26 for a kill.

[45.4] MANUEVER ADJUSTMENT CHART

(See Charts & Tables)

[45.5] TACAIR KILL TABLE

(see Charts & Tables)

[45.6] PLANE AND PILOT

RECOVERY (see Charts and Tables)

[45.61] In certain cases fighters shot down during air-to-air, as well as planes strafed on the ground (40.63), may be recovered.

[45.62] For planes *strafed* on the ground, the Player expends 1 Stores Point and 1 Fuel Point for every plane he wishes to attempt to repair. He then throws one die; if die roll is "1" or "2", the plane is repaired. Any other number and the plane is permanently destroyed. In this case supplies are expended *only* if the repairs are successful! For the plane to be repaired, the supplies *must* be expended. Any strafed plane may be repaired in this fashion, and once repaired that plane may fly one full Game-Turn later.

[45.63] For fighters (only) shot down (by either Enemy planes or flak) the Player rolls only *one* die. In this case, a roll of '1' means the fighter actually was not shot down and has managed to limp back to base. (This roll is the same used for recovery of the pilot (45.66); thus if a '1' is thrown, both the pilot and fighter return. On a 2, only the pilot makes it back.)

[45.64] Once the fighter has returned (from 45.63), the Player *must* attempt to repair it. He uses the same system as in 45.62, but in this instance no die roll is needed to repair the plane; simply expending the supplies will do.

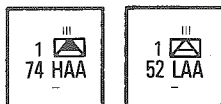
[45.65] Fighters recovered *and* repaired in this fashion are *not* "kills" for the purposes of 40.18.

[45.66] If a plane is shot down, the pilot may bail out and survive. For each fighter that is shot down, the owning Player rolls one die. If a "1" or "2" is obtained, the pilot survives and is recovered, regardless of where the plane was shot down. He is ready for action three Operations Stages later. Otherwise, the pilot is killed.

[45.7] EXTENDED RANGE

Certain aircraft have more than one maneuver rating. Such an aircraft must utilize the lower maneuverability rating if on a mission at extended range, unless the Player makes a decision to jettison the extended range fuel tank. In that case, the higher maneuverability rating may be used in air-to-air combat, but the plane cannot return to any air facility that is beyond its normal range. If out of range of any Friendly air facility, the plane must be crash landed (see Case 37.23)—if it survives the combat, that is. Fighter-bombers attacked by Enemy planes while on bombing missions may jettison their bombs to increase their Maneuver Rating for Air-to-Air. The jettisoned bombs have no effect.

[46.0] ANTI-AIRCRAFT FIRE (FLAK)



GENERAL RULE:

Planes flying missions in a given hex must undergo anti-aircraft fire, if there are Enemy units capable of such in that hex. Generally, AA fire is conducted before a mission is undertaken and any casualties taken before determining the effects of that mission. AA fire is given in points, based on capabilities of light or heavy AA (or flak) guns. The number of AA points for each unit is determined by multiplying the TOE strength by the AA rating. All AA units have Flak Points, and some other HQ, artillery and heavy weapons units also have flak/AA capability. A unit's flak/AA rating is listed in that unit's Characteristics Chart Table under the AA column. AA/flak fire consumes ammunition.

PROCEDURE:

Anti-aircraft Fire occurs after all Air-To-Air combat and Flak Suppression Missions have been completed. The Enemy Player divides the opposing planes into Target Groups (46.26) and indicates the number and type, but not Mission, of the planes in each Target Group. The Friendly Player may then attack the Target Groups in any order he wishes. Each attack is performed separately and all planes destroyed, aborted and Ammo expended requirements are completed before the next attack is performed. All anti-aircraft attacks are performed using the Anti-Aircraft Combat Results Table. The Friendly Player determines AA Points of Air Facilities and which TOE Strength Points of units with anti-aircraft Ratings are attacking a particular Target Group. The total Anti-Air points are then determined (do not divide by ten!)

All anti-aircraft attacks are performed using the Anti-Aircraft Combat Results Table (46.3).

All attacks on planes flying Fighter-type Missions are performed as follows: Roll two dice, reading them sequentially. Read down the rows of

results in the Planes On Fighter-Type Missions section of the CRT until one is found that contains that dice roll. Read along the row to the right to determine the number of planes in that Target Group destroyed by anti-aircraft fire. The actual planes in that Target Group that are destroyed should be determined at random. The players may use any method they wish as long as it is consistent and mutually agreeable.

Example: The players alternate picking which planes are destroyed. Example: Number the planes in the Target Group from 11 to 66 and roll the dice until enough different numbers are thrown to match the number of planes to be eliminated and then eliminate those planes whose number has come up.

All attacks on planes not flying Fighter-Type Missions (46.2.6a and 46.2.6b) are performed using the Planes on Other Missions section of the anti-aircraft CRT as follows:

Prior to determining the results of AA fire the firing Player may have to adjust his AA Point column to account for the density of enemy planes overhead. He consults the AA Density Adjustment Chart, noting the total number of enemy planes (other than fighters) in that hex. He then adjusts the column — in his favor — if warranted. Remember, Night Missions are done separately from others.

The firing Player rolls two dice twice using the same column, once to determine the number of planes destroyed (using the rows of the same name) and once to determine the number of planes that Abort involuntarily (using the rows of the same name). Determination of which planes are destroyed and Aborted should be made by the same method used for determining losses on Fighter Missions. Aborted planes are determined after all losses from destroyed planes are completed.

[46.1] FLAK/AA UNITS

[46.11] A large number of units have AA/Flak capability; that is, they may fire at enemy planes. A unit that has an AA Rating may so fire (however, see Case 46.22). Most of these units are called AA or Flak units, and are so delineated by their unit symbol on the counter. However, many other types of units have AA Ratings. These AA Ratings are multiplied by the TOE Strength of the unit to determine the Actual AA Points that can fire. (There is *no* dividing by 10, as with other strengths.) Any unit with an AA symbol on its counter is called a Pure Flak unit (See 46.17.).

[46.12] A flak unit having a (parenthesized) defense assault rating, defends against an assault only if alone in the hex. Players should note that the German 88's and Italian 90mms have a defensive assault rating and are treated as artillery units with assault ratings when attacked.

[46.13] Unless specifically stated otherwise, flak units are treated as artillery in terms of having their own transport. There are some flak units without transport, and to "motorize" these units requires trucks. Each Point of AA may be towed by a medium truck point or a Heavy Truck Point.

[46.14] Flak/AA units require supplies like any other unit. (They are treated as artillery). They will, of course, use ammunition when firing.

[46.15] Certain combat units—in addition to regular flak units—have AA/Flak Points. Certain tanks and recon units have AA/Flak Points, and these are points *per* TOE Strength Point (Thus, 5 TOE points of Sherman tanks would have 5 AA/Flak Points).

[46.16] Tanks may use their AA/Flak Points only against fighters strafing units in that hex or against dive bombers. They may not use their AA Points against any other planes/missions.

[46.17] Pure AA/flak units (i.e., units that have only AA Points) may stack freely in major cities; they do not count against the stacking limits. In addition, one pure AA/flak unit may stack freely in an air landing strip hex, and three may stack freely in an airfield.

[46.2] RESTRICTIONS ON ANTI-AIRCRAFT FIRE

[46.21] Anti-aircraft fire may only be directed at planes flying missions in the same hex as the units and air facilities possessing an anti-aircraft ability.

[46.22] Infantry, recon, and tank units may fire AA only at planes in the fighter mission, strafing and/or dive bombing target group (see Case 46.26).

[46.23] Anti-aircraft fire occurs after all air-to-air combat and flak suppression missions have been performed.

[46.24] The planes fighters flying flak suppression missions and those flying naval convoy reconnaissance missions may not be attacked by anti-aircraft fire.

[46.25] Opposing planes flying missions in a hex must be grouped into target groups for anti-aircraft fire purposes.

[46.26] The target groups are as follows:

- All Enemy planes not assigned to fighter-type missions and flying (normal) daytime missions by each mission.
- All Enemy planes not assigned to fighter-type missions and flying night missions by each mission.
- All Enemy planes assigned the fighter-type missions of combat air patrol. Scramble and interception flying daytime.
- All Enemy planes assigned night fighter-type missions of combat air patrol, scramble and interception.
- All planes flying fighter-type missions of strafing and/or dive bombing.

[46.27] AA may be fired at as many target groups as the owning Player wishes.

[46.28] Replacement Points may not use AA ratings until absorbed by a unit.

[46.3] ANTI-AIRCRAFT COMBAT RESULTS TABLE (see Charts and Tables)

[46.4] FLAK ADJUSTMENT CHART (see Charts and Tables)

[47.0] ABSTRACT LOGISTICS RULES

The Players use the Abstract Logistics and Air Rules (Section 32.0) except as modified below.

[47.1] SUPPLY UNITS AND SUPPLY

[47.11] All off-map (N. African and Mediterranean) air facilities possess unlimited supplies for aircraft maintenance.

[47.12] Cairo and Alexandria have unlimited supplies for Commonwealth aircraft maintenance for any SGSU occupying a Cairo/Alexandria hex or within supply range (see Case 47.13).

[47.13] An SGSU may draw supplies from any friendly supply unit per Case 32.16 (or Cairo/Alexandria) that is within 15 Capability Points.

[47.14] The Players receive regular (i.e., those described in Section 32.0) and Depot supply units. A Depot initially contains 180 Fuel and 40 Ammo Points. It retains its Depot status until it contains

no more than 60 Fuel Points, at which time it is treated as a regular supply unit. Depots may never absorb fuel or ammo points. Supplies may be withdrawn from a depot only for use by SGSU's and motorization points engaged in transporting that depot (i.e., supplies may not be redistributed to other supply units nor used by combat units or motorization points not actively engaged in moving that depot).

[47.2] SUPPLY EXPENDITURES

[47.21] Only Gun-class units are required to expend ammo in order to fire at enemy aircraft. A gun-class unit must expend 1 Ammo Point in order to perform anti-aircraft fire against a *target group*.

[47.22] Fighter and fighter-bomber squadrons do not require ammo expenditures to enable their planes to use their TacAir and Bombload ratings. Non-fighter squadrons are entirely rearmed with the expenditure of one Ammo Point.

[47.23] Fighter/Fighter-bomber squadrons require the expenditure of one Fuel Point to refuel the entire squadron. Axis non-fighter squadrons require the expenditure of two Fuel Points in order to refuel the squadron. Commonwealth non-fighter squadrons require the expenditure of three Fuel Points in order to refuel the squadron.

[47.3] MOVEMENT OF SUPPLY UNITS

[47.31] Fifty motorization points are required to transport a depot.

[47.32] A depot may only be transported in the Truck Convoy Movement Phase (i.e., it may not be attached to a combat unit).

[47.33] A depot counts as two supply units if being transported by rail. The Commonwealth Player may not transport a depot by sea.

[47.34] The Axis Player may transport a depot by Axis Coastal Shipping. Each depot requires 3,000 tons of coastal shipping.

[47.35] Regular supply units may be transported by air. The Axis Player may air transport regular supply units as follows: Any thirty planes with air transport capability may transport a supply unit containing up to 8 Ammo Points and up to 20 Fuel Points, any sixty planes with transport capability may transport a regular supply unit with any level of supplies. The Commonwealth Player may air transport regular supply units as follows: Any twenty planes with transport capability may air transport a supply unit containing up to 10 Ammo and up to 25 Fuel Points. Any fifty planes with transport capability may transport a regular supply unit with any level of supplies. Either player may transport a dummy supply unit with any ten planes with transport capability. Note that the supply unit *counter* must be physically transferred from one location to another.

[47.4] AVAILABILITY OF SUPPLY UNITS

[47.41] The Axis Player receives Depot supply units as an addition to the regular supply units produced by the Axis Simplified Supply Availability Table. He receives one Depot for each regular supply unit produced.

[47.42] The Commonwealth Player receives Depot supply units as an addition to the regular supply units produced by the Commonwealth Simple Supply Availability Table. During time period I he receives one Depot per game turn. During time period II he receives two Depots per game-turn. During time period III he receives three Depots per game-turn.

[47.5] MOTORIZATION POINTS

[47.51] Cases 32.57 and 32.59 are ignored and Case 32.56 is modified as follows: Motorization

Points may be attacked (and destroyed) as though they are trucks as per the Strafing (Cases 40.5 and 40.6) and Land Support Bombing (Cases 41.2, 41.3 and 41.5) rules in the Air Game.

[47.52] The initial motorization points available to the Players are calculated in the same manner as listed in 32.5 except that Second/Third Line trucks listed as available at air facilities are included in the calculations.

[47.53] Motorization points may be received as though they are truck replacements. The Players' ability to bring in motorization points is equal to their ability to bring in Medium plus Heavy Truck Points (i.e., each medium/heavy truck point is equal to one motorization point) as per the Axis Truck Production Chart of the Axis Replacement Pool and the Commonwealth Truck Production Table of the Commonwealth Production System.

[47.6] SIMPLIFIED AXIS NAVAL CONVOYS

These rules replace those listed in Case 32.6.

[47.61] In the abstract Logistics game system, Axis Naval Convoys consist of all Supply Units, Motorization Points and Replacement Points for that Game-Turn. These items may be divided and assigned to any Convoy Lane available, as per rules for such assignment. There are no tonnage limitations.

[47.62] The Axis Player may never assign more than three regular and three depot supply units and twenty-five motorization points to any one Convoy Lane in any one Naval Convoy Stage. Note that the Axis Player is limited by the Replacement Pool as to the maximum number of replacement points of each type of item that may be assigned to Axis Naval Convoys (total) for arrival during a particular Game-Turn.

[47.63] If the Commonwealth Player successfully performs a Axis Naval Convoy reconnaissance mission, he is informed as to the composition of the convoy in total number of: Regular supply units, depot supply units, motorization points, tank replacement points, infantry replacement points, gun replacement points and armored recon/armored car replacement points.

[47.64] Attacks against a convoy are carried out as per the Axis Naval Convoy Bombing rules.

[47.65] The result of an attack on an Axis Naval Convoy is modified as follows:

A/A result of 10% applies *only* to motorization points and replacement points, rounding fractions down for the former and rounding fractions up for the latter.

B/A result of 20% applies to all items except Depot supply units, rounding fractions down. However, at least one regular supply unit at one point of each class of replacement point is always eliminated.

C/A result of 30% or greater is applied as "B" above with the modification that at least one supply unit of any type must be eliminated (supply units are considered as a total for loss determination).

[42.7] BOMBARDMENT OF THE COMMONWEALTH FLEET

Ignore Case 32.7 Bombardment of the Commonwealth Fleet is carried out as per the Air Game Rules.

[42.8] STACKING MODIFICATIONS

[42.81] The Players ignore all Road and Track (only) Stacking Point limits as listed in Section 9.0.

[42.82] The Players may stack one Depot supply unit in addition to the regular supply unit stacking limits listed in Case 32.14.

LOGISTICS GAME

The Logistics Game allows the players to realistically experience the constraints that actually applied to the armies which fought in North Africa. Basically, there are four distinct "flavors" of supply in CNA: water, fuel, stores, and ammo. The Logistics Game requires that players keep track of the amount of each of these that each unit possesses and consumes. Also included are detailed naval convoy rules that cover the movement of the supply types to Africa for the Axis and rules covering the use of truck transport to move the supplies.

The Logistics Game constitutes the heart and soul of *Campaign for North Africa*. The entire design emphasis is on recreating the difficulties of supply in desert warfare. Overall commanders will soon find out that their most important weapon is not their tanks or their 88's or their long-range bombers; it is their Trucks.

Over a year of intensive testing has proven at least one thing: no one can do well in this game if he does not have a definitive grasp of the logistical situation and, moreover, a far-sighted plan on how to handle it. This goes double for the Axis Player, who is often so far away from his base of supply that it will seem to take forever just to "feed" his troops. Players should learn to use the three-tiered "Line" system for their trucks, and they must set up a system whereby trucks and the shipment of supplies is handled with a maximum of efficiency and speed. Any Stage in which supplies are not constantly being moved forward is a wasted three days (game-time). The thoughtful use of Supply Dumps is an absolute must.

The general feeling that players will get throughout the game, vis a vis the logistics, is frustration (albeit more so on the part of the Axis Player than the Allied Player). The Axis Player will have countless opportunities to smash weakly-held Allied positions, only to find he has neither the fuel the ammunition to carry out his plans; they are somewhere on trucks 200-300 miles to his rear. The player who best minimizes this problem will most often be the winner.

Much research was done on the logistics portion of the game. Statistics *are* available, but not in a form that is easily translatable to game terms. The supply needs, in terms of the points consumed and needed, are based on basic computations using weights and consumption figures (where available). These figures were then extrapolated to cover usage over the time scale in the game. This was especially important when determining ratios for fuel and ammunition for aircraft. Furthermore, complete information on fuel consumption was not available on a goodly number of the bigger planes. Another problem area was ammunition usage. But the biggest problem area, and the area that is most abstracted for game purposes, is the correlation between supplies used and weight of supplies convoyed (by the Axis). Exact figures here are rare, and much testing was spent on seeing whether the numbers we came up with were satisfactory.

All of the above is by way of giving the players a little background on the logistics game. The supply problem in CNA is extensive, and the work needed to run it—in terms of the game—is just as extensive. It is highly recommended that one player from each side be assigned to this area, and this area alone. Testing proved that rare was the player who could handle military matters, strategy, tactics, etc., and also coordinate supply for his troops.

[48.0] SEQUENCE OF PLAY (Logistics Game)

COMMENTARY:

It is possible to use the Logistics rules in concert with the Land Game and omit the Air Game rules. If that is the desire of the players, they should ignore all part of the following Sequence marked "Air Game Only."

GENERAL RULE:

Play of the game should proceed in accordance with the following outline.

I. INITIATIVE DETERMINATION PHASE
The Players determine who will have the initiative for the coming Game Turn. (See 7.0) The Player going first within the Operations Stage (see below) is known as "Player A" and the other is "Player B."

II. STRATEGIC AIR PLANNING STAGE (Air Game Only)

A. Designation Phase:

The Players assign their airplanes to fly Land Support or Strategic missions.

B. Axis Malta Availability Determination Phase
The Axis Player determines the amount of support he will receive from his abstracted N. African Theatre airforce for Raids on Malta.

C. Strategic Mission Assignment Phase
Planes designated as flying Strategic missions are assigned. The Axis Player assigns his planes to Raids on Malta or naval convoy protection. The Commonwealth Player assigns his planes to naval missions or Bombing Reserve.

D. Malta Raid Phase

The Axis Player resolves flak suppression, anti-aircraft fire and bombing missions against Maltese air facilities (Note that Commonwealth warships stationed at Valetta may only be attacked by Land Support Missions.)

III. NAVAL CONVOY STAGE

A. Naval Convoy Schedule Phase

The Axis Player refers to the Axis Naval Convoy Level Chart and then to the Axis Convoy Capacity Table and rolls one die to determine the total tonnage available for the *next* Game-Turn. He then plans what specific cargoes the convoys will carry and their routes.

B. Convoy Resolution Phase (Segments 1 & 2: Air Game Only)

1. Naval Convoy Reconnaissance Segment: The Allied Player resolves Strategic Convoy Reconnaissance missions.
2. Convoy Lane Assignment Segment: Axis convoy protection aircraft are assigned CAP missions in specific Convoy Lanes. Commonwealth Bombing Reserve aircraft are assigned CAP, Flak Suppression or Convoy Bombing missions in specific convoy lanes.
3. Convoy Bombing Segment: All air-to-air combat, Flak Suppression, anti-aircraft, and convoy bombing is carried out.

IV. STORES EXPENDITURE STAGE

Both Players distribute and expend stores (Section 51.0) to eligible units requiring them. Units not receiving required stores are noted. Players also adjust fuel and water storage levels for any losses as a result of spillage or evaporation.

V. FIRST OPERATIONS STAGE

A. Initiative Declaration Phase

The Player who gained the initiative in Stage I now states whether he will be "Player A" or "Player B" for this particular Operations Stage.

B. Weather Determination Phase

The player with initiative rolls for weather (Section

29.0). All adjustments for unusual weather are made at this time, including any additional evaporation of fuel or water from hot weather.

C. Organization Phase

The following Segments are undertaken in any order desired.

1. *Water Distribution Segment*: Players distribute water to eligible units requiring such.
2. *Reorganization Segment*: Attachment and/or assignment of reinforcement, replacements, or any other non-assigned units including trucks may be undertaken. Detachment of attached units may also be done.
3. *Attrition Segment*: Reductions of any units that have not been adequately supplied with water or stores is undertaken.
4. *Construction Segment*:

a. Construction Completion Step: Any work scheduled for completion is finished, markers being removed or added as required. Units that have completed construction may be freely moved.

b. Construction Initiation/Continuation Step: Any continuing or ensuing work on projects is noted. Any units involved in such work may not be moved (voluntarily) during the remainder of the current Operations Stage.

5. Training Segment:

a. Training Completion Step: Units and replacement points completing a level of training are noted and the effects applied.

b. Training Initiation/Continuation Step: Any units beginning or continuing training are noted. They may not be moved (voluntarily) in the remainder of the Operations stage.

6. *Supply Distribution Segment*: Supplies in the same hex as land units may be redistributed at this time, subject to the maximum innate capacity of the unit(s) and the hex. Trucks may be loaded/unloaded.

7. *Tactical Shipping Segment*: Both players execute the transport of cargo between African ports. *Note*: Axis coastal ships are represented in the game by counters. Allied coastal shipping is not represented by counters, and is limited only by the port capacities.

D. Naval Convoy Arrival Phase: All reinforcements, Replacement Points, and supplies scheduled for arrival (and actually arriving) appear in their designated ports of arrival or entrance hexes. Check Reinforcement Tables and convoy rules. The Axis Player plans for the arrival of future Replacement from the Axis Replacement. In his first Naval Convoy Arrival Phase of each month, the Commonwealth Player consults the Commonwealth Production System and determines how many Replacement Points he will receive two months hence and plans for their arrival.

E. Commonwealth Fleet Phase:

1. *Fleet Assignments Segment*: The Commonwealth Player assigns his ships to any sea or coastal hexes for bombardment purposes.
2. *Fleet Repair Segment*: Any repair work on ships is undertaken at this time.

F. Land Support Air Phase (Air Game only):

Only those planes assigned to land support missions may be used in this Phase.

1. *Mission Assignment Segment*: All planes that are fueled may be assigned missions by the Player.
2. *Mission Deployment Segment*: Counters representing each type of plane assigned a mission are placed in their hex of assignment.
3. *Air-to-air Combat Resolution Segment*: Combat between opposing aircraft, including any caused by scramble and interception, is resolved. Mis-

sion aborts may take place before combat resolution.

4. *Flak Resolution Segment*: Aircraft in the same hex as Enemy anti-aircraft units undergo flak fire.

5. *Mission Completion Segment*: All missions that can be completed (e.g., bombing, strafing, etc.) are completed. Abortions may take place here, too.

6. *Return to Base Segment*: All surviving aircraft are returned to their base of origin (excepting those on transfer missions and some other rare exceptions).

7. *Tactical Maintenance Segment*: Both Players may attempt to ready all planes eligible to fly land support missions.

Reserve

1

G. Reserve Designation Phase:

Player "A" designates which of his units he is placing in reserve status, indicating that with a reserve marker. (See Section 18.0.)

H. Movement And Combat Phase

The following four Segments comprise the Movement and Combat Phase. Player "A" may, if he so wishes, repeat Segments 1 through 4 as many times as he desires within the restrictions of the continual movement rules (8.2). Each repetition must include all four segment.

1. *Movement Segment*: All units, except trucks and tank recovery squadrons not in reserve and eligible to be moved may be moved. Non-Phasing Player ("A") may move units in accordance with the reaction movement rules (Case 8.6).

2. *Breakdown Determination Segment*: All vehicles and motorized units of both sides are checked for breakdown. Broken down vehicles are indicated with the appropriate marker.

3. Combat Segment:

a. Position Determination Step: Both Players determine the "position" (front or back) of all gun and armor class units.

b. Barrage Step: Both Players secretly plot and then execute any barrages.

c. Retreat Before Assault Step: Player "B" may retreat before assault any of his units eligible to be so moved.

d. Force Assignment Step: Both Players secretly assign TOE Strength Points to anti-armor or close assault. Player "A" determines which assaults will be probes (although he need not reveal this decision) and which TOE Strength Points will be withheld from assault.

e. Anti-Armor Step: Both Players simultaneously resolve any anti-armor fire, extracting any casualties. Destroyed tank markers are deployed as required.

f. Close Assault Step: Close assaults are resolved in any order determined by Player "A". He announces which of them are actually probes, after each is resolved.

4. *Reserve Release Segment*: Player "A" may release any of his reserves that he wishes to.

J. Truck Convoy Movement Phase: Player "A" may move his Second and Third Line (unattached) Trucks, and any POW's and Guards.

K. Commonwealth Movement Phase: If Player "A" is the Commonwealth Player, he may utilize rail movement (Case 8.9) to move land units or supplies. Otherwise this Phase is omitted.

L. Repair Phase:

1. *Towing Segment*: Player "A" may tow broken-down and recovered vehicles.

2. *Maintenance Segment*: Player "A" may at-

tempt to repair breakdown or destroyed vehicles which were not towed in this Phase.

M. Patrol Phase: If there has been no combat in the current Operations Stage, the Phasing Player may use patrols for reconnaissance purpose.

VI. Second Operations Stage:

Both Players repeat all facets of the First Operations Stage.

VII. *Third Operations Stage:* Both players repeat all facets of the First Operations Stage.

VIII. *Strategic Air Recovery Stage* (Air Game Only)

A. Return to Base Phase: All surviving planes from missions flown in Stage II are returned to their base of origin if possible.

B. Aircraft Maintenance Phase: Both Players attempt to ready planes that have flown missions during the Strategic Air Stage.

IX. End of Game-Turn

You have now completed one full game-turn, replete with airplanes fuel sandstorms and the lot! Congratulations. You may put in for a transfer, but before doing so, keep in mind that the Russian Front is the only other active theatre.

[49.0] FUEL

GENERAL RULE:

Almost every vehicle in the game consumes fuel when moving. Players will consider a vehicle to consume fuel unless it specifically states somewhere in the rules that it does not. Fuel is consumed at a consumption rate dependent on the vehicle and on the distance travelled. Fuel may also be consumed in certain types of construction. Players should consult the Section on Construction for the rates.

Players must keep track of the amount of fuel they have available. This may be done on the Supply Dump Sheets as well as the unit TOE Sheets. Fuel being transported by truck convoys will be noted on the Truck Convoy Sheet. As fuel is consumed (by usage or evaporation and spillage) or increased (by arriving convoys) the Player should record the changes in levels immediately.

[49.1] FUEL CONSUMPTION RATES

[49.11] All Fuel is considered in terms of Fuel Points. The Fuel Point is the basic unit of consumption. A Fuel Point is roughly equivalent to 250 pounds or 35 gallons of gasoline and auxiliary petroleum products. (This rate does not apply to the Air Game, where fuel usage has been abstracted because of the decrease in the true number of sorties flown.)

[49.12] Every vehicle in the game consumes fuel when it moves, except where specifically noted. (e.g. desert raiders and patrols among others, have slightly different rules.) Fuel users include HQ's with non-parenthesized TOE Strength Points and all gun class units. They do not include motorcycles or the towing of tanks and trucks back to repair facilities.

[49.13] Every TOE Strength Point of vehicles has a fuel consumption rate and a fuel capability rating. The fuel consumption factor is the number of Fuel Points consumed for every *five* Capability Points (or fraction thereof) expended by the TOE Strength Point for movement (or any type). **Note:** Fuel is *not* consumed when CP's are expended for combat or any other action that does not involve actual movement from one hex to another. **Example:** One TOE Strength Point of tanks with a fuel consumption rate of "4" that moved in such a way that expends 12 Capability Points would consume 12 Fuel Points—"4", the fuel consumption rate multiplied by "3", the number of groups of 5 CP's expended.

[49.14] The fuel capacity rating is the number of

Fuel Points a TOE Strength Point may have, so to speak, in its gas tanks. All vehicles' fuel capacity ratings are determined according to the following formula:

$$\text{Fuel capacity rating} = \frac{\text{CPA} \times 1/5}{\text{fuel consumption rate}}$$

Players will note that a TOE Strength Point always has a fuel capacity rating exactly sufficient to allow all its CPA to be expended on movement.

[49.15] For fuel to be consumed, it must be present in the same hex with the consuming unit/TOE Strength Point. Generally, movement of a unit will be "counted against" its fuel capacity rating—i.e., the fuel in the unit's gas tanks, which by definition is in the same hex at all times—and players will have to record how far each land unit has been moved since the last time its fuel supply was replenished. See also Case 49.16.

[49.16] In essence, a unit consumes fuel in the hex in which it *begins* Movement for any Movement Segment in the OpStage. It may draw fuel from any source in that hex, except for second-third line trucks. (Such fuel from convoying trucks must be off-loaded first). There is no cost in CP's to "gas up". The Player simply notes how far he is going to go in that Movement Segment and then subtracts the necessary fuel points from whatever source is available in the hex. He then moves the unit. If such unit were to move again in that OpStage, it would have to draw fuel from the new hex or from its own Fuel Capacity (49.14).

[49.17] In an emergency, Players may "siphon gas." That is, if two units are in a hex, one has fuel and the other doesn't, and there is no other fuel supply available, the Player may transfer some—or all—of the fuel from one unit to the other, up to the maximum Fuel Capacity of the receiving unit. Such transfer costs 3 CP's to both units. Siphoning may occur in any Movement segment. The definition of "emergency" is left to the Players. (Note that fuel may be siphoned from abandoned or Breakdown enemy vehicles; it may not be siphoned from a destroyed vehicle. The enemy player should supply the siphoning player with the number of fuel points so available—when requested.)

[49.18] Trucks carrying fuel will also consume fuel themselves while they are moving. They may consider the fuel they are transporting as a source of fuel. It is possible for a truck to consume half of its cargo getting from a port to a forward area.

[49.19] Fuel is non-denominational. It can be used by either player, making a supply dump a worthwhile objective.

[49.2] TRANSPORTATION OF FUEL

Fuel may be transported from one place to another by trucks, trains and, in some cases, planes. Fuel may never be "transported" by tanks or other such vehicles. Walking infantry may never transport fuel. (How would you like to cross the desert carrying a 20 pound jerry-can filled with gasoline?) See the Truck Characteristics Chart for the amount of Fuel Points that can be carried on a truck and Aircraft Characteristics Chart for the transport tonnage capacity of planes.

[49.3] EVAPORATION AND SPILLAGE

Fuel is subject to loss through evaporation and spillage, regardless of where it is kept. During the Stores Expenditure Stage of each game-turn, each Player reduces *all* fuel levels on the game-map (*not* in convoys at sea) by six percent (6%), rounded down. In addition, if the weather for an operations stage is "hot weather", an additional reduction of five per-cent (5%) is taken as soon as the hot weather is determined. There is one exception

to these rates: from Sept., 1940 until the last Game-Turn (inclusive) in August, 1941, the Commonwealth spillage and evaporation rate is **nine** percent (9%) per Game-turn. This is due to the poorly constructed containers used by the British; it wasn't until the British copied that German "jerry can" that their rate was reduced (This rule also applies to certain sources of water, see 52.44.)

[50.0] AMMUNITION

GENERAL RULE:

Land units that use their TOE Strength Points in combat expend ammunition, as do planes that carry bombs and/or engage in air-to-air combat. Ammunition is consumed in terms of Ammunition Points at rates of consumption dependent on the weapon-type firing. A unit without ammunition is useless in terms of combat. Players must keep track of the ammo supply of each TOE Strength Point. Each TOE Strength Point may carry (i.e., transport by itself without trucks) only enough ammo to fire once. However, units may always use ammo present in the hex (either in attached trucks or in a supply dump).

CASES:

[50.1] CONSUMPTION OF AMMUNITION

[50.11] All ammunition is considered in terms of Ammunition Points. Each Ammunition Point is (very) roughly the equivalent of *four* tons. Ammunition consumption rates are given in 50.2

[50.12] Whenever a land unit uses its barrage strength, its anti-armour strength, its AA points, or any of its assault strengths it consumes ammunition. A unit without ammunition of any type is useless in combat; it may not even defend against assault if it finds itself in an Enemy Zone of Control and/or it is fired upon by an Enemy unit, it surrenders and is captured. (Units without ammunition have NO ZOC.) A unit without ammunition may not enter an Enemy ZOC unless stacked with (or entering a hex occupied by) a Friendly unit with ammunition or a supply dump.

[50.13] Each combat function has an ammunition consumption rate, listed in 50.2. Units, such as tanks, that have several types of strengths (anti-armour as well as assault) use the ammunition consumption rate of on the combat function chosen.

[50.14] The ammunition consumption rate is the number of Ammunition Points expended for each *TOE Strength Point* being used in combat. Ammunition is expended by guns barraging (or firing flak), anti-armour fire, or offensive *and* defensive assault. Thus, four TOE Strength Points of German 150mm guns would expend 16 Ammunition Points (4 TOE times the artillery consumption rate of 4 = 16), regardless of how many Actual Barrage Points are being applied. Thus, a Player may choose to conserve ammunition by only firing, say, 3 out of 6 available TOE Strength Points.

[50.15] Ammunition is consumed only if present in the hex. However, ammunition *on* truck convoys (2nd or 3rd line trucks) may not be used until it is off-loaded. Ammunition is expended the *instant* it is used, and the ammunition level for that hex/unit/dump must be reduced accordingly.

[50.16] Ammunition is specialized for both players. Only one-third of all captured Ammo Points (rounded up) may be used by the opposing Player. The rest are lost. If a supply dump is recaptured, only one-third of all remaining Ammo Points may be used by the re-capturing Player.

[50.17] Ammunition may be transported by trucks or aircraft; it may be airdropped. It is available for use when in first line trucks or supply dumps. Ammunition with truck convoys may not

be used until off-loaded. Combat units may carry enough ammo to fire once.

[50.2] AMMUNITION CONSUMPTION RATES CHART

(see Charts and Tables)

[51.0] STORES

GENERAL RULE:

The term "stores" encompasses everything that isn't fuel, ammunition or water. This includes food (which is what most playtesters ended up calling stores), paper, uniforms, and whatever else was used and needed to keep going. Stores are different from other types of supply in that they are distributed at the beginning of the *Game-Turn*, rather than during each Operation Stage, and that units may get along without them, albeit with limited effectiveness and with the possibility of attrition.

CASES:

[51.1] USAGE OF STORES

[51.11] Every TOE Strength *Point* in play requires four Stores Points per Game-Turn, except as noted below.

POW
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[51.12] Prisoners use stores, at a rate of one Stores Point per five Prisoner Points per Operations Stage. (See 28.15). Prisoners must be supplied with Stores before any other units. For this purpose only, if there are no Stores in the hex, Stores may be deducted from the nearest supply dump, regardless of how far away it is.

[51.13] HQ and engineer units require only *one* Stores Point per Game-Turn.

[51.14] Stores may have to be expended during construction. See 24.0.

[51.15] Each Stores Point weighs one ton. Stores may be transported by trucks or planes, and they may be airdropped. Stores must be present in the hex to be used. Stores on truck convoys cannot be used until off-loaded.

[51.16] Fifty per cent of any captured Stores Points may be used by the opposing Player; the rest are lost.

[51.17] Guard points require two Stores points per Game-Turn each. Guards may also draw from the nearest dump as per 51.12.

[51.2] EFFECTS OF LACK OF STORES

[51.21] If any TOE Strength Point of a unit (including parent formation-size) does not receive required stores, that unit earns *one* Disorganization Point, per Game-Turn of lack of stores.

[51.22] For every two *consecutive* Game-Turns units in a hex are without stores, they lose 2% of their TOE Strength Points, rounded to the nearest whole number. This loss is *progressive and cumulative* (as long as it is consecutive). Thus, at the end of the fourth consecutive Game-Turn, the rate rises to 4%, and at the end of six turns, 6% and so on. TOE Strength Points may be eliminated only if they are infantry-type TOE Strength Points. Guns and tanks are not affected by this.

[51.23] A Player in desperate straits may place units on *half rations*. He cuts his stores requirements from four to two for units, with the following effects:

1. No half-ration unit may voluntarily exceed its CPA;
2. A half-ration unit may not voluntarily enter an Enemy Zone of Control.

Players may *not* use this case to "save" Stores.

This is used only if there are not enough to go around.

[52.0] WATER

GENERAL RULE:

A somewhat ironic fact of the Desert Campaign was that, although water is the single most important item in the desert, it was never really a decisive factor, *operationally*. Players should find that they rarely run out of water, but that it is a nuisance as it is *absolutely necessary*. Water comes from wells (called "birs" in North Africa; hence, the expression "Hey, Cold Bir Heah" was first heard at an inter-divisional Soccer Match.) Most wells are located close to the coast, and thus most of the water supply is away from the inland areas. Wells may be "poisoned" or depleted through usage. Depleted wells are replenished by rains, and poisoned wells may be "sweetened." Players may construct water pipelines from major sources to facilitate water usage and availability.

CASES:

[52.1] WELLS

[52.11] Water is found in wells—and only wells. (It may be extended from wells through the use of water pipelines.) Wells are located in major cities, villages, and birs. Water is also found in oases (See 52.3). Each of the above automatically contains a well; no other locations or hexes have wells. (The Tripoli/Tunisia boxes each have unlimited wells.)

[52.12] The amount of water available from a given source is given in the Water Availability Table, (52.7). The Table also gives the depletion possibilities for each source.

[52.13] To obtain water, a unit moves into a hex with a well. The Player then expends *one* CP and throws a die to draw water (the die roll is unnecessary in Major Cities and oases.). He consults the Water Availability Table under the correct column (village or bir) and cross-references the die roll with the source. The result is the number of Water Points drawn, as well as whether the source is depleted or not. Players should note that wells in major cities and oases have *unlimited* water; they may not be depleted and they may *not* be poisoned.

[52.14] When a well becomes depleted, the Player who drew the water notes this on a piece of paper. He does not have to tell the opposing Player that the well is depleted. However, if an opposing Player's units enter a hex with a depleted well, and attempts to draw water the Player then tells him that it is depleted (after he has expended the one CP!). When a depleted well is revealed, a depleted marker is placed in that hex.

[52.15] Depleted wells remain depleted until a rainstorm occurs. All depleted wells on a game-map section with a rainstorm are automatically replenished at the instant the rainstorm occurs.

[52.16] Wells in villages and birs (*not* in major cities or oases) may be "poisoned". Actually, poisoning is a fanciful term. No one actually dropped fifty pounds of arsenic down a well; the folks in Geneva looked askance at that sort of thing. Rather, the wells could be salted, as there was plenty of salt around. To poison a well, therefore, any unit in a hex (village/bir) with a well may expend one CP and roll one die. If the die roll is a '1', the well is poisoned. If the die roll is 2-6 the attempt fails (for any number of reasons) and no further attempt on that well may be made that Operations Stage. Poisoned wells may be kept secret as in 52.14.

[52.17] Poisoned wells may be sweetened (unpoisoned) by having *any* land unit expend 5 CP's

and roll a die. If the result is a 1, 2, or 3 the well is OK. Any number of attempts—at 5 CP's a shot—may be undertaken, except that a unit may not exceed its CPA trying to sweeten wells.

[52.2] WATER PIPELINES

[52.21] Players may try to extend their access to fresh water by constructing water pipelines. Pipelines extend from a *specific* source of water—only wells in major cities.

[52.22] Water pipelines may extend from any well in a major city any length the Player desires. The Commonwealth Player may consider *any* operating Railroad hex to be a pipeline for water, and he may construct pipelines *from* any railroad hex. The Axis Player may *not* use the defunct Barce-Benghazai railroad for this purpose.

[52.23] Any water pipeline (or RR hex) is considered as a source of water similar to a major city. It may not be poisoned or depleted, and it is unlimited. It may be destroyed.

[52.24] Pipelines are built by engineer battalions or CW HQ units in the engineering capability. It takes one Construction Phase, plus the expenditure of 10 stores, to build a hex of pipeline. Only one pipeline hex may be built in any given Operations Stage.

[52.25] Water Pipelines are susceptible to enemy attacks. Pipelines—except for railroads acting as pipelines—may be destroyed by desert raiders, the presence of an Enemy unit in the hex, or strafing. Railroad pipelines are destroyed if the railroad is destroyed. If a pipeline is destroyed, water may be drawn only from the farthest connected hex. Pipelines are repaired in the same fashion and cost as they are built. Pipeline and destroyed pipeline counters are provided.

[52.3] OASES

Oases are treated the same as the terrain of the hex they are in, with one major exception: oases are "natural" supply dumps for water and stores (in terms of food). Oases wells are never depleted and have an endless supply; they may not be poisoned. Furthermore, oases provide as many stores as needed to supply units with non-construction stores (or stores to maintain normal levels; 51.1) Units sitting in Oases have all the stores and water they need to last them the entire game. Pipelines *cannot* be built from Oases.

[52.4] WATER USAGE RATES

[52.41] Each infantry battalion or company regardless of its TOE Strength, requires *one* Water Point per Operations Stage. See also 52.6.

[52.42] *Each* TOE Strength Point of Vehicle (Tank, Recce, Artillery, etc.) or Truck Point requires *one* Water Point each Operations Stage, if it uses *any* of its CPA.

[52.43] Hot weather (29.3) requires additional water usage.

[52.44] Water—except for water in wells and pipelines—is subject to evaporation and spillage (See 49.3).

[52.45] Water may be transported by trucks at the rates given in the Truck Capacity Tables (40.2).

[52.5] EFFECTS OF LACK OF WATER

[52.51] Vehicles without water may not move or close assault offensively. Vehicles without water defending against close assault halve their total raw strength before determining actual strength.

[52.52] Infantry units (all units moving without benefit of vehicles) may not voluntarily exceed their CPA for the Operations Stage they are deprived of water. Furthermore, they may not engage in offensive close assault; they defend at half strength.

[52.53] For every consecutive operations Stage *after* the first Operations Stage without water, each infantry unit loses one TOE Point.

[52.6] THE ITALIAN PASTA RULE

One of the biggest mistakes the Italians made during the entire Desert Campaign was to provide their troops with a diet which was composed, in large part, of spaghetti and macaroni. Aside from providing insufficient protein (this wasn't Buitoni Brand), pasta has one serious drawback in the desert: you need water to cook it! Therefore, each Italian battalion, when it receives its Stores, must receive an additional 1 Point of Water when Stores are distributed. Any battalion-sized unit that does not receive their Pasta Point (one Water point) may not voluntarily exceed their CPA that Turn. Furthermore, Italian battalions not receiving their Pasta Point that have a Cohesion Level of -10 or worse immediately become Disorganized, as if they had reached -26. As soon as such units get their Pasta Point, they regain the original Cohesion Level (i.e., the level they had before they disintegrated).

[52.7] WATER AVAILABILITY TABLE (see Charts and Tables)

[52.8] POISONING AND SWEETENING WELLS TABLE (see Charts and Tables)

[53.0] TRUCKS AND TRANSPORT

COMMENTARY:

Transport is the lifeline of the Player. The Player who fails to grasp the overriding importance of transport (or, as it is called throughout the rules, trucks) to his military effort will be a losing Player. Trucks *are* the game; without a well-organized and efficient truck convoy system your entire military effort will fall apart.

GENERAL RULE:

Trucks are used for one of two purposes: carrying supplies or transporting men (motorizing infantry-type units). Each one "point" of trucks is equivalent to ten actual trucks. *Heavy Trucks* are those of the (approximately) three ton or heavier variety, including heavy-duty lorries and partially-tracked vehicles. *Medium trucks* are the equivalent of all transport vehicles of around 30 cw. *Light trucks* are all transport lighter than the 30 cw limit. Trucks consume fuel and water when they move, and they suffer breakdown.

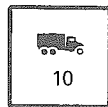
CASES:

[53.1] FIRST, SECOND, AND THIRD- LINE TRUCKS

There are three truck formations: First, Second, and Third Line Trucks. These formations (or rather definitions) delineate what the Trucks in that formation are doing and where they belong. These definitions are provided not so much in the form of rigid rules but rather as a system—based on one actually used in the campaign—so that the player may better visualize what he can accomplish and what he has to do it with.

[53.11] First Line Trucks

These are trucks attached directly to the parent combat unit (division, brigade, battalion, etc.). They are used to carry men and/or supplies of that parent combat unit. First line trucks are *not* represented by counters; the Player notes on his TOE Sheet for a parent combat unit the number and type of truck attached as first line and assigns men or supplies to each truck unit. First line trucks may be detached (in the Organization Phase), in which case they will be represented by a counter and are no longer considered to be first line trucks.



[53.12] Second Line Trucks

These truck units essentially operate between combat units and the forward supply dumps from which they are retrieving supplies. These trucks are not limited to carrying supplies; they may, like all trucks, carry anything not specifically forbidden. Second line trucks are represented by counters. Players keep track of all second line trucks on their truck and Major End Items sheets. They move during the Truck Convoy Movement Segment of the Operations Stage, as they are not attached to a regular unit.

[53.13] Third Line Trucks

These trucks operate between the major ports of entry (e.g., Alexandria, Tripoli, Cairo, Tobruk, etc.) where supplies and reinforcements originate, and the forward supply dumps. Third line trucks rarely transport directly to combat units, although there is no rule restricting such transport. Third line trucks are similar in all ways to second line trucks, with the sole exception of the name, which distinguishes them in any supply network you may devise.

[53.14] *Three Tier Organization:* The most efficient system for hauling supplies and reinforcements is a three-tiered effort: third line trucks pick up supplies from the port of entry and transport them to forward dump areas; second line trucks pick up supplies at those forward dump areas and drive them directly to the combat units, where the supplies are then loaded on to that unit's first line trucks. They are now available for instant use by the unit. This system is recommended by the designers and the playtesters; it is not, however, a rule.

[53.2] RESTRICTIONS ON THE USE OF TRUCKS

[53.21] Trucks acting as convoys may move only in the Truck Convoy Movement Segment (See 8.18). This applies only to trucks not *attached* to a combat unit or HQ. Thus, truck counters on the game maps are, by definition, convoys and will move only in the Truck Convoy Movement Segment.

[53.22] Trucks moving in a Convoy Movement Segment have an extended CPA: heavy and medium trucks have a CPA of *thirty* (30), and light trucks have a CPA of *forty* (40). However, Trucks moving in a Truck Convoy Movement Segment may never, under any circumstances, exceed their extended CPA. If, for some reason, they are forced to exceed their extended CPA they are considered to have suffered breakdown (21.0) at the instant they exceed their extended CPA. First line trucks use their basic CPA (20 or 25) and may exceed their CPA as their parent unit does (See 6.1).

[53.23] Any combat unit with a cohesion level of *minus five* (-5) or worse may not voluntarily detach its first line trucks. Conversely, second and third line trucks attached as first line trucks assume the cohesion level of the parent unit.

[53.24] There is a Capability Point cost to load and/or off-load supplies and men. (See 6.3). Supplies for one truck unit must have been off-loaded by that truck before another truck unit may load and carry them. This applies even if both truck units are attached to the same parent combat unit or convoy. Furthermore, supplies and/or men carried by second and third line trucks that have reached their full extended CPA's may not be off-loaded (and picked up by another truck) if such action will cause that truck unit to exceed its extended CPA. **Exceptions:** During the Supply Distribution Segment, trucks may freely be loaded and

unloaded; during the Combat and Movement Segment, ammunition and fuel loaded in trucks that hex may be expanded with no unloading cost.

[53.25] "Leapfrogging" is *not* permitted. Leapfrogging is defined as having truck "A" carrying supplies to a hex, then having truck "B" pick up the same supplies in the same Movement Segment and move them on to another truck, which then picks them up, and so on. If this were allowed, a given cache of supplies could be moved from one end of the game-map to the other in one Segment! Thus, for this purpose, supplies and men being transported by trucks are considered to have and to have used the same number of CP's that the trucks carrying them expended, and they may never exceed the CPA of the first truck they started with in the Operations Stage.

[53.3] ABANDONED VEHICLES

[53.31] Trucks, or any prohibited vehicles, that are in a salt marsh hex and not on a road or track are automatically abandoned (see 8.44). If the abandoned vehicle was carrying infantry-type units or is an artillery unit *and* there is an Enemy combat unit within *three* traversable hexes (counted from Enemy unit to salt marsh hex) *and* there are no Friendly combat units (at the end of the Movement Stage) within those same three hexes, those infantry units surrender and the guns are captured. If there is no Enemy around, the infantry is now on foot and walking (starting next Movement Stage) and the guns are abandoned.

[53.32] Abandoned vehicles and guns may be recovered by any unit in an adjacent non-salt marsh hex with engineering capability (23.1). The cost to recover such vehicles (or guns) is 10 CP's to both units. (**Note:** The abandoned vehicle is considered to have its full CPA.) Recovery may not be performed if either the recovering unit or the abandoned unit is adjacent to an Enemy combat unit. Recovered vehicles become attached to the recovering unit.

[54.0] SUPPLY CO-ORDINATION

COMMENTARY:

Many of the rules in this Section are covered elsewhere, but they are spread throughout the rules booklet. This Section is provided to consolidate an important facet of the game: coordinating your supply effort and logistics system. This involves the rules needed for setting up a supply network of dumps and trucks and maintaining the network so that whatever supplies you do have flow smoothly.

CASES:

[54.1] SUPPLY DUMPS

[54.11] Supply dumps are used to hold large amounts of supplies — ammunition, fuel, stores, and water — in places that can be easily reached and used by combat units. Any hex can be used as a supply dump, and all major cities are natural supply dumps.

[54.12] Dummy supply dumps hold no supplies. They are revealed only when an Enemy unit enters the hex or bombs/strafes it. The Dummy supply counter is removed upon revelation.

[54.13] Supply Dump Capacity Chart (see Charts and Tables)

[54.14] Players may attempt to "blow" supply dumps and their supplies. Only non-gun units may attempt to blow dumps. Only one Phasing unit per

[continued on page 23]

[8.37] TERRAIN EFFECTS CHART

Terrain Type	CP Cost to Enter or Cross		Breakdown Value	Combat Adjustment			Stacking ¹ Limit
	non-Mot	Mot		Barrage	Anti Armor	Close Assault	
Clear	2	2	4	-	-	-	6
Gravel	2	2	6	-	-	-	6
Salt Marsh ²	3	2	6	-	-	R1	6
Heavy Vegetation	3	3	3	-	L1	L1	6
Rough	3	4	8	L1	L1	L2	6
Mountain	4	6	12	L2	L2	L3	3
Delta	2	4	2	-	-	-	6
Desert	3	4 ³	24	-	-	-	6
Major City	1	½	½	See Fortifications			8
Swamp ⁴	May enter only on road or railroad			-	-	-	6
Village/Bir/Oasis	Same as terrain in hex for all purposes			-	-	-	-
Railroad ⁵	Same as terrain in hex for all purposes			-	-	-	-
Road	1 ⁶	½ ⁶	½ ⁶	-	-	-	5 ⁷
Track	1 ⁸	1 ⁸	- ⁸	-	-	-	5 ⁷
Ridge	+2	+4	2	-	L2	L2	-
Up Slope	+2	+4	2	-	L1	L2	-
Down Slope	+1	+2	2	-	L1	R1	-
Up Escarpment	+6	P	-	-	P	L3	-
Down Escarpment	+4	+8 ⁹	+6	-	L2	R1	-
Wadi	+1 ¹⁰	+4 ¹⁰	+8	-	-	L1	-
Major River	+8	P ¹¹	-	-	-	L6	-
Minor River	+3	+6	+1	-	-	L2	-
Fortifications							
Level One	Same as terrain in hex			L1 ¹²	L1	L2	-
Level Two	Same as terrain in hex			L2 ¹²	L2	L3	-
Level Three	Same as terrain in hex			L2 ¹²	L2	L4	-
Friendly Minefield ¹³	+1	+4	0	-	L1	L1	-
Enemy Minefield ¹³	+4	+CPA	+2	-	-	-	-

CP = Capability Point; Mot = Motorized; non-Mot = non-Motorized; P = Prohibited; L or R = Column Shift(s) to left or right, respectively.

Notes

1. A garrison unit in its assigned hex occupies zero stacking space. In addition, an unlimited number of anti-aircraft units may occupy a Major City hex at zero stacking space; three anti-aircraft units may occupy an Airfield hex at zero stacking space while one anti-aircraft unit may occupy an Air Landing Strip hex at zero stacking space.
2. Vehicles *other* than Light Trucks, Recce-type units, and motorcycle infantry may only enter/leave the hex on a track. Motorized units may not engage in combat into/out of a Salt Marsh (Case 8.44).
3. Light Trucks, motorcycle reconnaissance units, and motorcycle infantry units may not enter, even via a track.
4. Alexandria and Cairo hexes are Level Three Fortifications, all others are Level Two.
5. Commonwealth units may be able to use Rail Movement (Case 8.7).
6. Negates all hexside terrain feature entry costs and Breakdown Point Values.
7. Applies only to the units/truck points in that hex at that time.
8. Halves all terrain feature entry costs and all hex/hexside Breakdown Point values *except* the Capability Points expended and the Breakdown acquired by moving a vehicle down an escarpment.
9. Motorized units and trucks may only cross on a track.
10. May only be crossed during a Rainstorm if the hexside is crossed by a road or railroad (which is treated as a road for this purpose) at a cost of two additional Capability Points.
11. May only be crossed if the hexside is crossed by a road or railroad. A motorized unit may so cross at no additional cost in Capability or Breakdown Points.
12. Armor targets receive the defensive benefit only if the hex is a Major City hex.
13. Engineer units reduce cost to enter (Case 26.2). If *assaulting* forces are in an Enemy minefield, the non-Phasing forces receive L1 shifts for Anti-Armor and Close Assault if not already receiving them for occupying a Friendly minefield.

[36.53] ALLIED OFF-MAP AIR FACILITIES CHART

Entry/Exit Hex	Air Facility	Distance to Entry/Exit Hex	Maximum Squadrons
E4033	Port Said*	12	3
E3433	Abu Seier	7	9
E3433	Ismailia	9	9
E1833	Fayid	11	9
E1833	Deversoir	12	9
E1833	Kabrit	13	9
E0127	Ethiopia†	60	Unlimited

*Flying Boat Basin, †No planes may be based here until the first OpStage of Game-Turn 35.

[8.89] OFF-MAP LAND UNIT MOVEMENT DISTANCE CHART

	Tunis	Gabes	Tripoli	Tripolitania
Tunis	-			
Gabes	1/2/5	-		
Tripoli	2/3/8	1/1/3	-	
Tripolitania	3/4/11	2/2/6	1/1/3	-
Nofilia	4/5/14	3/3/9	2/2/6	1/1/3

Numbers represent number of Operations Stages of continuous movement required for a unit to complete its movement between indicated places. Number refer respectively to units with a CPA of 25 or more/15 to 20/ less than 15. Note that a unit may not exceed its Basic CPA during off-map movement.

[37.4] AIR DISTANCE CHART

A. Axis Off-Map N. African Areas

	Tripolitania	Tripoli	Gabes	Tunis
Tripolitania	-			
Tripoli	32	-		
Gabes	68	40	-	
Tunis	91	60	40	-
Malta	60	42	58	52
Nofilia	32	64	104	120
Benghazi	44	88	124	136

B. N. Mediterranean Off-Map Areas

	Crete	Malta	Sicily	Italy
Malta	105	-	24	72
el Nofilia	P	80	P	P
Benghazi	62	84	96	138
Derna	27	108	105	147
Tobruk	48	126	P	P
Bardia	50	144	P	P
Mersa Matruh	66	P	P	P
Alexandria	84	P	P	P

All distances are in hexes. P=Prohibited; not possible to traverse air distance directly between indicated points.

[56.26] ROAD DISTANCE TABLE

	Nofilia	Marble Arch	El Agheila	Benghazi	Derna	Tobruk	Bardia	Mersa	Alexan
Nofilia	-								
Marble Arch	9	-							
El Agheila	19	10	-						
Benghazi	61	52	42	-					
Derna	98	89	79	37	-				
Tobruk	121	112	102	60	23	-			
Bardia	138	129	119	77	40	17	-		
Mersa Matruh	174	165	155	113	76	53	36	-	
Alexandria	213	204	194	152	115	92	75	39	-
Cairo	239	230	220	178	141	118	101	65	31

Determining Shortest Distance

The shortest distance (not the actual flight path) in hexes between any two map locations is determined as follows: Choose one hex to be the origin hex and the other to be the destination hex. Trace a path of hexes through the origin hex along any of the three 60° hex axes (i.e., east-west, northeast-southwest, or northwest-southeast) and trace a similar path using a different hex axis through the destination hex. At some point, these two paths will converge at a 60° not 120° angle. Count the number of hexes from the hex of origin to the hex in which the convergence occurs to determine the shortest distance. **Example 1: Tobruk to Ben Gania.** Either 31 hexes west to hex B4809 or 31 hexes southwest to hex B1725 as Ben Gania lies along the northwest-southeast axis that intersects both paths. **Example 2: Agedabia to Ismailia** (Allied off-map airfield). East 33 hexes to B2629 + 33 hexes to C2629 + 33 hexes to D 2629 + 33 hexes to E2629 + zero hexes northwest to E3425 + 8 hexes to E3433 + 9 hexes east to Ismailia. Total distance is 149 hexes.

[37.42] LAND DISTANCE CHART

	Nofilia																			
Nofilia (A2703)	-	El Agheila																		
El Agheila (A1816)	18	-	Benghazi																	
Benghazi (A4827)	35	30	-	Agedabia																
Agedabia (A2629)	27	17	22	-	Soluch															
Soluch (A4130)	34	25	7	15	-	Barce														
Barce (B5504)	48	39	13	29	14	-	Ben Gania													
Ben Gania (B3715)	50	41	26	24	20	20	-	Mechili												
Mechili (B4921)	62	53	27	36	28	20	12	-	Derna											
Derna (B5925)	71	61	36	45	37	23	22	10	-	C2803										
C2803	67	57	52	41	46	46	26	26	31	-	Tobruk									
Tobruk (C4807)	81	71	46	55	47	40	31	20	21	20	-	Giarabub								
Giarabub (C1014)	86	68	72	59	66	66	46	46	49	20	38	-	Ft. Maddalena							
Ft. Maddalena (C3019)	84	75	67	58	61	61	41	41	42	17	21	20	-	Bardia						
Bardia (C4321)	92	83	62	67	59	56	42	36	37	25	16	33	13	-	Bir-el-Qatrani					
Bir-el-Qatrani (D2903)	100	91	84	74	77	80	58	58	59	33	38	31	17	22	-	Mersa Matruh				
Mersa Matruh (D3714)	115	106	91	89	85	85	65	65	66	48	45	46	31	29	15	-	Gerawla			
Gerawla (D3516)	116	107	94	90	88	88	68	68	69	49	48	47	32	32	16	3	-	Alexandria		
Alexandria (E3613)*	147	138	124	121	118	118	98	98	99	79	78	78	63	62	47	33	31	-	Wadi Natrun	
Wadi Natrun (E2418)	149	137	135	122	130	129	109	109	110	83	89	76	68	73	51	44	41	12	-	
Cairo (E1829)*	163	145	149	136	143	143	123	123	124	98	103	85	82	87	65	58	55	25	14	

*One major city hex. Number represents the shortest distance in hexes from one location (exclusive) to the other (inclusive). Note that this includes cutting across the Mediterranean (assuming an all-sea hex grid) where necessary. A specific hex on a map section is always 33 hexes from a hex on an adjacent map section bearing the same hex number (e.g., D4102 is 33 hexes from C4102 and E4102). See Determining Shortest Distance above.

[56.18] AXIS NAVAL CONVOY AIR DISTANCE CHART

		Rally Points						
Nr.	Shipping Lane	Sicily	Italy	Crete	Malta	Benghazi	Derna	Tobruk
1	West Sicily to Bizerta	20	56	-	42	126	145	165
2	West Sicily to Tripoli	45	82	134	35	100	135	160
3	Italy to Benghazi	41	79	80	50	51	57	80
4	Greece to Benghazi	83	117	38	80	27	22	42
5	Greece to Tobruk	96	137	28	125	44	18	26
6	Italy to Tobruk	57	101	62	70	39	38	60

Nr. = The Axis Convoy Lane's description number. Sicily, Italy, Crete = Airplanes flying from any of those boxes. Malta = Airplanes flying from any hex on Malta. Values represent distance in hexes from that location to the Shipping Lane. Note that in order for a plane to fly a Mission to a Shipping Lane from North Africa it must possess a Range at least equal to the number of hexes from its Air Facility to a Rally Point and from that Rally Point to the Shipping Lane.

SUMMARY OF IMPORTANT LOCATIONS

Place	Location	Major City	Port	Assigned Garrison	Victory Condition
Aboukir	E3815	-	X	-	-
Agadabia	A2629	-	-	-	X
Alexandria	E3613/ 3714	X	X	-	X
Bardia	C4321	X	X	X	X
Benghazi	A4827	X	X	-	X
Benina	A4829	-	-	X	-
Bir Scheferzen	C3419	-	-	X	-
Bizerta	off map	-	X	-	-
Cairo	E1730/1829/ 1830/1930/ 1931	X	-	-	X
Derna	B5925	-	X	X	X
el Agheila	A1816	-	-	-	X
el Grein	C1715	-	-	X	-
Fort Maddalena	C3019	-	-	X	X
Giarabub	C1014	-	-	X	X
Halfaya Pass	C3922/3821	-	-	-	X
Helwan	E1430	X	-	-	-
Jalo	B0513	-	-	-	X
Marble Arch	A2109	-	-	-	X
Mechili	B4921	-	-	X	-
Mersa Brega	A2021	-	-	-	X
Mersa Matruh	D3714	-	X	X	X
Nofilia	A2703	-	X	-	-
Ras el Ali	A2010	-	X	-	-
Rosetta	E4019	-	X	-	-
Sidi Barrani	C4131	-	-	-	X
Sidi Omar	C3618	-	-	-	X
Siwa	C0127	-	-	-	X
Sollum	C4021	-	X	-	X
Soluch	A4130	-	-	X	-
Tobruk	C4807	X	X	X	X
Zuetina	A2928	-	X	-	-

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hex may attempt to blow a particular dump per Player-Turn. To blow a dump, the attempting Player must expend Capability Points equal to one-third (rounded up) of the attempting unit's basic CPA. The attempting Player may never expend more than the unit's basic CPA attempting to blow a dump moreover, a unit may not exceed its CPA to blow a dump.

To blow a dump, the Player expends the necessary CPAs and rolls one die and modifies the die roll if necessary for unit size, dump ownership, location, etc. The Player may adjust the die roll by announcing, before rolling the die, that he is expending an additional one-third or two-thirds of the attempting unit's basic CPA. Cross index the modified die roll with the percentage of supplies row on the Supply Dump Demolition Table (54.17) to determine the percentage of each type of supply in the dump that is destroyed. In addition, a unit occupying the same hex as a supply dump may attempt to blow that dump as part of a retreat before combat. The Player may only expend one-third of the unit's basic CPA and must vacate the hex regardless of the result. Supply Dumps may be blown in any segment of an OpStage.

[54.15] Dumps may be used by any Player as supply sources. The supply dump counters have ID numbers on them so that the Player may facilitate record-keeping, including which dumps are Dummies.

[54.16] It is suggested that Players set up their dumps so that each is within a one-OpStage truck ride (including loading and unloading time) from the next. Otherwise you are asking for trouble. Establishing a viable dump network should be top priority for logistics commanders.

[54.17] **Supply Dump Demolition Table**
(see Charts and Tables)

[54.2] **TRUCK CHARACTERISTICS CHART** (see Charts and Tables)

The use and mechanics of trucks is discussed in many places. Let it suffice to repeat that it is of primary importance that the Player set up a system whereby his trucks are constantly in use, ferrying supplies forward as fast as they can. The carrying capacities of one Truck Point for each type of truck are listed on the Chart; see Charts and Tables.

*It requires two light trucks to carry one TOE Point of infantry; it is *suggested* that light trucks not be used for this task.

[54.3] **THE COMMONWEALTH RAILROAD: Supplies**

[54.31] As stated previously, the Commonwealth railroad may be used to move personnel in any direction. It may also be used to move supplies. It may move either supplies or personnel — it cannot move both at the same time.

[54.32] The Commonwealth supply capacity of the railroad is 1500 tons per Operations Stage in either direction. The RR may move personnel in one direction and supplies in another. Players should check the various Sections on supplies to see the equivalent weight in tons for each type of supply.

[54.33] The railroad may transport only one type of supply at a given time. It may move fuel, ammunition, or stores — not any combination of the three. (Water need not be transported by RR — the railroad hexes are pipelines in and of themselves.)

[54.34] For the duration of *one* Operations Stage per month (calendar month), the railroad may not be used for anything. It is transporting water forward for railroad use. Players must state each

month which Operations Stage they are not using the railroad.

[54.35] Like personnel, supplies may be moved from any one spot and dumped in another spot. Supplies are considered unloaded when they reach a specific hex. They may not be moved that Operations Stage.

[54.4] **AXIS USE OF THE COMMONWEALTH RAILROAD**

Under certain conditions the Axis Player may make use of the coastal railline extending from Egypt to Libya (the Commonwealth railroad system). In essence, he must control the rail hexes and he must import rolling stock. Remember, the Barce railroad east of Benghazi is not usable under any conditions.

[54.41] At any time the Axis player controls five or more contiguous rail hexes he may use such rail hexes to transport equipment and personnel. To control a rail hex the Axis player must be the last player to have a land combat unit of any type pass through that hex.

[54.42] The Axis Player may never build new rail line. However, he may repair rail hexes he controls at the same rate as the Commonwealth Player.

[54.43] To make use of controlled Commonwealth Railroads the Axis Player must convoy Rolling Stock (locomotives, tenders, stock cars, flat cars, etc.) from Europe to Africa. For each 250 Stores and 100 Fuel Points that the Axis Player imports to Africa and brings to any controlled and operative rail hex he may activate all such hexes under his control (as long as they are contiguous) to the extent of hauling 300 Tons of Supplies in any *one* direction during an Operations Stage. (Rail movement occurs in the Convoy Stage).

[54.44] In order to transport one Stacking Point of any type of unit the Axis Player must have active enough Rolling Stock to transport the equivalent of 900 Tons of Supplies in any one direction.

[54.45] Once Stores and Fuel is applied as Rolling Stock the points are used up; they may not be recovered. If at any time the Axis Player loses control of enough rail hexes so that he does not have the necessary five contiguous hexes the Rolling Stock is considered to have been destroyed. The Axis may not use Commonwealth Stock, nor *vice versa*.

[54.46] All rules concerning the movement of troops/supplies and the use of the railroad that apply to the Commonwealth apply equally to the Axis.

[54.5] **EQUIVALENT WEIGHTS CHART** (see Charts and Tables).

[55.0] **PORTS AND SUPPLY**

[55.1] **PORT CAPACITY AND USE**

[55.11] There are two types of ports: *major ports* and *minor ports*. Major ports may receive both men and supplies; minor ports may receive only supplies. Both types of ports may ship *out* men and supplies.

[55.12] Every port in the game has an Efficiency Level, which is an abstract number assigned to that port. For example, Tobruk has an Efficiency Level of 5. This Efficiency Level indicates the level of efficiency at which the port is operating. The Efficiency Level of the port may be reduced by various Enemy activities, all discussed previously.

[55.13] In addition to its Efficiency Level, each port has a specific limit (capacity) as to the number

of TOE Strength Points and/or tons of supplies that may enter and leave that port; this is called the port capacity. (See the Port Capacity and Efficiency Level Chart, 55.3.)

[55.14] The amount of troops and supplies that can be brought into or out of a port depends on its current Efficiency Level and its capacity. A port operating at peak efficiency (its level is at maximum assigned) uses its listed capacity. A port that has been reduced by bombing or mines operates at a lower efficiency directly proportionate to its reduced Level. Thus, a port with an assigned Efficiency Level of 5 that has had its Level reduced to 3 operates at 3/5 (or 60%) or its assigned capacity. Therefore, all incoming and outgoing shipments are reduced to 60% of original level. **Example:** Benghazi has a supply tonnage capacity of 2500 tons per OpStage. It has a maximum Efficiency Level of 3. If this Level has been reduced to 1, its supply receiving capacity is reduced to 1/3 of 2500 (or 834 tons). Round all reductions upward to the nearest ton.

[55.15] In addition to any reductions brought about by Enemy action, the fact that troops are being debarked at a port — as opposed to supplies (fuel, ammo, etc.) — may reduce its Efficiency Level *vis a vis* supplies for the OpStage or Game-Turn in which the troops arrive. The arrival of *scheduled* reinforcements (those that arrive mandatorily, via the Reinforcement Schedule), as opposed to those which the Players plan, etc., *do not* affect a port's capacity or Efficiency Level.

[55.16] Some ports may receive shipments only in a Strategic Phase, while others may be used each OpStage. Players should be careful to adjust all capacities and Efficiency Levels to cover *both* uses. A port's capacity applies to *all* shipments received in a Game-Turn (including OpStages), except as stated in Case 55.15.

[55.17] The port of Bizerta (Tunis Box) is an unusual case. Although nominally open to the Axis via Vichy France, on Hitler's request it was never used (for various convoluted reasons). To add to that, it is quite a long way to the front — even if it is out of range of most Commonwealth planes. In any case, Bizerta is non-operable until the start of the June 1, 1941 Game-Turn. Starting with that Game-Turn, the Axis Player throws two dice each Game-Turn. If he throws a 12, he may use Bizerta from that turn on. It remains inoperative until the Game-Turn that he does throw a 12.

[55.18] For every OpStage that a port does not lose any Efficiency Levels to Enemy bombs, it regains one point. **Exception:** Cases 55.26 and 55.27. It can *never* go above its maximum assigned level (see Case 55.3).

[55.2] **BLOCKING HARBORS**

Either Player may reduce the effectiveness of a port he actually occupies by scuttling ships in the harbor to block other ships from entering. This was done notably at Benghazi by the Axis, thereby reducing the effectiveness of Benghazi (as a port) to almost zero during the entire war. Enemy Players may also mine opposing ports.

[55.21] A Player who controls a port may desire — for whatever reason occurs to him — to block up the harbor. Each harbor being different, the ability to do this varies with the port. The ports at Bizerta, Tripoli, Alexandria, Aboukir (E3815) and Rosetta (E4019) may not be so blocked.

[55.22] In order to block a port harbor, the Player must have an engineer battalion or CW HQ with engineer capacity present in the port hex. He then expends 25 Ammunition Points and 10 Stores Points, which must be actually present in the hex,

in the Construction Phase. The Efficiency Level of the port is reduced by *one* (see Case 50.14).

[55.23] All ports are reduced at the same rate, with one exception: it requires the expenditure of 50 Ammunition Points and 25 Stores Points to reduce Tobruk port one Level.

[55.24] A port may be reduced one Level per Operations Stage; it may never be reduced more, regardless of the expenditure of supplies.



[55.25] Players will note that at the start of the game, the Italian cruiser *San Giorgio* is partially blocking the harbor of Tobruk. The *San Giorgio* reduces the efficiency level of Tobruk by three levels.

[55.26] The effect of blocking harbors may be undone only by Friendly engineer battalions or CW HQ with engineering capability. Engineers reverse the process of 55.22. However, for each port, it costs 50 Ammunition Points and 25 Stores Points to undo the damage, at one level capacity per Operations Stage. There are two exceptions to this: Tobruk requires only 25 Ammunition Points and 10 Stores Points to remove one level of blocking effect; Benghazi requires the expenditure of 100 Ammunition Points and 50 Stores Points *plus* the presence of *two* engineer units to remove one level of blocking effect.

[55.27] Harbors may also be blocked by Enemy bombers laying mines by air (see Case 41.3 for the mechanics). Each successfully laid mine lowers the Efficiency one level. Mines may be removed by engineer battalions. If there are mines in a harbor, and the Player controlling the harbor has an engineer unit (as in Case 55.26) in that harbor, he rolls a die for each mine in the harbor. A roll of 1-3 removes the mine; a roll of 4-6 and the mine remains. Mine removal is done in the Construction Stage, and the engineer unit may not move or voluntarily expend CPA's in that Stage. The engineers cannot be involved in any other construction or removal work.

[55.28] CW ships in an Axis port hex have no effect on Convoy arrivals. (This is somewhat of an abstraction; attempts to block the ports during the day would not have worked due to Axis air power.)

[55.3] PORT CAPACITY AND EFFICIENCY LEVEL CHART (see Charts and Tables)

[56.0] THE AXIS NAVAL CONVOY SYSTEM

GENERAL RULE:

Almost everything the Axis needs for its desert campaign must be shipped in from Italy and the Aegean. Supplies and personnel are sent in convoys. Convoys are planned one Game-Turn in advance and the route of the convoy (the shipping lane it will use) is also chosen at that time. Each convoy is given in tonnage, and the Axis allowable tonnage depends on the month, year, and roll of the die. It may be further depleted by Commonwealth bombers. In addition, the capacity of each port is limited as to the tonnage it may receive, and that facet will also limit the Axis planning. The Axis Player has a limited amount of coastal shipping which he may use to transfer small amounts of supplies from one coastal port to another.

CASES:

[56.1] THE SHIPPING LANES

[56.11] There are six Shipping Lanes in *CNA*, representing the various routes taken by Axis convoys. The Shipping Lanes are not printed on the game map but rather are identified by their ports of origin and destination (see the Axis Naval Convoy Air Distance Chart).

[56.12] An Axis convoy is defined as the total tonnage assigned to a Shipping Lane during a Game-Turn. The Axis Player may have as many as six convoys or as few as one during any Game-Turn. The Axis Player is restricted to the listed ports of origin and destination combinations (i.e., he may not create additional shipping lanes).

[56.13] Axis Naval Convoys may be attacked by Commonwealth bombers (see Axis Naval Convoy Strategic Bombing, 41.6). The Axis Player must therefore balance the risks in having a convoy arrive close to the front, thereby placing it within range of a greater number of Commonwealth N. African based planes or having it arrive far in the rear, trading time and supply expenditures for safety.

[56.14] Shipping lanes 4 and 5 (the routes originating from Greece) may not be used until the second Game-Turn of May, 1941. See also Case 55.17.

[56.15] All Axis convoys are assigned to a specific shipping lane in the Axis Convoy Planning Phase. The lane assignment may not be changed, regardless of any changes in the positions of the land units, with one exception: a convoy scheduled to arrive at a port that is captured by the Commonwealth is cancelled. It never sails.

[56.16] The Axis Player notes on his Convoy Control Sheet what he is shipping and through which lane. If that shipping lane is not attacked by Commonwealth planes, all items arrive at their designated port (which is stated as part of the lane). If the lane is under bombardment, the Axis Player totals the tonnage in that lane and divides by 1000 to obtain his Flak/AA Points for that convoy. Thus 15,000 tons of shipping would amount to 15 Flak/AA Points with which to fire at Commonwealth planes. Any losses are taken as per the strategic bombardment rules.

[56.17] There is no limit to the tonnage that can be carried by a convoy in a given lane, except as noted in Case 56.2.

[56.2] SHIPPING BY CONVOY

Many different methods of determining arrival rates and times of Axis convoys were considered in designing this game. Each one had several merits — all had drawbacks. We chose this method (using the actual tonnage figures for a given month) over the other less easy to determine methods. No one was completely satisfied with any of the methods. (The drawback with the chosen method is that the arrival rates may not coincide with Axis needs.) But both designer and developer were loathe to tackle the problem of Italian ship-building, availability of supplies in Italy, and a whole host of other problems. Thus the rules below reflect the arrival of Axis supplies as they actually occurred (within the parameters of the rules).

[56.21] Axis shipping of supplies by naval convoy uses several charts, under the heading of the Axis Naval Convoy Level Chart. The Axis Convoy Capacity Table refers the Axis Player, by month and year, to the Tonnage Capacity Table for that particular month. The figures given on the Tonnage Determination Table are the tonnage of supplies that the Axis may ship in that Game-Turn (for which he is planning).

Example: The Axis Player, at the beginning of Game-Turn 54 (October IV, 1941), makes his plans for his shipping and convoys for Game-Turn 55 (November I, 1941). (Remember, while Axis replacements are planned *two* turns in advance, the Axis Player always plans his convoys *one* Game-Turn prior to actually sending them out.) The Axis Monthly Shipping Table tells the Axis Player to use row E on the Tonnage Capacity Table for November 1941. The Player refers to that Table and column and rolls a die; he rolls a 4. That means he may ship 21,000 tons of supplies and Replacement Points (see 20.00) on Game-Turn 55. The choice of shipping lane and destination is up to him.

[56.22] Having determined the allowable tonnage for a given Game-Turn, the Axis Player may now plan to ship *any* amounts (within the limits of allowable tonnage) of fuel, ammunition, and stores that he wishes. They are available (for game purposes) in unlimited quantities in Europe. For example, with those 21,000 tons available on Game-Turn 55 (November I, 1941), the Axis Player may decide to ship out 10,000 tons of fuel, 10,000 tons of ammunition, and 1000 tons of stores.

[56.23] The tonnage equivalencies of supplies are listed on the Equivalent Weight Chart (54.5)

[56.24] The Axis Player may also have planned previously (see Case 20.6) to ship replacements from Italy/Greece. These take up space and weight, and their tonnage equivalencies are given in the Equivalent Weight Chart (54.5). Only Replacement Points *not* arriving via the Reinforcement Schedule affect shipping tonnage; reinforcements and scheduled replacements do not use convoys and do not affect shipping tonnage. Thus if the Axis Player had planned to send 10 Infantry Points in the Game-Turn 55 convoys, this would subtract 300 tons from the available tonnage for that Game-Turn.

[56.25] After determining what tonnage he has available and what he wishes to ship, the Axis Player then, at the same time, allocates his available tonnage to the lanes — and ports — he wants them to use. He cannot change lanes and destinations once chosen (see Case 56.1).

[56.26] Axis convoys — and their cargoes — may undergo Commonwealth bombardment.

[56.27] The Axis Player should remember that ports have maximum capacities; they may not receive supplies over that capacity (see Case 55.3).

[56.28] Supplies and personnel are unloaded the instant of their arrival, (if possible) and they may be used immediately. All ports of arrival (e.g., Tripoli, Bizerta, Tobruk) are considered to have supply dumps built in, as it were.

[56.3] AXIS COASTAL SHIPPING

The Axis had a small fleet of boats that they used for coastal transfer of small amounts of supplies. These were old shipping boats and aging tramp steamers that could ill afford to venture too far from land. They have a limited capacity.

[56.31] Axis coastal shipping is represented by several ship counters. Each ship counter has its tonnage capacity printed on it. Each ship has a CPA — for movement only — of 50. Each sea hex costs one point.

[56.32] Axis coastal shipping moves in the Truck Convoy Phase, and *only* in that Phase.

[56.33] Axis coastal ships cannot be attacked from air or land. They may not enter a port that has been neutralized (Capacity Level of zero). They may also never enter an Enemy-occupied port.

[56.34] To use coastal shipping, the Axis Player loads supplies (*no* personnel) on his boats at the beginning of the Truck Convoy Phase (at a cost of 5 CPs) and then moves the boat. Unloading supplies costs 5 CPs. Only one type of cargo may be carried at any one time by any one ship. Tanks, guns, etc., may not be shipped via coastal shipping

[56.35] Coastal ships may stop and unload in one port and then continue on to another port, etc., as long as they do not exceed their CPA. Coastal ships do not require fuel.

[56.4] AXIS NAVAL CONVOY LEVEL CHART (see Charts and Tables)

[56.5] AXIS CONVOY CAPACITY TABLE (see Charts and Tables)

[57.0] COMMONWEALTH SUPPLY BASE

GENERAL RULE:

The Commonwealth Player has an unlimited amount of supplies of all types — fuel, ammunition, water, and stores — in Cairo at *all* times during the game. His problem is solely to get it to where he wants it. He does not ship supplies in; if he wants something, it is in Cairo. Remember, Commonwealth ships need no supplies. For reinforcements and replacements for personnel and equipment, the Commonwealth Player should consult his Reinforcement Schedule and CW Replacement System (see Section 20.0)

[58.0] ABSTRACT AIR RULES

COMMENTARY:

This Section covers the rules required for the Players to play the Land and Logistics Games without utilizing the Air Game rules. *CNA's* play is relatively unaffected by the lack of an Air Game, the adjustments required being made in the Player's "trucking" ability.

CASES:

[58.1] EFFECT ON AXIS NAVAL UNITS

Attacks on Axis naval convoys are conducted using the same procedure listed in the Basic Logistics and Air Rules of the Land Game (see Section 32.0). The only difference is that the convoy being attacked is now more detailed as to what it is transporting. Note that the Axis coastal shipping and both Players' tactical shipping may not be attacked.

[58.2] COMMONWEALTH FLEET BOMBARDMENT

The Axis Player is entitled to make attacks on the Commonwealth fleet. Such attacks utilize the same procedure as that in the Basic Logistics and Air Rules described in the Land Game (see Section 32.0).

[58.3] EFFECT ON SUPPLIES

There is only one restriction — the Axis Player loses three-quarters of all fuel actually brought in to a port on the instant of its unloading (considered to be used for airplanes). This loss is incurred before any loss for evaporation is calculated.

[58.4] EFFECT ON TRUCKS

[58.41] The Players' initial set-ups will be reduced by a certain number of Truck Points (those assigned or that would be assigned the task of supplying the air facilities).

[58.42] Both Players are subject to truck losses in addition to those caused by land combat. These losses represent the abstracted effects of strafing and bombing the opposing Player's trucks. The losses are determined in the same manner as those for motorization points (see Case 32.57) using the percentage loss figures in the Abstract Motorization Point Loss Chart (32.59) but applying them to the Players' trucks.

The owning Player may remove the lost Truck Points from any place he wishes. However, losses must be distributed in proportion by truck type and by proportion of trucks in convoy and those attached. **Example:** A Player possesses 50 Light, 100 Medium, and 50 Heavy Truck Points distributed as 40 Light and 10 Medium in convoy and the remainder attached. If he is required to lose 10% of the total Truck Points, he must lose 4 Light and 1 Medium Truck Points from those in convoy and 1 Light, 9 Medium, and 5 Heavy Truck Points from those attached.

[58.43] Both Players are subject to a reduction in their total available truck force in order to account for the trucks that would be carrying supplies to their air facilities. Both Players lose 10% (rounding fractions up) of all truck points that arrive in N. Africa (the instant they arrive) via the Commonwealth Truck Production Table and the Axis Replacement Pool. Trucks listed on the reinforcement track are not affected.

[58.44] Abstract Truck Loss Chart (see Charts and Tables)

SCENARIOS

[59.0] INTRODUCTION TO SCENARIOS

COMMENTARY:

There are five scenario groups in *Campaign for North Africa*. Each scenario group contains at least one separate game. Of course, one is for the playing of a campaign game. The groups cover what the designer felt were the most important — and most accessible, in terms of information — areas of the campaign. Specific battles, such as Gazala, were avoided as they did not fit within the concept of the game (operational). Each scenario is a self-contained game and each highlights certain problems.

Those wishing to get the most out of *CNA* without becoming enmeshed in a lifetime occupation are advised to try the Rommel's Arrival scenario — it is fast-moving and a lot of fun. It is also not too long or difficult (the small number of units involved keeps paperwork to a minimum). For those interested in seeing why trucks are so important and want to keep all their work in a small area, try the Italian scenario. For those interested in seeing a lot of units sit around doing nothing, try El Alamein — a remarkably boring situation (in game terms, if not historically).

Players should take great care in setting up each scenario, as a great deal of information is provided. The bigger scenarios could take 10 to 15 hours to prepare. A further word to the Africa buffs: given all the information in the game, it should be fairly easy to concoct your own situations and scenarios. Your major problem would be logistical information.

On the subject of Victory Conditions, it is the designer's feeling that, in a game of this size and scope, few Players will patiently battle through 100-200 hours of playing time and feel they have actually lost (or won). Finishing is an event in itself. The Victory Conditions have been thought out carefully, in light of not only what was actually accomplished, but also what had to be accomplished. Yet the conditions *are* somewhat general, especially the campaign game objectives. This is because outside considerations that could not possibly have been simulated had to be taken into effect. Any Player worth his salt will know whether he has accomplished what he set out to do without checking what we've told him he must do.

CASES:

[59.1] SCENARIO ORGANIZATION

[59.11] The scenarios in *CNA* are divided into five groups. The first four of these groups deal with individual battles or offensives, and the fifth covers the campaign game.

[59.12] Each of the first four scenario groups contains the following information:

- Length of the scenarios in the group.
- Victory Conditions of the scenarios in the group.
- Initial location and strength of all land and air forces.
- Initial location and amount of all supplies.
- Initial location and number of all trucks.
- Forces available on Malta & Crete/Italy/Sicily.
- Abstractions necessary to play the scenarios without the Air and/or Logistics Games.
- Special rules for the individual scenarios.

[59.2] HOW TO READ THE SCENARIO SET-UPS

Basically, all air and land units, supplies, and trucks available at the start of a scenario are listed by hex location. Generally, the parent unit will be listed, and Players must refer to the appropriate OA Chart to determine which subordinate units are assigned to that parent unit. Whenever a given unit has a composition different from the one listed on the OA Chart, that is noted in parentheses.

Hex Location. If more than one hex is listed for a given group of units, the Player may divide the listed units between the hexes in any way he chooses, so long as stacking limitations are not exceeded.

Unit Type. If not apparent from the name, this is noted with an abbreviation. Where noted, it is always the first item enclosed in parentheses. I = infantry; R = reconnaissance (recce); T = tank; Ar = artillery; AT = anti-tank; AA = anti-air; HQ = headquarters (the headquarters counter *only*).

Parent of Listed Unit. If the listed unit has been assigned to a parent unit (whether or not it still is), that parent unit's designation is listed, in **bold type**, as it appears on the listed unit's counter. This is to facilitate identification of the listed unit.

Listed Unit Lacks Subordinate Units Originally Assigned. All such subtractions from what is listed on the OA Chart are noted after the indicator "Less."

Listed Unit Includes Subordinate Units Not Originally Assigned. All such additional units (which may be detached) are listed after the indicator "Assg."

Assigned Subordinate Units Are Detached. These are listed under the parent unit after the indicator "Det." Of course, they are listed separately in the hex(es) they themselves are in.

Listed Unit Has Entirely New/Different Roster of Assigned Units from Those Listed on OA Chart. All subordinate units listed after indicator "Consists of."

Note. Whenever a unit inside parentheses requires expanded explanation, that explanation is enclosed inside double parentheses: e.g., ((...)).

Listed Unit Exceeds Formation Organization Limits. Such units may operate normally, including being themselves assigned and/or attached, with the exception that no new subordinate units may be assigned or attached to them, and should any units be detached, no new ones may be attached in violation of the limits.

Listed Unit Includes Subordinate Units Attached But Not Assigned. All such units are listed after the abbreviation "Att."

Disposition of the Units in a Hex. All units listed to begin in the same hex may be organized as the owning Player chooses, within the limits of the Formation Organization rules and stacking limitations. In other words, units in the same hex may be freely detached or attached to each other at the start of the scenario, in any fashion deemed desirable.

Trucks. All listed trucks are first line trucks and must be assigned to the units in the hex, in any manner deemed desirable.

Abbreviations Used in Scenario Set-Ups.

Div = division; **Bde** = brigade; **Regt** = regiment; **Bn** = battalion; **Coy** = company; **In** = Indian; **Gds** = Guards; **RTR** = Royal Tank Regiment; **RHA** = Royal Horse Artillery; **Hus** = Hussars; **SA** = South African; **NZ** = New Zealand; **Cp** = Comps; **CLY** = County of London Yeomanry; **Gaf** = Garrison troops; **CCNN** = Blackshirts; ***** = no parent organization.

[59.3] INITIAL AIR STRENGTHS

[59.31] Each scenario lists the starting Axis and Allied plane types, numbers, pilots, air facilities, and squadron ground support units. This listing is divided into forces available in North Africa and Malta for the Allied Player and to forces available in North Africa and Sicily/Italy/Crete for the Axis Player.

[59.32] Planes are listed by type, with the ready planes indicated as a portion of the total available. All planes (refitted or not) are considered to begin a scenario fueled and armed (at no cost to the supplies available).

[59.33] Planes may be distributed among the squadron ground support units as the owning Player desires within the restrictions on squadron composition listed in Case 35.2.

[59.34] Pilots may be assigned to planes and squadrons as the owning Player desires.

[59.35] Squadron ground support units may be placed on any of the air facilities owned by the Player. However the number of SGSU's may not exceed an air facility's readying capability.

[59.36] Maintenance may not be performed on planes during the first OpStage of a Scenario.

[59.4] INITIAL TRUCK STRENGTHS

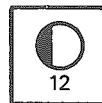
[59.41] Trucks initially available to each Player are divided into 1st line, 2nd-3rd line available for supplies, and 2nd-3rd line available for air facilities.

[59.42] First line trucks are listed by hex in the land forces initial deployment section of the scenario. These trucks must be assigned to units in their listed hex. The trucks may be divided among the units in that hex in any manner the Player desires.

[59.43] Second-third line (in convoy) trucks available for air facilities must be placed on an air facility.

[59.44] Second-third line trucks available to transport supplies must be placed (if the Axis Player is not starting them off-map) on a hex occupied by a Friendly combat unit, a supply dump, or a major city/town/oasis. These trucks may be distributed as the owning Player desires.

[59.45] Trucks may start a scenario transporting supplies or motorizing troops or anti-aircraft units at no cost or penalty. Any trucks starting with supplies may load any type of supply. These supplies are *in addition* to those listed in the supply section of the scenario.



[59.5] INITIAL SUPPLY

[59.51] Each scenario lists the location and amounts of a Player's initial supplies.

[59.52] The supplies are divided into those available at air facilities and those available at supply dumps. Air facilities automatically possess a supply dump. If a Player places supplies available at a supply dump in the same location as those available at an air facility, the totals are combined and it becomes one dump (see Case 36.17). Some scenarios separately list an aggregate number of supplies available at a player's air facilities. These may never be allocated to off-map facilities.

[59.53] Certain scenarios list a Player as receiving active and non-active supply dumps. Non-active dumps are dummy dumps and cannot, therefore, be placed in the same hex as an air facility.

[59.54] Axis coastal shipping begins all scenarios unloaded (i.e., empty).

[59.6] ABSTRACT LOGISTICS AND AIR

[59.61] If only the *Air Game* is abstracted, the Players simply ignore all Trucks and supplies available at/for Air facilities in the initial set-ups. Play is then governed by the rules in Section 58.0.

[59.62] If the Logistics Game is abstracted, whether or not the Air Game is also abstracted, Players will be using Motorization Points and Supply Units. The initial set-ups for the Supply Units and Motorization Points are given in the following cases.

[59.63] Initial Motorization Point Strengths: The initial motorization points available to each Player are calculated from the trucks listed in the scenario set-up.

A. Attached Motorization Points: These are calculated, hex by hex, from the 1st Line trucks available to the Player unless specifically modified by a scenario's instructions. Each Medium Truck and each Heavy Truck is equal to one Motorization Point. **Example:** In the first Scenario Group, the 4th CCNN is listed as beginning the scenario with 10 Light, 25 Medium and 10 Heavy Trucks. It possesses 35 Motorization Points.

B. Unattached Motorization Points: These are calculated from the second-third Line trucks available to each Player in the same manner as in "A." If the Air Game is being abstracted, ignore all trucks available at air facilities. Note that the initial distribution/placement of these trucks will affect their ability to start a scenario loaded with supplies in addition to those listed as available in supply dumps (see Case 59.66).

[59.64] Initial Motorization Point distributions may be, with one exception (see Case 59.66 "B"), freely altered once the game starts as per the Land Game rules on attachment and detachment.

[59.65] Supply Units: The Player's initial supply levels are listed as to the number, location, and

contents of regular and depot supply units available. Regular supply units are described in Section 32.0. Depot supply units — used only if the Air Game is being played and the Logistics Game is being abstracted — are described in Section 47.0. Initial supply unit placement may exceed stacking limitations only if unavoidable due to scenario distribution instructions. There is no penalty in such situations; however, no additional supply unit may be stacked in that hex. Once the supply unit distribution meets stacking restrictions all normal stacking restrictions for that hex must be met from that point on. (Remember that motorization points transporting supply units do not count against hex stacking limits.)

[59.66] Additional Supplies:

A. Motorization Points may begin a scenario loaded with supplies. These do not count against supplies listed in the scenarios supply dumps.

B. Each attached Motorization Point (see Case 59.63 "A") not initially transporting men or guns may start loaded with 1 Ammo Point or 3 Fuel Points. The unit must consume these supplies first (i.e., it may not draw on a supply unit) in order to meet any supply expenditure requirement. These supplies may not be transferred to a supply unit. The Motorization Points transporting these supplies may be detached from the unit *only* if remaining attached to a subsidiary combat unit of that parent unit (Note that the Motorization Point is under this restriction only while it is transporting these initial supplies).

C. Unattached Motorization Points (see Case 59.63 "B") are divided into those that are listed as initially available at air facilities (these appear only if playing the Air Game) and all other unattached Motorization Points. Air facility Motorization Points may begin by transporting additional supply units as follows: Each 50 Motorization Points in a hex may start with a depot supply unit; each 30 Motorization Points in a hex may start with a regular supply unit. Each 30 other unattached Motorization Points in a hex may start with a regular supply unit. **Example:** 80 air facility and 100 other Motorization Points may begin with the air facility points transporting one depot and one regular supply unit while the other points may begin with three regular supply units. Note that for these Motorization Points to begin with a supply unit they must be in groups of 30 (and/or 50 if air facility) points (i.e., 20 Motorization Points may not begin with a supply unit). The Tripoli/Tunisia boxes are hexes for the purposes of this Case.

[60.0] SCENARIO GROUP ONE: The Italians

CASES:

[60.1] GENERAL DESCRIPTION

The first scenario group covers the initial months of the desert fighting, from 15 September 1940 through the middle of February 1941. Although there was an actual state of war existing between the British and the Italians *vis a vis* Libya/Egypt, the months of June through early September 1940 have been ignored as there was little done other than aggressive patrolling. The actual campaign starts with the first Italian offensive and subsequent British counter-attack.

[60.2] SCENARIO LENGTH

[60.21] There are two separate scenarios in the Italian scenario group; each may be played separately to a specific conclusion.

[60.22] Graziani's Offensive

This scenario starts with Game-Turn 1, OpStage 1; it ends with the conclusion of Game-Turn 6, OpStage 3. The estimated time of play is anywhere from 25 to 50 hours. Victory Conditions are listed in Case 60.71.

[60.23] The Italian Campaign

This scenario starts with Game-Turn 1, OpStage 1, and proceeds through the end of Game-Turn 20, OpStage 3. It includes Operation Compass (O'Connor's counteroffensive). The estimated time of play is 90 to 150 hours. Victory Conditions are listed in Case 60.72.

[60.3] AXIS FORCES**[60.31] Italian Initial Deployment (Hex; Units)**

C4218: 1 CCNN Div (I; Att: I(M) [T; Ar/LTC] Trucks: 10 light, 25 medium, 10 heavy.

C4120: 63 Cirene Div (I; Att: LXIII (L) [T; Ar/LTC]; Trucks: 45 medium, 5 heavy.

C4020: 1 Libyan Div (I); Trucks: 10 light, 15 medium.

C3920: 2 Libyan Div (I; Att: IX(L) [T; Tvl/LTC]; Trucks: 15 light, 15 medium.

C3919: Aresca Regt (T; LTC, Det: all except XXI); Trucks: 5 medium, 5 heavy.

C3918: 62 Marmarcia Div (I; Att: LXII(L) [T; Ar/LTC]; Trucks: 40 medium, 5 heavy.

C3617: Maletti Div (I; Att: II(M) [T, Tvl/LTC]; Trucks: 5 light, 30 medium, 5 heavy.

C4707: 64 Catanzaro Div (I); Trucks: 30 medium, 5 heavy.

C4507: 4 CCNN Div (I); Trucks: 10 light, 25 medium, 10 heavy.

Bardia (C4321): Trivoli Regt (T; LTC, Det: II(M) and IX(L)); Bardia Garrison; Trucks: 20 medium.

Tobruk (C4807): HQ: Libyan Tank Command; Tobruk Garrison; Trucks: 5 medium.

Benghazi (B4827): Benghazi Garrison.

Barce (B5504): Libyan Parachute Regt (I); Trucks: 5 medium.

Derna (B5925): Derna Garrison.

Mechili (B4921): Mechili Garrison.

Giaribub (C1014): Giaribub Garrison.

Bir Scheferzen (C3419): Scheferzen Garrison.

Fort Maddelena (C3019): Fort Maddelena Garrison.

el Grein (C1715): el Grein Garrison.

Benina (A4829): Benina Garrison.

Soluch (A4130): Soluch Garrison; X Cp (Ar).

Anywhere in Libya: 1/1AR; 2/1AR; 3/1AR; 4/1AR; XXI Cp; XXVI Cp 4/1AR; XXICp (Ar).

Within 3 Hexes of C0716: Saharan Det (I); Trucks: 5 light.

Anywhere on Map A or B: XVIII Lib (I; *); XXX-I Lib (I; *); and (all Ar) 147; 131; 2/24; 5/1; 6 GaF; 16 GaF; 22 GaF; 42 GaF; 350 GaF.

Tripoli: 2 CCNN Div (I); 3 CCNN Div (I); XXII Cp (Ar); XXIII Cp (Ar).

Tripolitania: 4/10 Army (I).

In addition, two TOE strength Points of Autoblinda 40's (recce, 1D Code "WW") may be attached, each point to any unit (but not both to the same) ignoring all regular unit composition restrictions. Or they may be deployed in Tobruk as regular (i.e., trained and fueled) replacement points.

[60.32] Italian Air Strengths

The following planes may be placed at any Italian airfields, landing strips, flying boat basins, etc., within the capacities of those facilities. However, no planes start the game in Italy/Scily. Crete is unavailable (it is still British).

Plane	Total Available	Total Refitted
CR 42	65	25
CR 32	70	28
Ba 65	10	2
Ba 88	24	6
SM 79	133	35
SM 81	17	4
Ca 309	56	12
Ro37Bis	10	6
2S01	9	9

The Italian Player may not attempt to refit planes until Game-Turn 1, OpStage 2.

Italian Pilots Available:

Three's: 3

Two's: 10

One's: 15

Italian SGSU Available: 39.**[60.33] Italian Second-Third Line Trucks**

The Italian Player receives the following trucks (Points). He may assign them and place them as he wishes; within the restrictions listed. (Original assignments may, of course, be changed as the game progresses).

Location	Trucks Available (in points)
Tripoli	25 Light 140 Medium 40 Heavy
Anywhere in Libya	30 Light 100 Medium 25 Heavy
Any Air Facility	10 Light 50 Medium

The Italians have a major repair facility at Tripoli, and a temporary facility at Tobruk and Benghazi.

[60.34] Axis Initial Supply Status: The Axis Player receives Dumps with the following supplies at the following locations:

Location	Supply (in points)			
	Ammo	Fuel	Stores	Water
Tobruk (C 4807)	200	2000	500	-
Bardia (C 4321)	100	1000	200	-
Benghazi	100	250	100	-
Derna (B 5925)	-	250	50	-
Tripoli (box)	250	5000	250	-
Dump 1*	1000	1500	1500	200
Dump 2*	1000	1500	1000	200
C 0716	100	50	50	-

*These and two Dummy supply dumps may be placed in any hex on map section C in Libya that is not within four hexes of a Commonwealth unit.

In addition the Axis Player receives a total of 1200 Ammo, 850 Fuel, 100 Stores and 100 Water Points which may be freely distributed among his airfields.

[60.35] The Italian Player may use all Axis coastal shipping counters. They start the game in Tripoli.

[60.36] The Axis Player's Strategic attacks on Malta are restricted to forces called for by Availability Level I (see the Axis strategic Commitment Chart, case 44.41).

[60.37] The Italian Player must plan convoy runs for the remainder of September, 1940, prior to starting the game. He may only use lanes 2,3 and 6.

Consult the Convoy Availability Charts to see what shipping the Italian Player has. He receives all reinforcements as per the Reinforcement Track.

[60.4] COMMONWEALTH FORCES**[60.41] Commonwealth Initial Deployment (Hex; Units)**

C4131: 2nd Scots Guards (I); 31st Field Arty Regt (4 In); Trucks: 10 medium.

C3926: 1st Coy French Motor Marines; Trucks: 2 medium.

C3922: 3rd Coldstream Gds/(I); 1st Kings Royal Rifle Corps (I; 7Spt/7); 4th RHA (7Spt/7); 7th Medium Arty Regt; Trucks: 15 Medium, 5 Heavy.

C3721: 1st Royal Northumberland Fusiliers (I); 3rd RHA (AT; 7Spt/7); Trucks: 5 medium, 5 light.

C3520: 1st RTR (7/7); Trucks: 5 medium.

C3320: 2nd Rifle Bde (I; 7Spt/7); Trucks: 10 medium.

C3020: 11th Hus (R; 7Spt/7); Trucks: 3 Medium.

D3714: Matruh Garrison (I; Att: 1 Essex 23/70; 1st Durham Lt Inf 23/70; 1st South Stafforshires; Trucks: 5 light, 15 medium, 5 heavy.

D3612: 7th Armored Division (Det: all 7 Arm Div. units listed above); Trucks: 5 light, 5 medium, 5 heavy.

D3615: 4th Indian Division (Det: 31st Field Arty and 7th In Bde); Trucks: 5 light, 20 medium, 2 heavy.

E1829: 16th Inf Bde (70); Trucks: 20 medium.

Alexandria (E3613 &/or 3714): HQ: 2nd New Zealand Division; 4th NZ Bde (2 NZ); 27th NZ MG Bn (2 NZ); 2nd NZ Cavalry (2 NZ); 4th NZ Field Arty Regt (2 NZ); Trucks: 10 light, 15 medium, 5 heavy.

Cairo and/or Helwan (E 1430): In Training: 6th Australian Division (Less: 19 Aus Bdg; 2/1Afld Arty; 2/2Afld Arty).

Cairo: 8th Field Arty Regt; 1st Welch (I); 1st Cheshire MG Bn; 2nd King's Own Ryl Regt (I).

Anywhere on Map D or E: 8th Medium Arty Regt; 65th Anti-tank Regt; 149th Anti-Tank Regt; 9th Heavy AA; 15th Light AA; 57th Light AA.

Note: 1st Welsh counter bears the same coloration as Indian Units.

Broken Down Vehicles

Alexandria: 2 TOE of A9 Cruiser Tanks; 1 TOE of A10 Cruiser tanks.

[60.42] Commonwealth North African Air Force

The following planes, SGSU's and pilots may be placed at any Commonwealth N. African air facility, within the capacity of that facility.

Plane	Total	
	Available	Refitted
Lysanders	12	8
Bombays	15	12
Sunderlands	18	11
Blenheim Mk. I's	26	18
Blenheim Mk. IV's	12	12
Blenheim Mk. IVF's	9	5
Gladiators	36	30
Hurricane Mk. I's	6	4
Morane 406's*	4	2
Potez 63/11*	2	2
Valentia's	3	1

*May be in the same squadron (Free French).

Pilots Available: 1 Three, 5 Two's and 8 One's.

SGSU Available: 14.

[60.43] Commonwealth Second-Third Line Trucks

The Commonwealth Player receives the following Truck Points; He may assign them and place them as he wishes, within the restrictions listed. (These will change).

Location	Truck Points Available
Any hex in Cairo	40 Medium 10 Heavy
Alexandria	10 Light 20 Medium
Anywhere, maps	15 Light 40 Medium 5 Heavy
Any Air Facility	5 Light 30 Medium 20 Heavy

[60.44] INITIAL SUPPLY STATUS

Location	(Factors)		
	Ammo	Fuel	Stores
Mersa Matruh (D3714)	1000	3000	4000
Sidi Baranni (C4131)	250	500	100
Dump I	500	750	500

The Dump I dump and one dummy dump may be placed anywhere in Egypt on map C or D. Remember that unlimited supply points are available in Cairo and Alexandria.

Air Supply (Distribute amongst Air Facilities as desired)

Ammo: 200

Fuel: 250

Stores: 50

Unlimited Supplies in Cairo/Alexandria (as per rules)

The Commonwealth Player has his Major Repair Facility in Alexandria. He has one Temporary Facility in Mersa Matruh.

[60.45] Commonwealth Fleet: The Commonwealth Player has a total of one battleship, three cruisers and seven destroyers available at the beginning of the campaign. These are deployed according to the Commonwealth Fleet Schedule (see Section 30.0). The warships may not be moved until the first OpStage of the second Game-Turn. Their combat ratings are unaffected (i.e., they do not need repair and may fire).

The following units are available and in Port at Alexandria:

Number	Type
2	Battleships
2	Cruisers
4	A. A. Cruisers
6	Destroyers

The following units are in Valetta, in Malta:

Number	Type
1	Cruiser
1	A. A. Cruiser
2	Destroyer

No Fleet units may be moved until the September 1/IV Stage/Turn. All ships may use their AA Ratings.

[60.46] Malta

Planes:

15 Hurricane MkI's	(2 SGSU's)	(12 Ready)
12 Swordfish	(1 SGSU)	(9 ready)
3 Gladiators	(1 SGSU)	(all ready)
1 Sea Gladiator	(1 SGSU)	(all ready)

Pilots:

4 Two's, 6 One's

AA: 17 AA Points distributed as player sees fit

State of Air facilities: The Malta bases have an initial capacity of five SGSU's. These may be spread amongst the Malta facilities as the player wishes. Construction on increasing the capacity of Malta Air Facilities may begin in October, 1940.

[60.47] The Commonwealth Player uses his Reinforcement Track as printed, bringing in units where and when listed (Remember that some units need Training, which means that other units have to be assigned as Training Units!) No Replacement Points may arrive before November 1940.

[60.5] AIR FACILITIES

The following Air Facilities are in existence. Players will note that Air facilities may be used by anyone Who controls them. All facilities in Egypt

belong to the Commonwealth; all those in Libya belong to the Italians. at the start of the game.

Airfields	
City/Village	Hex
Abbassia	E2132
Aboukir	E3815
Abu Seier	Off-Map E(3433)
Alexandria	Both hexes, (E3613, 3714)
Almaza	E2133
Amiriya	E2212
Barce	B5504
Benina	A4829
Dekheila	E3512
Deversoir	Off-Map E(1833)
El Adem	C4507
El Berca	A4728
Fayid	Off-Map
Ft. Maddalena	C3019
Ismailia	Off-Map
Kabrit	Off-Map E(1833)
Martuba	B5526
Mersa Matruh	D3714
Soluch	A4130
—	B5825

The four Tripoli/Tunisia boxes (off-map).

Air Landing Strips

Augila	B0707
Bardia	C4321
Bir el Gubi	C4108
Burg el Arab	E3109
Buq Buq	C3926
El Agheila	A1816
El Hamman	E3007
Ft. Capuzzo	C4020
Fuka	D3323
Gambut	C4414
Gazala	B4933
Giarabub	C1014
Maraua	B5410
Matten Baggush	D3520
Mechili	B4921
Menastir	C4419
Qotifiya	D3227
Sidi Azeiz	C4218
Sidi Barrani	C4131
Sidi Haneish	D3418
Sidi Omar	C3618
Sollum	C4021
Siwa	C0127
Tmimi	B5229
Tobruk	C4807
Ztert	B5917
—	C4119
—	D3231
—	D3416
—	D3516
—	D3903

Note: Insufficient Air Landing Strip counters are provided. Use blanks.

Flying Boat Basins

Alexandria	E3614
Bomba	B5331

Port Said (off-map E4033).

Flying Boat Alighting Areas

Derna	B5925
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[60.6] INITIATIVE

For this Scenario Group, the Italian Player has the Initiative for the entire first Game-Turn. Starting with the second Game-Turn, players determine Initiative as per the rules for such.

[60.7] CONSTRUCTION ON THE MAP AT THE START OF THE GAME

There are no minefields, fortifications, or pipeline (other than those provided by the RR). The RR runs to Mersa Matruh (D3714) and ends there. All ports are at listed Efficiency with the exception of

Tobruk, which is at Efficiency Level 7 (the San Giorgio is partially sunk, blocking the harbor).

[60.8] FIRST SCENARIO GROUP VICTORY CONDITIONS**[60.81] Grazianni's Offensive**

Italian Strategic Victory: The Italian Player must take and hold either Alexandria or Cairo (any hex) with a combat unit that can be supplied (by convoy) from any hex on Game-Map D.

Italian Decisive Victory: The Italian must take and hold (occupy) both Mersa Matruh and Siwa with any combat unit that can be supplied by convoy from Tobruk.

Italian Tactical Victory: The Italian must take and hold (occupy) Sidi Birani and retain possession of Sollum (C4021) and Fort Maddalena and Giarabub in supply (as above, to Tobruk)

Commonwealth Tactical Victory: Must take and hold Sollum and Halfaya Pass (C3922) as well as Siwa (C0127) tracing supply by truck convoy back to Alexandria or Cairo.

Commonwealth Decisive Victory: Must take and hold Bardia, Fort Maddalena, Sidi Omar, Sollum and Siwa, supply as above.

Commonwealth Strategic Victory: Must take and hold Tobruk; supply as above.

[60.82] The Italian Campaign

Players are awarded points for holding (occupying) the following locations. Units holding such places must be combat units (i.e., units with non-parenthesized Close Assault Ratings) and they must be able to trace supply of not more than one Game-Turn's duration back to a Supply Dump or City which can eventually be traced back either to Cairo (Commonwealth) or Tripoli (Italian).

City/Village	Victory Points	
	Italians Hold	Commonwealth Holds
Mersa Matruh	100	0
Tobruk	50	25
Derna	20	30
Giarabub	10	10
Siwa (C0127)	10	10
Benghazi	10	50
Agadabia		
(A2629)	10	50
Mersa Brega		
(A2021)	0	50

If there are no Italian combat units that can trace a Supply Line on the game-map the Commonwealth Player wins a Strategic Victory. All other conditions produce a simple "Victory". Most points wins. These conditions are based on what actually happened, and less on actual objectives.

[60.9] AIR & LOGISTICS GAMES ABSTRACTIONS

This Case lists the adjustments needed in addition to those given in Case 59.6 in order to play this scenario without the Air and/or Logistics Game(s).

[60.91] Air Abstracted

(Land & Logistics Games): None.

[60.92] Air & Logistics Abstracted

(Land Game Only):

A. The Players' ability to bring in Motorization Points is equal to their respective abilities to bring in Medium Truck Points (i.e., each Medium Truck Point equal one Motorization Point).

B. Axis Initial Supply Status: Case 60.34 is replaced by the following:

Location	Regular Supply Units		Each with Ammo Fuel
	3	15	
Tobruk (C 4807)	3	15	50
Bardia (C 4321)	2	10	40
Benghazi (B 4827)	1	10	10

Derna (B 5925)	1	-	10
Tripoli (box)	8	5	60
Dump 1*	3	40	50
Dump 2*	3	40	50
C 0716	1	20	6

*In any one hex on map-section "C" in Libya, not within four hexes of a Commonwealth unit.

In addition, the Axis Player receives two groups of three dummy supply unit counters which may be placed according to the rules for Dumps 1 and 2.

C. Commonwealth Initial Supply Status: Case 60.44 is replaced by the following:

Location	Regular Supply Units	Each with Ammo	Fuel
Mersa Matruh (D 3714)	5	40	60
Sidi Baranni (C 4131)	1	40	60
Dump 1*	3	40	30

*All three in any one hex on map-section "D" west of Mersa Matruh.

In addition, the Commonwealth Player receives one group of three dummy supply units which may be placed according to the rule for Dump 1.

In addition, Alexandria and Cairo each possess three regular supply units, each supply unit containing 40 Ammo & 60 Fuel.

[60.93] **Logistics Abstracted** (Land & Air Games): A. The Players' ability to bring in motorization points is equal to their respective abilities to bring in Medium plus Heavy Trucks (i.e., each Medium and each Heavy Truck is equal to one motorization point).

B. Axis Initial Supplies: As in 60.82 "B" above and in addition a total of 5 regular supply units, each with 40 Ammo and 15 Fuel Points, to be distributed among and placed on any Axis air-fields(s).

C. Commonwealth Initial Supply Status: As in 60.82 "C" above and in addition a total of 2 regular supply units each with 20 Ammo and 12 Fuel Points to be distributed among and placed on any Commonwealth air facilities.

[61.0] SCENARIO GROUP TWO: THE DESERT FOX

[61.1] GENERAL DESCRIPTION

The Desert Fox scenario covers the arrival of the Germany Africa Korps and General Rommel. There are two scenarios that may be played from this set-up. The first is the initial arrival of Rommel and the race for Tobruk. The second is a complete campaign, starting from the same point.

[61.2] SCENARIO LENGTH AND DURATION

The race for Tobruk scenario begins with the 3rd OpStage of Game-Turn 26 and ends at the conclusion of the 3rd OpStage of Game-Turn 38. It should take from 75 to 125 hours to play.

[61.3] COMMONWEALTH FORCES

[61.31] Commonwealth Land Forces Initial Deployment (Hex: Units)

A2022: 3rd Armored Bde (2 Arm); (Assg: 6th RTR [4/7]; 3rd Hus); HQ: 2nd Armored Division; Trucks: 20 Medium.

A2021: 2nd Support Group (2 Arm); Trucks: 25 Medium, 5 Heavy.

C4507: 3rd In Motor Bde; Trucks: 5 Light, 25 Medium.

Msus (B4004): 1st Coy French Motor Marines Trucks: 3 Medium.

Any hex within the triangle formed by Tocra (B5501), Er Regima (A4830) and Benghazi: 24 Aus Bde (9 Aus; Less: 2/32 Bn); 20th Aus Bde (9 Aus Bde); 2/3 Aus AT Regt (9 Aus); 51st Field Arty Regt; Trucks: 5 Light, 35 Medium.

Jalo (B0513): One unit of the LRDG

Tobruk (C4807): HQ: 9 Aus Division; 26th Aus Bde (9 Aus); 1st RHA; 8th Field Arty Regt; 104th RHA; 65th AT Regt; 2nd Heavy AA Regt; 9th Heavy AA Regt; 15th Light AA Regt; 57th Light AA Regt Trucks: 5 light, 5 Medium.

Anywhere within one hex of:

Alexandria: 7th Armored Division (see 61.38); HQ: 70th Infantry Division; 22nd Gds Bde (I); ATT: 1st Buffs; 2nd Scots Gds; 3rd Coldstream Gds; 107th RHA; 149th AT Regt; 13th Light AA Regt; "Layforce"(I).

Mersa Matruh (D3714): 1st Royal Northumberland Fusiliers MG Bn; 7th Medium Arty Regt.

Any hex in Cairo: Polish Bde; 11 Czech Bn; 1st Cheshire MG Bn; 1st South Staffordshires (I); 2nd Kings Own Royal Regt. (I); 8th Medium Field Arty; 64th Medium Field Arty; 106th AT Regt; 74th Heavy AA Regt, 52nd Light AA Regt; 1st Essex (23/70)

Note: Several units in Egypt at this time have not been listed as they did not see action and were eventually shipped to Crete or Syria; e.g. 7th RTR, 1st Hampshires, 1st Welch; 1st Durham Light, infantry and the entire 16th Bde of the 70th Infantry division, among others.

[61.32] Tank

The following tanks (in TOE Points) are assigned to the given units (which start this scenario assigned to the 3rd Armored Bde):

Unit	Tanks
3rd Hussars	3 Points MkVI Light
5 RTR/3d Arm X	1 TOE Points A/9's; 4 TOE Points A/13's
6RTR/3rd Amr X	5 TOE Points of M13/40s (Captured)

Tanks of the 7th Armored Division are at maximum TOE, but all are broken down (see Case 61.38).

[61.33] Commonwealth North African Air Force

Planes & SGSU's: The following squadrons of planes, each being represented by one SGSU, are available with all planes refitted:

Location	Type	Plane Available
Benina (A 4829)	Hurricane I	15
Agadabia (A 2629)	Lysanders	3
Maraua (B 5410)	Blenheim IV	15
Barce (B 5504)	Lysanders	9
Tobruk (C 4907)	Hurricane I	16

Pilots: 1 TWO pilot and 4 ONE Pilots.

Air Facilities:

Flying Boat Basins @ Alexandria (E 3614) and Port Said (off-map, E 4033).

Flying Boat Alighting Areas = none.

Airfields @ Benina (A 4829), Barce (B 5504), Derna (B 5925), Martuba (B 5526), El Adem (C 4507), Mersa Matruh (D 3714), Amiriya (E 3212), Dekheila (E 3512), Alexandria (E 3613 & 3714), Abbassia (E 2132), Aboukir (E 3815), Almaza (E 2133) plus the four off-map fields to the east (i.e., all except Ethiopia in the south).

Landing Strips @ Agadabia (A 2629), Antelat (A 3233), Msus (B 4004), El Abiar (B 4901), Maraua

(B 5410), Gazala (B 4933), Tobruk (C 4807), Gambut (C 4414), Sidi Barrani (C 4131), Gerawla (D 3516), Sidi Haneish (D 3418), Maaten Baggush (D 3520), and Fuka (D 3323). In addition the Commonwealth Player may place three Landing Strips anywhere on the game map.

[61.34] **Malta:** The Commonwealth Player has an initial total air facility capacity of 8 which he may divide amongst the facilities. He has 26AA Points which he may divide amongst the facilities:

Pilots: 2 TWO's and 4 One pilots.

Planes (all are refitted): 24 Hurricane I, 12 Wellington I, 10 Swordfish, 5 Sunderlands, 4 Marylands.

He may not begin construction/repair of air facilities until the second game-turn of this scenario.



[61.35] COMMONWEALTH TRUCKS

The Commonwealth Player receives the following Second-Third Line Trucks for distribution to any location on game-maps A through D. (Not E). This includes trucks for Dumps and Air Facilities.

Type	Number
Light	45
Medium	100
Heavy	25

The Commonwealth Player places the following Trucks in Alexandria or Cairo:

Type	Number
Light	25
Medium	175
Heavy	35

[61.36] Supply

The Commonwealth Player receives *nine* Dumps and *three* Dummy Dumps. In addition, both Benghazi and Tobruk are supply dumps.

The Dumps may be placed at any of the following locations: Msus (B 4004), Tecnis (B 5407), Martuba (B 5526), Mechili (B 4921), Timimi (B 5229), El Magrun (A 3729), Barce (B 5504), Mersa Brega (A 2021), Derna (B 5925), Agadabia (A 2629), Tocra (B 5501) and Scedidima (A3933). All Dumps must be placed. (Historically, the first nine were actual dumps, with Msus and Tecnis being the major dumping areas.)

Benghazi holds 100 Fuel Points, 50 Ammunition Points, and 150 Stores. Benghazi also has 10 Points of Italian Prisoners, a POW Camp, and one Commonwealth Guard.

Tobruk has 500 points of Fuel, 1500 Ammunition Points, and 1000 Stores.

The Commonwealth Player has for distribution to his Dumps the following: 2550 Fuel Points, 1700 Ammunition Points and 1600 Stores. These may be distributed as the Commonwealth Player sees fit, with one limitation: no more than 25% of the total of each item may be placed in a single dump. Furthermore, all actual dumps must have at least 50 points of each item. All First Line Trucks *not* carrying men/equipment may be loaded with Fuel/Ammo/Stores and Water as the Players see fit. This is *in addition* to what is supplied above. Unlimited supplies in Cairo/Alexandria as per the rules.

The Commonwealth Player has a Temporary Repair Facility in Tobruk and one in Mersa Matruh. Alexandria (either hex) and Cairo have Major Repair Facilities.

Air Supply

The Commonwealth Player has 180 Fuel Points, 250 Ammo Points, and 50 Stores for distribution amongst Air Facilities.

[61.37] COMMONWEALTH FLEET

The Commonwealth Player receives all his fleet counters in Alexandria. They may not move out of Alexandria before the first OpStage of Game-Turn 27. All are armed and undamaged.

[61.38] REINFORCEMENTS/ REPLACEMENTS

Except as listed below, the Commonwealth Player takes his Reinforcements as they arrive, via The Reinforcement Track. Replacement Points arrive in the normal manner; the Commonwealth Player may not *plan* any such Replacements and must start any such planning in the normal course of events. There are no replacements allowed for March 1941.

There are several Commonwealth units being refitted, returned from assignments elsewhere, etc. These units had been either on the way to Greece/Syria, or in the shops. Plans had to be changed hastily, yet the scenario must adjust for the time in which that can be done. In essence, no units in Cairo or Alexandria may move during the third OpStage of Game-Turn 26. After that, only one battalion may be released per OpStage. (Thus it would take several stages to release the entire Polish Brigade; you would have to wait the required time to release all the battalions.) Remember, also, that battalions will be needed for Training purposes! In addition to the above, the 18th Australian Brigade appears in Alexandria on the first OpStage of April, 1941. It may move (by rail, sea or land) on the succeeding OpStages of April, 1941 (in addition to any of the above units).

The 7th Armored Division presents a special case. It had been chewed up pretty badly in the previous six months' action; its tank battalions were a mess and its infantry and support units worn to the bone. The unit was in Alexandria for a complete refit. At the start of the scenario, all 7th Armored tanks are considered Brokendown. The 7th's Tank battalions may not be released until they are at least 75% repaired, and then they may be released only one per OpStage. In addition, one more non-tank battalion may be released and sent westward per OpStage, all starting with the first OpStage in April. HQ units may be released at any time the player wishes. At the time of this scenario the 7th Armoured contained the following units: 4th Armrd Brigade (6 RTR reassigned, ARTR assigned); 7th Armrd Brig; 7th Support Grp.

For the purposes of the above Release Schedule, the 22nd Guards Brigade counts as one battalion.

[61.4] AXIS FORCES

[61.41] Axis Land Forces

All German land units that are scheduled to arrive *before* the third OpStage of Game-Turn 26 (See Reinforcements Schedule) may be placed within one hex of El Agheila (A1816) but not east of that village. The Axis Player may place them in any formations he wishes (using Battle Groups, if he so desires). The Italian Player has six Divisions on the game-map: the Ariete Division may be placed within two hexes of El Agheila, but not east of that village. The Pavia, Bologna, Brescia and Savonna divisions may be placed anywhere between Ras el Ali (A 2010) and Nofilia (A2703) with no division less than two hexes from another. The Trento Division is in Tripoli. Also in Tripoli are the XXI Corps Artillery, the 350th GaF Artillery Regiment, the 2nd Artillery Raggruppamento, and the Sabratha Division. The Sabratha Division has lost its 6 infantry battalions (although the 85th and

86th Regimental Headquarters still exist). Its MG Bn, and Arty Regiment have been reduced to 1 TOE Strength Point each. NOTE that the six infantry battalions and two "support" units may be rebuilt by Replacements.

All units (except the Sabratha) are as listed on the OA Sheets (including Tanks).

One Oasis Company (German) is at Maaten Groter (A 1318) and one at Magadah (A 0817).

[61.42] Axis Airforce

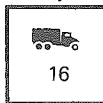
Planes: The Axis Player receives all of the German planes listed in the Axis Airplane Reinforcement Chart as scheduled to arrive prior to the 3rd OpStage of Game-Turn 26. All of these planes are considered refitted. The Axis Player receives the following Italian planes:

Type	Total Available	Refitted
C.R. 42's	61	27
G. 50's	62	33
B.R. 20's	1	0
SM 79	30	14
Z 1007bis	1	1

Pilots: Italian = 3 THREE, 11 TWO's and 12 ONE pilots. German = The Axis Player rolls once to determine the number of ONE thru FOUR rated pilots he receives (i.e., the March, 1941 die roll).

SGSU's: 13 Italian. German SGSU's are available as per the Air Game rules (because of Air Facilities, below).

Air Facilities: The Axis Player may base any portion of his airforce at Italy/Sicily within the minimum German plane restrictions of Case 43.1 (Crete is still British). The Axis Player may place one airfield and one air landing strip in any hex west of El Agheila (A 1816).



[61.43] Axis Trucks

All German first line Trucks (which are indicated on the Reinforcement Schedule as being attached to arrived German units) may be distributed among those units freely.

The Italian Player receives the following First-Line Trucks, which may be assigned as he sees fit to his units:

Type	Number
Light	45
Medium	220
Heavy	50

The following Second/Third Line Trucks are available (for any and all purposes):

Type	Number
Light	95
Medium	280
Heavy	50

It is firmly suggested that the Axis Player use these as Second/Third Line Trucks, otherwise he will have a hard time moving his Supply around.

Additional second/third line trucks: 10 Medium Trucks at air facilities.

[61.44] Supply

The Axis Player receives five Dumps, two of which are Dummies. These dumps may be placed at any hex between El Agheila and Nofilia.

For division between the three active dumps the Axis Player has 9600 Fuel Points, 2500 Ammo Points, 950 Stores, and 1100 Water Points. No more than 50% of each may be located at any one dump.

In addition, Tripoli holds 3000 Fuel Points, 1500 Ammunition Points, and 500 Stores.

For distribution to Air Facilities (remember, Tripoli/Tunisia air facilities have unlimited Fuel and Ammo) there are 50 Fuel Points and 50 Ammo Points. The Axis player does not receive a Convoy during the first game-turn. He must plan his convoys for the second and third game-turns.

[61.45] All Axis Reinforcements are taken normally, as they appear on the track.

[61.5] INITIATIVE

The Axis Player has the Initiative through the end of Game-turn 27. Players roll for Initiative starting with Game-Turn 28.

[61.6] CONSTRUCTION ON THE MAP AT THE START OF SCENARIO

The railroad runs to Mersa Matruh (D3714) and ends there. There are no minefields or pipeline (other than provided by the RR). All ports are of full (listed) Efficiency Level except Benghazi (at zero) and Tobruk (at seven—and San Giorgio is still there). The Commonwealth Player may place a total of four Level 1 Fortifications within two hexes of Tobruk. The *unfinished roads* connecting Sollum (C4021) and Sidi Barrini (C4131) and the entire network around Tobruk have been built.

[61.7] AIR & LOGISTICS GAMES ABSTRACTIONS

This Case covers the adjustments needed in addition to those given in Case 59.6 in order to play this scenario without the Air and/or Logistics Game(s).

[61.71] *Air Abstracted (Land & Logistics Games):* The 2nd-3rd Line Trucks available to the Players are modified as follows:

A. Axis Player: 2nd-3rd Line are now a total of 90 Light; 270 Medium and 40 Heavy.

B. Commonwealth Player: 2nd/3rd Line Trucks in Alexandria/Cairo are now a total of 25 Light, 150 Medium and 25 Heavy. Other 2nd/3rd Line Trucks are now a total of 40 Light, 90 Medium and 20 Heavy.

[61.72] *Air & Logistics Abstracted (Land Game Only):*

A. The Players' ability to bring in motorization points is equal to their respective abilities to bring in Medium Truck Points (i.e., each Medium Truck Point equal one Motorization Point).

B. Initial unattached motorization points: The Axis Player has 290; the Commonwealth Player has 150 in Alexandria/Cairo and 90 at other locations.

C. Axis Initial Supply Status: Case 61.43 is replaced by the following: The Axis receives 10 regular supply units, each with 40 Ammo and 60 Fuel Points and 3 Dummy supply units which may be distributed freely (within stacking restrictions) in any of the road hexes between El Agheila (A 1816) and Nofilia (A 2703) inclusive. This does not include the road from E 1716 the Maaten Groter (A 1318). In addition, the Axis Player receives 8 regular supply units, each with 40 Ammo and 25 Fuel Points, in the Tripoli box.

D. Commonwealth Initial Supply Status: Case 61.35 is replaced by the following:

The Commonwealth Player receives one regular supply unit with 20 Ammo and 30 Fuel Points in Benghazi (A 4827) and 5 regular supply units, each with 40 Ammo and 15 Fuel Points, in Tobruk (C 4807). In addition, he receives 9 regular supply

units, each with 40 Ammo and 25 Fuel Points, and 4 Dummy supply units to be divided among the following locations with the restriction that no more than 2 regular supply units may be placed at any location: Mersa Brega (A 2021), Agadabia (A 2629), Sceledima (A 3933), El Magrun (A 3729), Msus (B 4004), Tecnis (B 5407), Barce (B 5504), Tocra (B 5501), Martuba (B 5526), Mechili (B 4921), Timimi (B 5229) and Derna (B 5925)

In addition, Cairo and Alexandria each contain two regular supply units: each supply unit contains 40 Ammo and 60 Fuel Points.

[61.73] Logistics Abstracted (Land & Air Games):
A. The Players' ability to bring in motorization points is equal to their respective abilities to bring in Medium plus Heavy Truck Points (i.e., each Medium and each Heavy Truck Point is equal to one Motorization Point).

B. Axis Initial Supply Status: As in 61.62 "C" above and in addition 1 regular supply unit with 20 Ammo and 30 Fuel Points may be placed at any one Axis air facility.

C. Commonwealth Initial Supply Status: As in 61.72 "D" above plus 1 regular supply unit with 40 Ammo and 60 Fuel Points which may be placed at any one Commonwealth air facility.

[61.8] VICTORY CONDITIONS

Victory Conditions for The Race for Tobruk Scenario are fairly simple. The Player who holds Tobruk at the end of the scenario wins. If the Axis Player also holds Bardia (C 4321) he wins a Decisive Victory. If he holds Benghazi-Tobruk-Bardia plus any two villages/cities in Egypt he wins a Strategic Victory. If he holds anything less than the above, it is a Commonwealth Victory. Furthermore, if the Commonwealth holds all the above (including Benghazi) it is a Smashing Victory.

[62.0] SCENARIO GROUP THREE: OPERATION CRUSADER

[62.1] GENERAL DESCRIPTION

Operation Crusader covers the events of the Winter of 1941-2, in which the Commonwealth forces attempted to relieve Tobruk (under siege by the Axis) while, at the same time, Rommel had planned to strike eastward at the Allied forces in Egypt. The result was an extended series of battles in the area between Sollum and Tobruk.

[62.2] SCENARIO LENGTH AND DURATION

The scenario starts with the 3rd OpStage of the 57th Game-Turn. The scenario ends with the conclusion of the 1st OpStage of the 65th Game-Turn. The estimated playing time for this scenario is between 200 and 250 hours.

[62.3] COMMONWEALTH FORCES

[62.31] Commonwealth Forces Initial Deployment (Hex: Units)

Within 1 hex of

C4807: 32 Army Tank Bde (2; Assg: 1st RTR [7/7]; 4th RTR and 7th RTR); Polish Bde; 70th Infantry Division (Less: 1 Argyll & Sutherland; Assg: 2nd Kings Own Rifles [in place of the 1 A&S], 1st Royal Northumberland Fusiliers MG Bn, 1st RHA, 104th RHA, 107th RHA; ATT: 144th Field Arty, 2/13 Aus Inf Bn [20/9 J]; 12/15 Aus In Bn [TOE strength of 4; 0/9]; 13th Light AA; 14 Light AA; 2nd Light AA and 89th Heavy AA. Trucks: 20 Light, 135 Medium, 53 Heavy.

C4111: 7th Armored Bde 7; Assg: 7th Hus [4/7], 6th RTR [4/7], 2nd RTR; Less its own 1 RTR and 8 Hus; Att: 4th RHA [7Spt/7] Trucks: 30 Medium, 15 Heavy.

C4131: Kings Dragoon Gds (R; 2Spt/2); 67th Medium Arty; 16th Light AA; Trucks: 6 Light, 10 Medium, 3 Heavy.

C 3814: 4th South African Armored Car Bn Trucks: 2 Light, 3 Medium, 1 Heavy. 4th Armored Brigade (7; Independent Bde: no longer considered assigned assigned to the 7th Armored Div; Less: its 6 RTR and 7 Hus; Assg: 8th Hus [7/7]; 3rd RTR [1/2]; 5th RTR [3/2]; Att: 102nd AT Regt; 2nd RHA [2Spt/2]; Trucks: 45 Medium, 15 Heavy.

C3823: 11 In Bde (4 In; Less: 4/7 Rajp; Assg: 2/5 Mahratta Light Inf [in its place]); Trucks: 5 Light, 20 Medium, 5 Heavy.

C3721: Central India Horse Regt (R; TOE Strength of 3; 4 In)

Trucks: 3 Light, 3 Medium. **C3714:** 1st South African Division (Det: the three Inf Bdes and 4th SA Field Arty); Trucks: 5 Light, 22 Medium, 6 Heavy.

C3614: 7th Support Group (7; Det: 11 Hus [R] and 4th RHA; Assg: 1st Light AA and 60th Field Arty) Trucks: 30 Medium, 15 Heavy.

C3621: 4th In Division (Det: All three Inf Bdes and the Recce Regt); 8th Field Arty; 65th AT Regt; 57th Light AA; Trucks: 5 Light, 20 Medium, 8 Heavy

C3532: 2nd South Africa Div (Det: 7 SA Armored Car, 2nd SA Field Arty and 2nd SA AT Regt); Trucks: 12 Light, 70 Medium, 14 Heavy.

C3521: 1st Army Tank Bde
Trucks: 30 Medium, 17 Heavy.

C3530: 73rd AT Regt; 7th Medium Arty; 64th Medium Arty; 68th Medium Arty; Trucks: 3 Light, 25 Medium, 5 Heavy.

C3419: 7th In Bde (4 In); Trucks: 5 Light, 22 Medium, 5 Heavy.

C3426: 5th In Bde (4 In; Less: 1 Royal Fusilier; Assg: 1 Buffs [in its place]); 2nd South Africa AT Regt (2 SA); Trucks: 5 Light, 20 Medium, 5 Heavy.

C3410: 11th Hus (7Spt/7); Trucks: 10 Medium, 3 Heavy.

C 3414: HQ: 7th Armored Division
Trucks: 2 Light, 2 Medium

C 3318: 2nd New Zealand Div
Trucks: 15 Light, 75 Medium, 21 Heavy.

C 3212: 22 Armored Bde (now Assg to the 7th Armored Div)
Trucks: 15 Medium, 10 Heavy.

C 3111: 1st South African Bde (ISA)
Trucks: 5 Light, 20 Medium, 3 Heavy.

C 3015: 22 Gds Bde (I; Assg: 1st Tower Hamlet Rifles Spt/2, 3rd Coldstream Gds; Att: 51st Field Arty)
Trucks: 20 Medium, 10 Heavy.

C 2712: 5th South Africa Bde (ISA)
Trucks: 5 Light, 20 Medium, 3 Heavy.

C 1014: 6th South African Armored Car Bn; 7th SA Recce Bn (2 SA); 29th In Bde (5 In; Less: 6/13 FFR; Assg: 1st Royal Fusiliers 5/4 In; in its place#; Att: 2 South African Field Arty Regt 2 SA)
Trucks: 7 Light, 25 Medium, 6 Heavy.

D 3714: 2nd South African Bde (I SA); 4th SA Field Arty (I SA)
Trucks: 5 Light, 22 Medium, 6 Heavy.

D2001: 1st South African Road Construction Bn.
Trucks: 3 Light.

[62.32] Tanks

The units listed below have the following TOE of tanks (Unit: Tanks).

1 RTR (32nd Army, Bde 12): 4 Mk VI Light

4 RTR (32nd Army, Bde 12): 10 Matildas

7 RTR (32nd Army, Bde 12): 4 Matildas

8 RTR (1st Army, Tank Bde): 10 Valentines

42nd RTR (1st Army, Tank Bde): 3 Matildas (the following OpStage, the first of game-turn 58, the unit receives 7 more TOE of Matildas if it does not move. If it does, it receives them the second OpStage of game-turn 58.)

8 Hussars (4/7): 10 Stuarts

3 RTR (4/7): 10 Stuarts

5 RTR (4/7): 10 Stuarts

7 Hussars (7/7): 4 Crusader I's, 3 A/13's, 4 A/10's

6 RTR (7/7): 10 Crusader I's

2 Hussars (22nd Armor Bde): 9 Crusader I's, 1 A/10

4 CLY (22nd Armor Bde): 9 Crusader I's, 1 A/10

In addition to the above tanks, the following tanks (in terms of TOE Points) are considered to be broken down in Repair Shops in Alexandria.

3× Crusader I; 15× Crusader II; 18× Stuart; 7× Matilda; 8× Valentine.

Repair on these tanks may not start until the scenario's second OpStage.

[62.33] Commonwealth North African Air Force

Planes: These may be distributed freely amongst the available air facilities:

Plane	Total	
	Available	Refitted
Hurricane I's	82	64
Hurricane IIA's	66	60
Hurricane IIB's	42	36
Blenheim IVF's	24	16
Tomahawks	88	64
Beaufighters	22	16
Blenheim IV F's	142	100
Wellington IC's	110	80
Bostons	20	16
Marylands	32	12

Pilots: 1 FOUR pilot, 5 THREE pilots, 15 TWO pilots and 20 ONE pilots.

Air Facilities: 1 Flying Boat Basin anywhere on map-section D or E, and Port Said (off-map).

No Flying Boat Alighting Areas.

Airfields at Maaten Baggush (D 3520), Fuka (D 3323), Amirya (E 3212), Kabrit (off-map) and Tobruk (C 4807).

Air Landing Strips at any two hexes in the Giarabub Oasis (C 1014) complex, Gerawla (D 3516), Sidi Haneish (D 3418), El Daba (D 3329), Qotafiya (D 3227), Abu Seier and Fayid (off-map, i.e., both have been reduced to the ability to perform maintenance on a single squadron).

SGSU's: As per Air Game rules.

[62.34] Commonwealth Trucks

The Commonwealth Player receives the following Second/Third Line (Convoy) Trucks for distribution to any location on game map-sections C thru E. However, no more than 10 Light, and/or 20 Medium and/or 10 Heavy Trucks may be placed within five hexes of Tobruk.

Type	Number
Light	150
Medium	400
Heavy	150

The Commonwealth Player receives the following Trucks for distribution to any Air Facility (except that in Tobruk): 80 Medium and 20 Heavy.

[62.35] Commonwealth Supply

The Commonwealth Player possesses dumps at the following locations containing the listed amounts of Supplies:

Location	Ammo	Fuel	Stores	Water
Tobruk	800	1,500	550	—
Sidi Barrani	500	1,000	300	700
Giarabub Oasis	200	400	300	—
Mersa Matruh	1,000	2,000	400	800
Bir Khamsa (C 3029)	200	600	50	300

In addition, the Allied Player receives dumps at Oxford Circus (C 3532), Bir al Sheferzen (C 3419), Sidi Suleiman (C 3721) and any two locations on game map section D and any one location on game map section C. Only four of these six dumps are active. A minimum of 20% the total of each type of the following total available Supplies must be placed in each active dump: 400 Ammo, 400 Fuel, 200 Stores, and 700 Water Points.

In addition, the Commonwealth Player receives 150 Ammo, 350 Fuel, 25 Stores, 50 Water Points at each Airfield and 10 Ammo 150 Fuel, 10 Stores and 20 Water Points at each Air Landing Strip, Flying Boat Basin, or Flying Boat Alighting Area. (This does not include off-map facilities.)

[62.36] Malta

The following Planes are Available on Malta

Type	Total Avail	Total Refitted
Swordfish	3	3
Hurricane IIB's	28	24
Beaufighter IC's	8	6
Hurricane I	10	9
Hurricane IIA	12	11
Wellington IC	13	13

The Malta air facilities have an initial capacity of 14 Squadrons. There are 24 AA Points to place where desired. Malta receives 1 Three Pilot, 4 Two Pilots, 5 One Pilots.

[62.37] Commonwealth Fleet

The Commonwealth Player may place all his warship counters in Alexandria. In addition, note that the Italian 10th Light Flotilla is active this scenario (See 30.4). However, the 10th Light may not raid Alexandria until December.

[62.38] Reinforcements/Replacements

These are taken as normal, from the tracks and production.

[62.4] AXIS FORCES**[62.41] Axis Forces Initial Deployment**

(Hex: Units)

C 4014: 33rd Aufklarungs Bn (R, 15) Trucks: 2 Light, 5 Medium.

C 4015: 3rd Aufklarungs Bn (R; 5 Le) Trucks: 2 Light, 5 Medium.

C 4108z: 32 Ariete Div Trucks: 20 Light, 42 Medium, 11 Heavy.

C 4216: 21st Panzer Div (consists of: 5th Panzer

Regt 5 Le; 104th Rifle Regt 15; which consists of: 2nd Bn of the 104th Inf Regt (15), 8th MG Inf Bn (5 Le) and the 708th Heavy Weapons Coy; 39th Panzerjager Bn AT; 5 Le; 155th Arty Regt; and the 200th Pioneer Bn 513)

Trucks: 12 Light, 45 Medium, 12 Heavy.

C4306: 26th Arty Regt (12 Pav); 57th Arty Regt (27 Bre)

Trucks: 4 Light, 4 Medium, 4 Heavy.

C 4406: RECAM (Italian Armored Car Regt) Trucks: 5 Light, 15 Medium.

C 4407: 17th Pavia Div (Det: 26th Arty Regt)

Trucks: 8 Light, 30 Medium, 4 Heavy.

f3C 4408: 155th Rifle Regt (90 Le)

Trucks: 5 Light, 18 Medium, 5 Heavy.

C 4411: 361 Afrika Regt (90 Le)

Trucks: 5 Light, 6 Medium, 3 Heavy.

C 4507: XXI Corps Arty; 1/2AR; 2/2AR; 3/2AR; 4/2AR; 1st and 2nd Articerlere Regts (all are Arty)

Trucks: 5 Light, 20 Medium, 6 Heavy.

C4606 and/or C4706: 102nd Trento Div(I) Trucks: 25 Light, 55 Medium, 15 Heavy.

C4608: 25 Bologna Div (I) Trucks: 8 Lights, 25 Medium, 7 Heavy.

C 4610: 90th Light Div (Det: 361 Afrika and 155 Regts, 3rd Bn of the 255 Regt and 3rd Bn of 347th Regt; Less: 190th Arty, 580th Recce, and 190th AT. Ergo, the counter consists of the HQ and the 613th Flak Bn); 200th Rifle Regt (15; consists of the 2nd MG Bn 5 Le and the 15/200 Kradschutzen Bn 15. Trucks: 5 Light, 21 Medium, 1 Heavy.

C 4613: 15th Panzer Div (Det: 33rd Recce; Less: 200th Rifle Regt, 104th Rifle Regt and 140th Arty Regt; 33rd Arty Regt has a TOE Strength of 6).

Trucks: 15 Light, 50 Medium, 16 Heavy.

C 4804: 27th Brescia Div (Italian Inf; Det: 57th Arty). Trucks: 8 Light, 30 Medium, 4 Heavy.

Divided approximately equally amongst B 4432, C 4303 and C 4102: 101st Trieste Div (I); Trucks: 20 Light, 50 Medium, 15 Heavy.

Anywhere within the triangle formed by C 3916—Sidi Omar (C 3618)—C 3922: 55th Savona Div (Det: II/15 and III/16 Inf Bns); I/104 Rifle Bn (15 Panzer); Trucks: 10 Light, 25 Medium, 5 Heavy.

Within one hex of C 4321: 2, 5, 6, 12, and 13th Coys of the Oasis Bn; III/255 Rifle Bn (90 Le); III/347 Rifle Bn (90 Le); IV Genova MG Bn (Italian); 350 GaF Arty Regt; II/15 and III/16 Inf Bns (55 Sav); Trucks: 5 Light, 7 Medium, 10 Heavy.

Attached to any Italian Inf Div(s): III(M) and LII(M) tank Bns; Trucks (each tank battalion): 2 Light, 6 Medium.

[62.42] Axis Tanks

The Axis Player assigns German tank TOE Strength Points as follows:

Unit	Tanks
I/8	4 PzII, 2PzIII(E) 5 PzIII(H), 2 PzIV(D),
II/8	4 PzII, 3PzIII(E) 5PzIII(H), 4 PzIV(E)
I/5	3 PzII, 7 PzIII(H), 1 PzIV(D)
II/5	3 PzII, 6 PzIII(H), 2 PzIV(E)

The Axis Player assigns Italian tank TOE Strength Points as follows:

Unit	Tanks
VII(M)	6 M13/40's
VIII(M)	6 M13/40's
IX(M)	6 M13/40's
LII(M)	6 M13/40's
III(M)	5 M13/40's
I(L)	5 CV33's
II(L)	5 CV33's

(Note: Over 100 additional Light Tanks—usually CV 33's and their ilk—were spread around other Italian units, assigned to Recce units, Bersagliere regiments, HQ's, etc. This information is "factored" into the strengths/TOE of those units.)

[62.43] Axis Air Strength

The Axis Player receives the following planes. The Axis Player receives 23 SGSU's.

Aircraft	Available	Refitted
MC 200	35	20
Cr 42	23	8
G50	28	20
BR 20	29	15
Ju87D	9	5
SM79	21	10

The Axis Player receives the following Italian Pilots:

Rating	Number
Two	5
One	11

The Axis Player receives 33 German SGSU's and the following planes:

Aircraft	Available	Refitted
Ju88D	83	45
He111	27	15
Ju87D	94	90
Bf109E	19	13
Bf109F	40	27
BF110C	23	13

The German Player receives the following pilots:

Rating	Number
Six	1
Four	2
Three	3
Two	6
One	12

The Axis Player has Airfields at Benina (A4829), Derna (B5925), Timimi (B5229), Gazala (B4933), Gambut (C4414), Martuba (B5526), Benghazi (A4827), Barce (B5504) and Maraua (B5410). The Axis Player has 3 Air Landing Strips anywhere on maps A thru C. The Axis Player has one Flying Boat Basin anywhere on Map A or B. In addition, he may base planes in Crete/Italy/Sicily within the restrictions of Case 43.1

[62.44] Axis Trucks

The Axis Player receives the following Second/Third Line (In Convoy) Trucks:

	Light	Medium	Heavy
Anywhere on game maps A & B	45	120	20
Tripoli	20	35	10
Nofilia	20	25	10
Derna	15	20	10

The Axis Player receives 30 Light and 60 Medium trucks for distribution among his air facilities.

[62.45] Axis Supply

The Axis Player possesses dumps at the following locations containing the listed amounts of Supplies:

	Ammo	Fuel	Stores	Water
Tripoli	200	1,300	350	—
Nofilia	500	800	500	250
el Agheila	150	450	350	400
Benghazi	650	2,200	700	—
Mechili	500	1,700	150	600
Derna	300	2,500	100	300

In addition the Axis Player receives dumps at Gazala (B4933) Rotunda Segnali (B 4432) and any three locations on game map section B. Only three of these five dumps may be active. A minimum of 20% of the following total available Supplies of 700 Ammo, 1,350 Fuel, 150 Stores and 400 Water Points must be placed in each active dump.

In addition, the Axis Player receives 150 Ammo, 350 Fuel, 25 Stores and 50 Water Points at each Airfield (excepting Tripoli/Tunesia).

[62.46] All Axis Coastal Shipping units are available, and may be distributed among the parts of Maps A, B and C.

[62.47] Axis Player must plan Replacements for all November OpStages in advance of starting the game. After that, he plans normally.

[62.5] CONSTRUCTION AT THE START OF THE SCENARIO

There are no minefields. All ports are at full (listed) Efficiency. All hexes adjacent to Bardia contain Level 1 Fortifications. All hexes adjacent to Tobruk contain Level 2 Fortifications.

The railroad extends past Mersa Matruh and ends in hex C 3430. From that point, the pipeline extends in a straight line as far as Birkhamsa (C 3029). All *unfinished roads* are built with the exception of the Siwa Track, which is completed to only twenty hexes from Mersa Matruh (i.e., the hex the 1st South African Road Construction Battalion occupies, D2001).

There is a Temporary Repair Facility in Tobruk. The Commonwealth Player may also place an additional Temporary facility in any village/city he wishes. As usual, Alexandria and Cairo contain Major Repair facilities.

Temp
Repair
Facility

There is a Temporary Repair facility in Derna, plus one in any other village or city. There is a Major Repair facility in Benghazi.

[62.6] AIR & LOGISTICS GAMES ABSTRACTIONS

This case covers the adjustments needed in addition to those given in Case 59.6 in order to play this scenario without the Air and/or Logistics Game(s).

[62.61] Air Abstracted

(Land & Logistics Games): None.

[62.62] Air & Logistics Abstracted

(Land Game Only):

A. The Players' ability to bring in Motorization Points is equal to their respective ability's to bring in Medium Truck Points (i.e., each Medium Truck Point equals one Motorization Point).

B. Axis Initial Supply Status: Case 62.45 is replaced by the following:

Location	Regular Supply Units	Each with Ammo	Fuel
Tripoli (box)	2	20	60
Nofilia (A2703)	3	30	25
el Agheila (A1816)	1	30	50
Benghazi (A4827)	4	30	60
Mechili (B4921)	3	30	60
Derna (B5925)	4	15	60
Group 1*	3	30	50

*These, and in addition 2 dummy supply units, are to be placed in any of the following locations in any distribution at the Axis Player's choice: Gazala (B4933), Rotunda Segnali (B4432) and any location(s) on game map-section "B".

C. Commonwealth Initial Supply Status: Case 62.35 is replaced by the following:

Location	Regular Supply Units	Each with Ammo	Fuel
Tobruk (C4807)	4	40	40
Sidi Baranni (C4131)	3	30	50
Mersa Matruh (D3714)	5	40	30
Giarabub Oasis (C1014)	1	40	50
Bar Khamsa (C3029)	1	30	60
Group 1*	4	20	15

*These, and in addition 2 dummy supply units, may be distributed among the following locations at the Commonwealth Player's choice with the restriction that no more than one regular supply unit may be placed in any hex: Oxford Circus (C3532), Bir al Scheferzen (C3419), Sidi Suleiman (C3721) and any two hexes (other than Mersa Matruh) on game-map-section "D". In addition, Cairo and Alexandria each contain two regular py units, each with 40 Ammo and 60 Fuel Points.

[62.63] Logistics Abstracted (Land & Air Games):

A. The Players' ability to bring in Motorization Points is equal to their respective abilities to bring in Medium plus Heavy Trucks (i.e., each Medium and each Heavy Truck is equal to one Motorization Point).

B. Axis Initial Supplies: As in 62.62 "B" above plus 1 regular supply unit at each Axis airfield (except those off-map) with 30 Ammo and 40 Fuel Points.

C. Commonwealth Initial Supplies: As in 62.62 "C" above plus 1 regular supply unit at each airfield (except Kabrit) with 30 Ammo and 40 Fuel Points and 1 regular supply unit at each remaining air facility (except off-map) each with 5 Ammo and 15 Fuel Points.

[62.7] INITIATIVE

Historically, the Commonwealth Player should have the Initiative for the first OpStage. However, the Players may roll in the normal manner to determine who has initiative in that OpStage if they prefer. In each OpStage after the first, the Players roll the die as per normal procedure.

[62.8] VICTORY CONDITIONS

Players amass points for controlling geographical locations. The Player with the most points subtracts the other player's points from his and then determines the level of victory, if any.

The Players receive Victory Points for controlling certain cities/villages. The number of points depends on the location and the player controlling. Control of a city/village means occupying that city/village with a combat unit of any strength. That unit must have sufficient supplies to exist for one month and Ammo for one week (one round each OpStage).

City/Village	Points for Axis Control	Points for CW Control
Tobruk	6	4
Bardia	2	3
Sollum	2	2
Halfaya Pass (both hexes)	1	1
Ft. Maddelenna	1	2
Sidi Biranni	3	0
Giarabub	1	1
Siwa	2	1
Ft. Capuzzo	1	1

Point Differential Level of Victory

10+	Strategic
6-9	Decisive
1-5	Tactical
0	Draw/Standoff

Players may wish to compare losses in men and tanks (as well as air, although that is not that revealing) to see how well they did, in comparison to each other, in gaining or not gaining their objectives. Players may wish to subjectively raise or lower their victory level as per this. However, losses can be misleading, so this type of condition is "officially" left out of the conditions here.

[63.0] SCENARIO GROUP FOUR: EL ALAMEIN

[63.1] GENERAL DESCRIPTION

There are two scenarios which can be played from this scenario Group. The first is a short scenario in which the Axis Player has one week (one game-Turn) to try to break through the Commonwealth lines and reach Alexandria (a relatively impossible task). The second scenario is one of survival: can the Axis army survive the end of the campaign intact? For those familiar with El Alamein they will realize that both scenarios are, militarily (and in terms of game-playing) quite stultifying. They are included for historical reference, mostly. Playtesting proved them to be as equally dull as their gaming predecessors. The first scenario has the advantage of being remarkably short; the other scenario is for those who like to retreat or chase fleeing troops.

[63.2] SCENARIO LENGTH AND DURATION

Both scenarios begin with the 1st OpStage of Game-Turn 10Z. "The Last Chance" Scenario ends with the completion of the 3rd OpStage of Game Turn 102 1st OpStage of Game-Turn 111. "The Long Retreat" ends at the completion of the December 3/IV, 1942 Game-Turn. "The Last Chance" should take about ten hours to play, excluding set-up; "The Long Retreat" should take from 50 to 100 hours.

[63.3] COMMONWEALTH FORCES

[63.31] Commonwealth Land Forces Initial Deployment

(Hex: Units)

E 3101: 9th Aus Div (Det: 24 Aus Bde; Att: 40 RTR 23/8)

Trucks: 30 Light, 55 Medium, 15 Heavy.

E 3001: 24th Aus Bde (9 Aus)

Trucks: 5 Light, 25 Medium, 6 Heavy.

D 3033: 51st Inf Div (Att: 50 RTR 23/8).

Trucks: 40 Light, 70 Medium, 21 Heavy.

E 2901: nd New Zealand Div (2nd NZ Cav is now TOE Strength of 5 Recce Points and 6 Stuart tank Points; Less: 4 NZ Bde; Att: 9th Armored Bde 10; Att: 3 Hus, 14th Sherwood Forresters I)

Trucks: 30 Light, 90 Medium, 20 Heavy.

E 2801: 1st South African Div (Less: 50 SA Bde; Assg: 3 SA Bde #SA; in its place#; ATt: 8th RTR #1 Army#, 1st SA Field Arty #2 SA#, Die Mid-delandse MG Regt #SA#, and 2nd Botha #5/1SA) Trucks: 35 Light, 85 Medium, 25 Heavy.

E 2701: 5 In Bde (4 In; Less: 1 Royal Fusilier and 4/6 Rajp Rifles; Assg: 1/4 Essex* and 3/10 Baluchies in their places); 6th Rajput Rifles MG Bn; 1st Field Arty Regt (4 In); 149th AT Regt; 11th Field Arty Regt; 32nd Field Arty Regt; 57th Light AA Regt HQ: 4th In Div (Less: 11 In Bde, 31st Field Arty, and Central In Horse R; Det: 5th and 7th In Bdes and 1st Field Arty). Trucks: 10 Light, 25 Medium, 10 Heavy.

E 2601: 7th In Bde (4 In; Less: 4/11 Sikh; Assg: 1/2 Gurkha Rifles in its place) and 161st In Motor Bde (consists of: 1st Argyll & Sutherland 16/17, 1/1 Punjab Regt, 4/7 Rajputanis 11/4 In). Trucks: 5 Light, 25 Medium, 5 Heavy.

2501: 50th Inf Div (Less: 150 Bde, 2nd Cheshire MG Bn and 72 Field Arty; Assg: Greek Bde and 111th Field Arty Regt in place of the Inf Bde and Arty Regt respectively; Att: 154th Field Arty Regt, 102nd AT Regt 2Spt/2 and 34th Light AA Regt). Trucks: 35 Light, 65 Medium, 21 Heavy.

E 2302: 7th Armored Div (Consists of: 4th Armored Bde #consists of 4/8 Hus**, Royal Scots Greys (8/10) and 1KRR (7 Spt/7); 22nd Armored Bde consists of 1 RTR (7/7), 5 RTR 3/2, 4 CLY (22 Amr Bde), and 1 Rifle Bde (2/1); 11th Hus r; 7; TOE Strength of 8 3 RHA. Which now is considered a regular artillery unit, Possesses 6 TOE Strength Points of 25lbs; 4th Field Arty Regt; 97th Field Arty Regt 20/10 In; 65th AT Regt; 15th Light AA Regt; AH: 2nd Derby Yeomanry R, 8; Strength of 6; 44th Recce Regt 44. Trucks: 75 Medium, 40 Heavy.

E 3005: 1st Armored Div (Consists of: 2 Armored Bde Less: 1st RflBde, 7th Motor Bde 7; Which consists of: 2nd Rifle Bde (7Spt/7), 7th Rifle Bde (23/8) and 2nd KRR (1Spt/1), 2nd RHA #Spt/2, 4th RHA 7Spt/7, 11th RHA self-propelled counter, 78th Field Arty, 76 AT Regt Spt/1, 1st Royal Northumberland Fusiliers MG Bn, 42nd Light AA, and 12 Lancers R; 1Spt/1; Att: An ad hoc Bde—level formation known as "Hammerforce"; consisting of: 4/6 SA Armored Car Regt*** with a TOE Strength of 7, 146th Field Arty, 73rd AT Regt and 56th Light AA Regt). Trucks: 10 Light, 90 Medium, 30 Heavy.

E 3007: 8th Armored Div (Det: 40 RTR and 50 RTR of the 23rd Armored Bde, 2nd DerYoe R; Less: 24th Armored Bde, 7th Rifle Bde of the 23rd Armored Bde); 121st Field Arty. Trucks: 5 Medium.

E 2805: 10th Armored Div (consists of: 8 Amr Bde 10; Less: ScTsGray; Assg: 3rd RTR (1/2; in its place) and 1st Buffs (I), 24th Armored Bde [8], 133rd Inf. Bde [44], 1st Royal Dragoons R, 1 RHA, 104 RHA, 98th Field Arty, 5th RHA self-propelled counter, 84th AT, and 53rd Light AA). Trucks: 75 Medium, 40 Heavy.

E 2904: 1st Army Tank Bde (Det: 8 RTR) Trucks: 5 Medium.

E 2401: 44th Inf Div (Det: 44th Recce; Less: 133Inf Bde; Att: 53rd Field Arty). Trucks: 20 Light, 55 Medium, 6 Heavy.

Within one hex of E 2403: 1st Free French Bde Trucks: 10 Light, 35 Medium, 5 Heavy.

2nd Free French Bde. Trucks: 5 Light, 12 Medium, 1 Heavy.

Within two hexes of el Alamein

(E 3002): 14th Light AA; 16th Light AA; 27th Light AA; 88th Heavy AA; 94th Heavy AA. Trucks: 5 Medium.

21st In Bde (10 In; consists of: 1/6 Rajput Rifles 11/4 In, 3/7

Rajputana Regt 61 In Mtr Bde and 2/8 Gurkha Rifles); 7th Medium Arty Regt; 64th Medium Arty; 69th Medium Arty

Trucks: 10 Light, 25 Medium, 7 Heavy.

*This is a British unit whose counter bears the Indian units' coloration.

**Formation created by combining the 4th and 8th Hussars. Use the counter for the 8th Hus (7/7). The unit's max TOE Strength is still 10.

***Formation created by combining the 4th and 6th SA Armored Car Battalions. Use either counter. The unit's max TOE Strength is now 8.

Certain of the above units may have their equipment and TOE upgraded, using the rules sections for such as references.

- A. Each non-motorized infantry battalion (CPA of 10) may add 1 TOE of 6-pounder AT guns; and
 B. Each motorized battalion of infantry (10+ CPA) may add 2 TOE of 6-pdr AT guns.
 C. 40% (rounded up) of all AT units may replace their 2-pdr AT guns with 6-pdr AT guns.
 D. Armored car (not armored recce) units may be upgraded as per the rules for such. (It is not necessary to use Replacement points; simply upgrade them as if they had used Replacement Points)
 E. Any artillery units with 18-pounders or 18/25 pounders may replace them with 25 pounders.

[63.32] Commonwealth Tank Battalion Composition

The following tank battalions are listed as to their assigned parent formatin(s) for this scenario and their starting compositions, in TOE Strength Points, if they differ from their listing on the OA Sheet.

Tank Battalion	Parent Formation(s)	Composition
40 RTR	23/8	8 Valentines
46 RTR	23/8	10 Valentines
50 RTR	23/8	9 Valentines
8 RTR	1 Army*	10 Valentines
3rd Hussars	†	2 Grants, 2 Shermans, 3 Crusader I's
Ryl Wi Yeomanry	9/-‡	3 Grants, 2 Shermans, 3 Crusader I's
Warwick Yeo		3 Grants, 3 Shermans, 3 Crusader II's
9th Aus Div Cav	9 Aus‡	3 Crusader II's, 1 Stuart 6 ID Code "11"
Royal Scots Greys	4/7	4 Stuarts, 3 Grants
4/8 Hussars	4/7	9 Stuarts
1 RTR	22/7	4 Stuarts, 5 Grants
5 RTR	22/7	4 Crusader II's, 5 Grants
4 CLY	22/7	6 Crusader II's, 2 Grants
Queens Bays		7 Shermans, 3 Crusader II's
9th Lancers	2/1	6 Shermans, 4 Crusader II's
10th Hussars	2/1	5 Shermans, 5 Crusader II's
7th Motor Bde HQ		1 Churchill
3 RTR	8/10	3 Crusader III's, 2 Shermans, 3 Grants
Nottingham Yeo	8/10	3 Crusader III's, 2 Shermans, 4 Grants
Staffordshire Yeo	8/10	3 Crusader III's, 2 Shermans, 4 Grants
41 RTR	24/10	3 Crusader III's, 6 Shermans
45 RTR	24/10	3 Crusader III's, 6 Shermans
47 RTR	24/10	3 Crusader III's, 7 Shermans
42 RTR	1 Army	10 Scorpions
44 RTR	1 Army	10 Scorpions

*Starts attached to the 1st SA Div.

†Starts attached to the 2nd NZ Division.

‡This armored unit is a tank battalion for these scenarios.

[63.33] Commonwealth Air Strength

Planes (Type and #'s Available)		
Location	SGSU's (Type and #s Available)	
Burg el Arab (E 3109)	3	Spitfire VB (18); Hurricane IIA (12); Tomahawks (6); Marylands (4); Baltimores (4)
Wadi Natrun (E 2418)	1	Marylands (12)
Deversoir	*	Mitchell II (39)
Amiriya	28	Boston III (48); Baltimore II (24); Baltimore III (24); Tomahawk (16); Kittyhawk I (56); Kittyhawk II (91); Kittyhawk III (39); Hurricane IIB (48); Hurricane IIC (64); Hurricane IID (32); Hurricane IIE (16); Spitfire VB (48)
Idku (E 3817)	2	Beaufighter IC (16); Beaufighter VIF (16)
Abu Seier	*	Wellington IC (32)
Jebel Hamzi (E 2024)	2	Wellington IC (32)
Kabrit	*	Wellington IC (16); Wellington II (16)
Degheila (E 3512)	2	Albancres (24)
Ismailia	*	Hudsons (32); Bombays (22)

* = SGSU's not needed; off-map base.

All Commonwealth Planes are Refitted!

Pilots Available

Rating	Number Available
Four	1
Three	5
Two	22
One	31

Air Facilities

The Commonwealth Player places six Airfields within one hex of Amiriya (The Amiriya complex). He has an Airfield at Burg el Arab (E 3109), Alexandria, and Cairo. He has a Landing Strip at Wadi Natrun (E 2418). The Commonwealth Player may also place 1 additional Airfield and 3 Landing Strips, 1 Flying Boat Basin and 1 Flying Boat Alighting Area anywhere on Map E, but not within five hexes of an enemy unit, or in the Qatarra Depression. The Commonwealth Player's seven off-map facilities are all undamaged.

[63.34] Commonwealth Trucks

Second-Third Line Trucks (In Convoy):

Anywhere on game maps D & E: 200 Light, 725 Medium, 225 Heavy

The Commonwealth Player receives 100 Medium and 60 Heavy trucks for distribution among his Air Facilities.

[63.35] Commonwealth Supply

The Commonwealth Player receives 10 Supply Dumps plus 3 Dummies, for a total of 13 Dumps. As many as five of these dumps may be placed within one hex of the Front Line (i.e., any hex east of and adjacent to an Axis unit.) One dump each must be placed at el Alamein, E2805, and E2403. The remaining dumps may be placed anywhere on Map E, but not within one hex of the front.

Front Line Dumps receive (amongst them) 2500 Ammo points, 2000 Fuel Points, 1500 Stores, and 1500 Water Points. The other dumps split (in any

way the Player wishes) the following: 1000 Ammo Points, 3000 Fuel Points, 500 Stores, 2000 Water Points.

Remember, there is unlimited supply available in Alexandria and Cairo. All Trucks not carrying men may be loaded with supplies as the Player wishes. The Commonwealth has no temporary repair facility. Alexandria and Cairo are Major Repair Facilities.

[63.36] Commonwealth Fleet

The Commonwealth Player receives the following warship counters in Alexandria: 2 Heavy Cruisers, 3 Light Cruisers, 4 Destroyers.

[63.37] Malta

All Air Facilities are operating at full capacity. The following planes are available in Malta:

Type	Number	Refitted
Wellington V	5	5
Baltimore I	4	4
Baltimore II	3	3
Beaufighter IC	12	5
Beaufighter VIF	8	6
Spitfire VB	54	33
Spitfire VC	18	10
Hurricane IIC	8	8
Albacore	3	3
Swordfish	3	3

48 AA Points are Available for placement as desired. The Following Pilots are available:

Rating	Number
Three	1
Two	5
One	5

[63.38] Reinforcements and Replacements

All Reinforcements are taken as per the Reinforcement Schedules. All Replacements are taken normally, except that all available points are reduced to 10%, rounded up, of their listed maximum. However, the Infantry and Truck Production Tables' output are unaffected. In any case, no Replacement points may arrive before the first OpStage of Game-Turn 105, and these may be planned before the scenarios starts. Replacements for the 106th Game-Turn may also be planned at the start of the scenario. There are no Replacements or Reinforcements for the "Last Chance" scenario.

[63.4] AXIS FORCES

[63.41] Axis Land Forces Initial Deployment

(Hex: Units)

D3328: 90th Light Div (Det: 580 Aufklarungs Bn [R]); Trucks: 10 Light, 40 Medium, 7 Heavy

D3132: 15th Panzer Div (Less: 200th and 104th Regts; Det: 33 Recce; Assg: 2 MG Bn [5 Le]); Trucks: 15 Light, 40 Medium, 10 Heavy.

D3032: 164th Light Div; Trucks: 5 Light, 45 Medium, 5 Heavy.

D2931: 3 Aufklarungs Bn (R; 2I); Trucks: 2 Light.

D2831: 21st Panzer Div (Less: 2nd MG Bn; Assg: 200th Rifle Regt [15]; which begins the scenario detached); Trucks: 7 Light, 20 Medium, 10 Heavy.

D2733: Ramcke Bde; Trucks: None.

D2630: 104th Rifle Regt (Now assigned to the 21 Panzer); Trucks: 5 Light, 25 Medium.

D2532: 33 Aufklarungs Bn (R; 15).

D2431: 580th Aufklarungs Bn (R; 90 Le); Trucks: 5 Light.

D 3327: 136th GGFF Div (I) Trucks: 25 Light, 15 Medium.

D 3326: 101st Trieste Div (I; Det: 21 Arty Regt; ATT:III(M) Trucks: 5 Light, 55 Medium, 5 Heavy.

D 3325: 16th Pistoia Div (I) Trucks: 30 Light, 30 Medium.

D 3322: 133rd Littorio Div (T; Det: 12th Bersaglieri Regt; Less: X(M) and XII(M) tank Bns and V and VI Arty Bns self-propelled: Assg: in their places, respectively IV(M) and LI(M) tank Bns and DLIV and DLVI Arty Bns self-propelled) Trucks: 25 Light, 25 Medium, 6 Heavy.

E 3133: 12th Bersaglieri Regt (I; *Littorio*) Trucks: 6 Light, 20 Medium.

D 2933: 102nd Trento Div (I; Det: 7th Berg Regt) Trucks: 68 Medium, 7 Heavy.

D 2932: 7th Bersaglieri Regt (102 Trn) Trucks: 20 Medium.

D 2833: 25th Bologna Div (I); 21st Arty Regt (101 Tst) Trucks: 12 Light, 60 Medium, 10 Heavy.

D 2732: 27th Brescia Div (I) Trucks: 10 Light, 65 Medium, 7 Heavy.

D 2632: 132 Ariete Div (T; Less: VII(M) tank Bn; Assg: LII(M) tank Bn in its place, V and VI Arty Bns self-propelled; *Littorio*; Att: XIII(M) tank Bn; The 132 Arty Regt now contains 3 TOE Strength Points of 105/28 guns and 2 TOE Strength Points of 90mm Flak guns heavy AA Trucks: 25 Light, 45 Medium, 10 Heavy.

D 2533: 185th Folgore Div (I) Trucks: 30 Light, 30 Medium.

D 2432: 17th Pavia Div (I) Trucks: 10 Light, 65 Medium, 7 Heavy.

Anywhere on Map D: Italian XXIV Cp Arty; XX-VI Cp Arty; 1/2AR Arty Bn; 2/2AR Arty Bn; 3/2AR Arty Bn; 4/2 AR Arty Bn; RECAM (R); III/133 Fanterie Inf Bn I(L) and II(L) tank Bns. German: Sondervverband 288 Regt (I); 707th and 708th Heavy Weapons Coys; 13th Coy of the Bandenburg Regt (I).

Anywhere on Map A thru D: Any four Oasis Coys and the 778th Naval Engineers. All remaining unassigned German independent (i.e., no parent unit) artillery and anti-aircraft units. This includes the 1st and 2nd African Arty HQ's and the 155th and Arko 104 Arty Hq's and their subsidiary units.

Modifications to units strength: All Axis Infantry-type units are reduced to 2/3 (round fractions up) of their maximum TOE Strength.

[63.42] Tank Deployment

The following are exceptions to the OA Chart Listing.

Unit	Strength
I/5 Panzer	2 PzII: 4 PzIIII; 1 PzIIIE; 4 PzIIIIJ Specials
II/5 Panzer	2PzII; 2PzIIIE; 4PzIIII; 4PzIIIIJ Specials; 1 PzIVE
I/8 Panzer	1PzII; 4 PzIIII; 5 PzIIIIJ Specials; 3 PzIVF ² Specials
II/8 Panzer	1PzII, 3PzIIII, 1PzIIIE, 5PzIIIIJ Special, 1PzIV6, 3PzIVF ²
LII(M)	6 M/13/40's
VIII/M	6 M/13/40's
I(L)	6 L6's
II(L)	4 L6's

Broken down in Tobruk's Repair Facility: 3 PzIIIIH's; 1 PzIVD.

[63.43] Axis Air Strength

German Planes

The following planes may be located at any air facility in Africa. Planes not refitted may not be readied until the second OpStage of the scenario (they are not ready for the first OpStage).

Type of Plane	Total Avail	Refitted
Ju87D	70	44
Me Bf109F	83	40
Me Bf109G	90	50
Me Bf109E	16	6
Me Bf110C	16	10

SGSU's available: 21

Pilots Available:

Rating	Available
Four	4
Three	4
Two	11
One	16

The Axis Player has available in Crete the following planes: 75 Ju88's (60 refitted) and 15 He111's (10 refitted). No SGSU's or pilots are needed.

Italian Planes

Type of Plane	Total Avail	Total Ready
Cant.Z.1007	14	14
Fiat CR42	125	75
Fiat G.50	12	0
Macchi C.200	58	40
Macchi C.202	139	84
S.M.79	34	24

SGSU's Available: 27

Pilots Available:

Rating	Available
Four	1
Three	3
Two	20
One	18

Air Facilities

The Axis Player has nine Airfields, eight one Flying Boat Basin and One Flying Boat Alighting Area. He may place these wherever he wishes, on Maps A-D.

[63.44] Axis Trucks

Second-Third Line Trucks (In Convoy):

May be placed anywhere on game maps A thru D: 115 Light, 350 Medium, 75 Heavy.

The Axis Player receives 50 Light and 100 Medium trucks for distribution among his Air Facilities.

[63.45] Axis Supply

The Axis Player receives ten Supply Dumps, three of which are Dummies. At least five of those dumps must be placed on Game-Maps A, B or C. The other five may be placed on game-map D (or A-C, if desired). In addition, the Axis Player receives supplies at several cities/villages, as listed below. He may use Supply Dumps for those villages listed, in addition to those given to him.

Location	Ammo	Fuel	Stores	Water
Map D Dumps*	2700	4000	1500	800
Other Dumps*	500	500	500	500
Ghazal(D3328)	200	500	0	500
Mersa Matruh	500	1500	250	500
Sidi Barrani	200	500	0	500
Bardia	300	1000	250	0
Tobruk	800	3000	250	0
Derna	200	500	0	500
Benghazi	300	1000	250	0
Marble Arch	200	500	250	500

*No one dump may contain more than 33% of the total supplies of each type listed.

Each Airfield has 150 Ammo points, 350 Fuel points, 25 Stores and 50 Water except Tripoli/Tunisia.

Each Strip, Flying Boat Basin/Alighting Area has 20 Ammo points, 25 Fuel points, 10 Stores and 10 Water Points.

Trucks may be assigned to the dumps, cities as needed from the 2nd-3rd Line Trucks available.

The Axis has a Major Repair Facility in Tobruk, and Temporary Repair Facilities in Bardia and Mersa Matruh.

Axis Coastal Shipping (all units/tonnage) is available. Ships may be placed at any port between Tobruk and Tripoli.

[63.46] Axis Mediterranean Air Force Deployment

The Axis Player may transfer 10% of his available force to Italy or Sicily but no more. He may not have more than 10% of his air power in Italy or Sicily. Strategic air attacks against Malta may be made only in the Long Retreat scenario (see 44.41).

[63.47] Replacements and Reinforcements

For the "Last Chance" Scenario, ignore all Reinforcements and Replacements.

For the "Long Retreat" scenario, Reinforcements as normal. As for Replacements, Axis Player must reduce all his Points Available to 5% (rounded down) of what is listed in the Axis Replacement Pool. Moreover, any Replacement Points sent must be sent to Tripoli and must stay there, unless the following occurs: If the Axis Player still holds Tobruk, Bardia, or Mersa Matruh by the third OpStage of Game-Turn 105, then 25% of Replacement Points in Tripoli may be released the following OpStage. After that, 25% of all available Replacement Points (by type, and rounded up) may be released in the first OpStage of Game-Turn 108.

This rule takes into effect the Allied Landings (feared and actual) to the West, and the historical fact that almost all manpower at this time was being diverted to Tunisia.

[63.48] Upgraded Axis Equipment

A. All eligible Axis armored car and armored recon units may be upgraded as per rules for such. It is not necessary to use Replacement Points; simply upgrade the units as if they had used Replacement Points.

B. Any two of the four following Italian infantry divisions may have their assigned artillery units' OA Sheet listing of two or three 75mm heavy anti-air TOE Strength Points replaced with two 90mm heavy anti-air TOE Strength Points: 16 Pistoia, 17i Pavia, 25 Bologna and 27 Brescia.

C. The following German anti-tank units have their weaponry upgraded replacing their OA Sheet listing: The 605th SP of the 21st Panzer has 6 TOE Strength Points of Marder SP anti-tank guns. The 3.7mm anti-tank guns in the 15th and 21st Panzers are replaced by the same number of 5cm guns. The 5cm guns in the 15th and 21st Panzers and 90th Light are replaced by the same number of 7.62 Pak(R) guns.

D. Any four eligible Italian brigade-level HQ's are ad hoc anti-tank batteries each with 4 TOE Strength Points of Italian AT guns. Any two eligible German brigade-level HQ's are ad hoc anti-tank batteries, each with 4 TOE Strength Points of 2.8cm anti-tank guns.

E. All Italian *artillery* units originally assigned to a division (i.e., as listed in the OA Chart) have their 75/18 Gun-Howitzer TOE Strength Points replaced by an equal number of 75/27 guns. The two self-propelled artillery units in the Littorio and the two in the Ariete each possess 4 TOE Strength Points of Semovente 75/18 SP guns.

F. The following German artillery units are upgraded: The III/220 and III/220 artillery battalions of the 164th Division each consist of 3 TOE Strength Points of 15cm K18 guns and 3 TOE Strength Points of 15cm sFH18 guns (replacing their OA Sheet artillery). The 149, 526 and 533 Coastal Defense artillery battalions replace their artillery with, respectively, 3 TOE Strength Points of 15cm sIG33 guns, 3 TOE Strength Points of 17cm K18 guns and 3 TOE Strength Points of 21cm Mrs18 guns.

[63.5] SPECIAL RULES

The following rules may not be used:

- A. 10th Italian Light Flotilla
- B. Raid on Rommel

[63.6] INITIATIVE

Historically, the Germans had the initiative during the first OpStage of this scenario. However, the Players may at their discretion roll the die to determine who has the initiative on the first OpStage. (If the Commonwealth Player gains the initiative for the first week, it will be even more depressing than usual for the Axis Player, of course.)

[63.7] CONSTRUCTION

[63.71] All *unfinished Roads* have been built.

[63.72] The entire unfinished railroad complex has been built. However, the RR in the three hexes immediately west of el Alamein are destroyed.

[63.73] The only pipeline available is that in railroad hexes. The Axis Player may draw water through it from Tobruk while the Commonwealth Player may draw water through it from Alexandria.

[63.74] All ports are at full (listed) Efficiency except Benghazi, at Efficiency Level One, and Tobruk at Efficiency Level Five.

[63.75] Level One Fortifications are placed in the four hexes adjacent to Tobruk. In addition, the Axis Player may place a total of three Level One Fortifications among the four hexes adjacent to Bardia.

[63.76] Commonwealth Player places minefields in the following hexes: D3033, and on map-section E3101, 2901, 2801, 2701, 2702, 2601, 2501, 2401, 2302 and other five hexes (on map-section E). Of the fifteen, ten are Real and five are Dummy; the distribution is at the Commonwealth Player's choice. All start the scenario at unknown status (i.e., face-up).

[63.77] The Axis Player places minefields in the following hexes of map-section D: 3233, 3133, 3032, 2933, 2833, 2733, 2632, 2633, 2533, 2429, 2433, 2831, 2832, and 2731. Of the fourteen, eleven are Real and three are Dummy; the distribution is at the Axis Player's choice. In addition the Axis Player may place up to five Real and two Dummy minefields in any eligible hexes within three hexes of Tobruk and a maximum of two Real and two Dummy minefields in any eligible hexes on game map-section A thru C that are not within five hexes of Tobruk. ("Eligible" refers to the ability of the Players' to construct a minefield in that hex under the normal game restrictions.) All start the scenario at unknown status (i.e., face-up).

[63.8] AIR & LOGISTICS GAMES ABSTRACTIONS

This case covers the adjustments needed in addition to those given in Case 59.6 in order to play these scenarios without the Air and/or Logistics Game(s).

[63.81] **Air Abstracted** (Land & Logistics Games): The Commonwealth initial supplies are reduced as follows: The Front Line Dumps now total 2250

Ammo, 1750 Fuel, 1500 Stores and 1250 Water Points. The other dumps now total 500 Ammo, 2000 Fuel, 400 Stores and 1600 Water Points (see Case 63.35).

[63.82] Air & Logistics Abstracted (Land Game only)

A. The Players' ability to bring in Motorization Points is equal to their respective abilities to bring in Medium Trucks (i.e., each Medium Truck equals one Motorization Point).

B. Axis Initial Supply Status: Case 63.45 is replaced by the following:

Location	Regular Supply Units	Each with Ammo	Fuel
"Other Dumps"			
Supply units*	14	40	20
"Map D Dumps"			
Supply units†	2	40	25
Ghaza (D 3328)	1	40	60
Mersa Matruh (D 3714)	1	40	60
Sidi Baranni (C 4131)	1	40	60
Bardia (C 4321)	2	20	60
Tobruk (C 4807)	3	40	60
Derna (B 5925)	1	40	60
Benghazi (A 4827)	2	25	60
Marble Arch (A 2109)	1	40	60

*These plus 2 dummy supply units may be placed in any hexes on maps A thru D at the Axis Player's discretion except that no more than a total of four supply units (i.e., regular plus dummy) may be placed on maps A through C.

†These, and in addition 1 dummy supply unit, may be placed in any hexes on Maps A through C.

C. Commonwealth Initial Supply Status: Case 63.35 is replaced by the following: The Commonwealth Player receives 13 regular supply units, each with 40 Ammo and 20 Fuel Points, to be placed at his Front Line in any fashion with the restriction that at least one regular supply unit must be placed in each of the following hexes: el Alamein (E 3002), E 2805 and E 2403. The Front Line is defined as hexes E 3002, E 2805, E 2403 and any hex that is east of and adjacent to an Axis unit. In addition, the Commonwealth Player receives 2 regular supply units, each with 40 Ammo and 60 Fuel Points, which may be placed anywhere on map E except that neither may be placed within two hexes of the Front Line (this includes Front Line hexes not occupied by supply units).

[63.83] Logistics Abstracted (Land & Air Games):

A. The Players' ability to bring in motorization points is equal to their respective abilities to bring in Medium plus Heavy Trucks (i.e., each Medium and each Heavy Truck is equal to one Motorization Point).

B. Axis Initial Supplies: As in 63.72 "B" above plus 2 depot supply units, each with 180 Fuel and 40 Ammo Points, one at Mersa Matruh and one at Tobruk. In addition, a total of 9 regular supply dumps, each with 30 Ammo and 50 Fuel Points, to be distributed among and placed on any Axis air facilities other than Mersa Matruh and Tobruk.

C. Commonwealth Initial Supplies: As in 63.72 "C" above plus a total of 2 depots, each with 40 Ammo and 180 Fuel Points, one each at any Commonwealth air facility not within three hexes of a Front Line hex.

[63.9] VICTORY CONDITIONS

[63.91] "The Last Chance" Scenario

A. The Axis Player wins a Strategic Victory if he

can occupy at least one hex of Alexandria with at least a Brigade-equivalent unit. Supply doesn't matter.

B. Failing the above, the Axis Player totals the number of TOE Strength Points (any type) on Game-Map E, the Commonwealth Player totals those TOE points he has on Game-Map D (exclusive of Layforce and SAS units). The Player with the most TOE points (in comparison to the other) is the winner of a tactical victory.

[63.92] "The Long Retreat" Scenario

The name of the game is survival here. How good is the Axis Player at delaying, at giving territory grudgingly. He cannot win in a military sense (the Allies are landing, Africa is lost—at least in retrospect). However, the better he fights, the more attention the Allies will have to pay to Africa. Here the Axis Player is playing against actual history. Can he do better than Rommel on this retreat? Rommel didn't do badly; see if you can best his record (if you don't fall asleep first). In essence, the Axis Player gains points for occupying certain locations. Occupying means that all combat units in that City/Village must have at least one Game-Turn's worth of Stores, be able to fire all weapons twice, and have enough fuel for all units to move 20 CP's.

Location	Axis Pts for Occupying
Mersa Matruh	1
Bardia	3
Tobruk	5
Derna	2
Benghazi	3
El Agheila	1
Marble Arch	1

If the Axis Player can get 10 or more points he wins a Strategic Victory.

If he can get 4-9 points, he wins a Substantive Victory.

If he gets 1-3 points, he wins a Marginal Victory.

If he get no points, the Commonwealth Player wins.

[64.0] SCENARIO GROUP FIVE: THE CAMPAIGN FOR NORTH AFRICA

[64.1] INTRODUCTION

If you have gotten this far you will by now have surmised that the full campaign game of *CNA* is, far and away, the most ambitious undertaking a gamer could ever want to get involved with. The campaign game requires not so much a commitment in time (which is always relative), but more so a commitment to planning and foresight. A campaign game will need at least ten players, perhaps more. Ideally, each land division should have a commander, plus an Air Commander and a Logistics Commander. On top of this should be one Player as Supreme Commander. This means a lot of players, and, again, this is the ideal situation. The fewest number of players would be three to a side. Lastly, it would be hard to underestimate the amount of paperwork and planning that are involved in a campaign game. Keeping on top of the situation is half the battle . . . and more than half the fun. And speaking of fun, the campaign game can be a lot of fun. If you are interested in a challenge, interested in wrestling with the problems of organizing and running a campaign, and if you simply want to experience the complexities of modern warfare at all levels, you will get much satisfaction from *CNA*.

[64.2] SCENARIO LENGTH

There are two campaign game scenarios in *CNA*. The first covers the entire war in Northeast Africa and begins in the third week of September, 1940 (i.e., the Italian Offensive, 1st OpStage of Game-Turn 1). The "shorter" campaign begins with the arrival of General Erwin Rommel and the Deutsches Afrika Korps during the fourth week of March, 1941 (i.e., the Desert Fox, 3rd OpStage of Game-Turn 26). Both campaign games end with the completion of the third OpStage of Game-Turn 111.

[64.3] INITIAL DEPLOYMENTS

If the entire campaign is being played, the Players use the information provided in Section 60.0 for the initial placement and distribution of their land, sea, and air forces, supplies and construction. If the Players are starting the campaign with the arrival of Rommel, the information in Section 61.0 is used.

[64.4] INITIATIVE

The Players determine Initiative using the information provided in Section 60.0 or 61.0 depending upon where they are starting their campaign. Initiative is determined normally once the restrictions in the appropriate section expire.

[64.5] SPECIAL RULES/ADJUSTMENTS

[64.51] There are no restrictions on normal game actions (i.e., Raids on Rommel, the Commonwealth Fleet, etc. may be performed as per the appropriate rules).

[64.52] The Axis Player's Malta Air Availability is determined using the appropriate campaign game row on the Axis Strategic Airforce Commitment Chart.

[64.53] If the shorter campaign game is being played the Axis Replacement Pool must be adjusted as follows:

- A. German Production Chart: No change.
- B. Axis Truck Production Chart: The total trucks the Axis Player may bring in are reduced to 800 Light, 2400 Medium and 500 Heavy.
- C. Italian Production Chart: One-half of all replacement points (rounding fractions up) that could be planned up thru (i.e., including) Game-Turn 24 are no longer available with the following exception: the Axis Player still has 60 CV L.3 & CV 33/35 tank replacement points available.

[64.54] If the shorter campaign game is being played the Commonwealth Production System must be adjusted as follows:

- A. Commonwealth Truck Production Table: No change.
- B. Commonwealth Infantry Production Table: No change.
- C. Commonwealth Production Chart: The replacement points listed below are reduced to (not by) those numbers, all other replacement points are unaffected:

Item	Number
Armored Car/Armored Recce	75
Light Anti-air	70
25-pounder artillery	230
2-pounder AT guns	50
Mk VI Light tanks	5
A9 crusiers	3
A10 crusiers	3
A13 crusiers	5
Matildas	20

[64.6] AIR & LOGISTICS GAMES' ABSTRACTIONS

Why anyone would play a campaign game without the Air and/or Logistics Game(s) is beyond me. However, the Players should pick their start point (Italian Offensive or Desert Fox) and use the rules in the appropriate section(s) (32.0, 47.0, 58.0) and the abstracted set-up for that scenario group and the replacement point adjustments in Case 64.5.

[64.7] VICTORY CONDITIONS

It would be hard to imagine anything more exasperating than playing a game for five years, getting to the end, and finding out that you have lost by one point or two hexes, or something of that ilk. Victory conditions for a game of this size and scope are almost meaningless: survival itself is a victory. However, you have to have some sort of objective, and the Designer has to make some sort of decision. Our decision here is to use Victory Conditions that compare to what actually happened, *not* what each side needed to do. The reason for this is that the Axis situation was a foregone conclusion by 1942, what with the pending Allied invasion of Africa from the West. Thus, the Axis had to take Alexandria (and the Delta) or else be caught in the desert between two armies. And Alexandria is thus the Axis objective, historically. As for gaming, it becomes a matter of points, or who can hold onto what.

[64.71] If the Axis Player occupies all hexes of Alexandria and Cairo for one full Game-Turn, and such occupying units can trace a line of supply (i.e., convoy route) back to a Supply Dump which in turn can be supplied from Tobruk or Tripoli in any way, and that line is 90 movement points by truck or less, the Axis wins the game—regardless of the turn or date. Failing that, we count points, as below.

[64.72] Starting with the first OpStage of Game-Turn 35, if there are no Axis Combat units that can trace a line of supply of 60 Movement Points (Truck) to a Supply Dump and thence to Tobruk or Tripoli as per case 64.71, game-map, the Commonwealth wins the game automatically. This does not include air or coastal shipping units.

[64.73] Failing the above two cases, players receive points for doing (or not doing) certain things. The first of these points are Geographical Occupation Points. The Player occupying the given cities/villages gains the points listed. Occupation for these purposes means having a combat unit of at least 1 TOE Strength in the hex. That combat unit, at the end of the game, must have enough Stores and Water for one Week, and enough Fuel and Ammunition to fire its weapons three times and move 20 CP's. Any units failing these "tests" do not occupy for victory conditions.

City/Village	Axis V.P.'s	C'wlth V.P.'s
Mersa Matruh	100	10
Sidi Barrani	50	10
Siwa	20	10
Jalo	10	20
Giarabub	15	10
Bardia	100	50
Sollum	25	10
Tobruk	200	100
Derna	25	50
Benghazi	75	100

[64.74] The next series of points are replacement Victory Points. Each Player receives *one* point for each unused Replacement Point allotted to him in his Production Charts, not including planes and/or Trucks and, for the Commonwealth Player only, Infantry. Thus, if the Commonwealth Player had a total of 35 Crusader I

Tank Replacement Points available for the game, but he uses only 18, he gains 17 points.

[64.75] The final type of Victory Points apply solely to the Commonwealth Player: Withdrawal Points. (Unit, not Drug). These apply to battalions withdrawn from the game, *other than* those Withdrawn via The Reinforcement Track. These are voluntary withdrawals, not mandatory withdrawals. They are earned as follows:

- A. For every combat battalion (not company) or battalion-equivalent of infantry, armour, artillery or anti-tank guns (not AA) that is at least 75% TOE Strength that is Withdrawn from the game the Player receives ½ point for each week that unit is gone, to a maximum of three points per unit. To be Withdrawn a unit must start the Stage in Alexandria or Cairo.
- B. Every time that a battalion withdrawn under A. above is returned to the game, the Commonwealth Player loses two points. A unit voluntarily Withdrawn and then returned to the game may not be Withdrawn until six months from the date of return. (This may not be entirely accurate, but it will stop players from doing the One-Point Two Step, shuffling a single unit in and out to get one point).

[64.76] The levels of Victory are as follows. Again, anyone who actually finishes this game can consider himself a winner (much as the soldier who fights two-three years does when he survives.) Questions about the balance of the campaign scenario received before 1981 will be considered with the seriousness which they deserve. The following schedule is consulted after both sides total their points, comparing the totals as a ratio of most to least.

Ratio	Result
Even (1 to 1)	Draw (My God, what a waste of time!)
Better than 1-1, up to 1½-1	Marginal Victory
Better than 1½-1 up to 2½-1	Decisive Victory
Better than 2½-1	A Smashing Victory

[65.0] IN CONCLUSION

We have attempted, in *Campaign for North Africa*, to provide the hobby with the most accurate and most stimulating simulation yet undertaken. Very little is abstracted (mostly the naval game), and the Players are given as much to do as we considered humanly feasible. *CNA* is a difficult and complex game; it is not for the novice, to be sure. It is our (both designer and developer) opinion that we have, in the main, succeeded in our endeavor. *CNA* was playtested for nearly two years an extensive list of "veteran" testers, most of whom were chosen for their insight and gaming acumen. Many researchers were used—including the inimitable Richard Garczynski—more than on any other project. The amount of work put into the game was enormous; the amount of information available is almost mind-boggling. We hope that the amount of fun is just as great.

DESIGNER'S NOTES In Brief

A game such as *CNA* certainly could use — nay, demands — extensive background information as to where and why. Unfortunately, to do this subject justice would require a veritable book. And you already have *enough* to read with this game. I'll simply try to cover a few subjects generally.

The *general* research was done mostly by me; the specifics were done by a host of others. The task was divided into areas — Italian OB, German air, logistics, etc. It was done over a period of two years, and provided many hours of confusion and some laughter. (We felt that being provided with a complete OB for German bakery companies was a bit much...although the temptation to use them was great.)

The scale of the game was decided by its scope and size — and the map configuration. The sources are listed in the bibliography, and suffice to say the game-maps are borrowed from several conflicting sources. Often the designer had to make difficult decisions as to which tracks existed and which simply filled in the blank spaces on an Italian map. The tracks proved most difficult — I don't think there are two maps that agree — but defining things such as sand-gravel, rough, and salt marsh was equally taxing. The map of the game is as accurate as we could get it on the scale presented. The greatest liberties, or shall we say, abstractions, are with ridges, slopes, and escarpments. Again the effort was, at the scale given, to present effect rather than strict geographical accuracy.

The system itself was developed rather quickly. The first problem tackled was the theory and application of mobile desert warfare. Several approaches were tried, most were unsatisfactory. What I wanted was a system that would enable units to move as far as the player wanted them to, but within the limits of fuel and fatigue. And I found the germ for such an idea in an unlikely place: Mick Uhl's *Gettysburg '77*. From that game came the general idea for the Continual Movement System. Actually, the system did not take full effect until I came up with the idea for Cohesion Points — and, of course, a workable set of numbers for such. Both of these provided the game the one thing I wanted to prove — that mobile units were more powerful not because they had more "combat points" but because they were more mobile.

After the movement system fell into place it took several weeks to come up with a combat system. The idea was to provide a tactical feel with an operational game. (The system for *CNA* is remarkably similar to Frank Chadwick's *Operation: Crusader*.) Of course, the backbone of the combat system is the OB work — and especially the values given to each unit. Lord knows where some of these came from, but I do know that I have a mountain of yellow sheets in my basement with a lot of mathematical computations. Tanks and guns were done strictly on relative strengths — using the Italians as a base for the lowest. The same with infantry. I'm sure questions will arise as to why some infantry is 2/2, some 1/2, some 1/1, etc. A lot has to do with weaponry, some with background and training, and some is fairly subjective. The strengths themselves are not the balancing factor, the Morale Level is. And Morale Level was based on performance, training and general esprit de corps. (The morale system was taken from my own *Veracruz*, another unusual source.)

Combat, however, is not what this game is about. Logistics is the subject, and trucks are the thesis. And from the very beginning that is what I concentrated on. Most of the design work went into mak-

ing the logistics system work. Hundreds of computations on breakdown rates, fuel consumption, tonnage capacities, etc., were done using available statistics and charts. One of the hardest jobs was the Replacement Charts. It is one thing to determine how much a man weighs, and thus a battalion; it is another to determine how much ship space he takes up. The same can be said for guns and other equipment. We had a fairly accurate picture of Axis shipping tonnage, but there was little information available as to what was shipped on what and how. The real problem was not coming up with all this information, it was integrating it so that everything meshed. Luckily, the logistics system proved to be relatively trouble-free.

Not so for the air system. That was hell. We were working with individual planes — a conceit of the designer, admittedly — and thus the combat system had to be tactical. However, this was an operational game and I had no desire to get involved in an African version of *Air War*. It took a long evening of fiddling with a calculator to come up with the complicated series of computations that we used to arrive at our air strengths. This was possible only when I came up with a combat system. If you ask me, the whole thing is a bit much — but then you could say that of the entire game. I would have scrapped the whole shebang had not the playtesters enjoyed it so much. As it is, the air system is a game within itself, with all its logistical problems and planning. Oh, yes — having the air commander roll for refit for each plane individually is a good way of passing time while the land game is on.

The main question with *CNA* is, "Why?" The answer, superficially, is because that's what the feedback said you wanted. But that's a cop-out. I did *CNA* as a definitive and informationally overpowering simulation because I wanted to see how far I could take simulation gaming and still survive. *CNA* is virtually unplayable — as a game. It is, for the most part, a source of information. Yet, if it is approached from the point of view of an experience, a lesson in history, then *CNA* can provide remarkable insights into a fascinating campaign. One does not "play" *CNA*, one immerses himself in it. Some of you will, no doubt, drown; most will just tread water. But a few of you will find the swim refreshing.

A word as to the accuracy of all this information. Let's face it, I am not the definitive scholar on North Africa — it's not even my main area of interest. There are bound to be some mistakes, some all-too-subjective assumptions, and some pure baloney. But all of it is based on hard research. As I have said elsewhere, if you have better information, use it. (And let me know, too....) This is a system for your enjoyment; it is not an ironclad contract between gamer and designer. One thing I do know, from all the research we did, this — *CNA* — is the first commercially available source to publish a complete battalion-level Order of Battle for all sides at all times during Africa. Just think how many games you can devise from that alone!

Some brief words of thanks to those involved. *CNA* took a long time (as those who ordered the game two years ago will no doubt attest to). Along the way many people gave freely (and not-so-freely) of their time and energy. To Redmond Simonsen goes the credit for coming up with the idea for this game — and he knows how I feel about that brainstorm. To our fearless leader, Jim Dunnigan, a devotee of the Torquemada School of Persuasion, goes some dubious thanks for assigning me to this project. An honest vote of thanks to my very good friend Howard Barasch, for spending many hours with his voluminous notes on the campaign. And to my two OB researchers —

David Ryan and, especially, the incredible Richard Garzcynski, I can only say that they saved my proverbial rear-end more than once. But most of all, I would like to thank the game's developer, Marty Goldberger. Marty stuck with this project through worse than I had to do. I may have designed it, but *he* had to make heads or tails of it...and he was the one who took all the grief when we screwed up. A remarkably dogged individual, Marty G. provided a sense of structure and parameter that was sorely needed. And finally, much gratitude goes to my wife Karen, who can now stop saying, "Africa? Are you still working on that damn game!"

BIBLIOGRAPHY FOR CNA A Brief Note

The books and references listed here are not an extensive or complete list of all works consulted. Some works were too general to be of use, and while they provided interesting background you could not say they were used as a source. Other works are not listed simply because, in the long delay in putting the game together, we just lost track of the exact title (the book having been returned to its owner, etc.) Other works, such as the Commonwealth histories — including New Zealand, Australian, Indian, and South African — were definitely used but can usually be found in any standard work on the subject. The same can be said for the famous Playfair volumes, the four-book work on the British effort in the Mediterranean. Playfair is an incredible work, but it is flawed. It is difficult to read, having been written by committee, as it were, and there is much information that is missing, as well as some that is wrong. Altogether, however, it is still the single most important source.

Finally, it must be noted that our two main OB researchers, the estimable Messrs. Garzcynski and Ryan, made use of a large number of unit histories, military studies, microfilms, and private letters and correspondence.

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The complete countersheets for *Campaign for North Africa* are reproduced on the following pages to provide a record for players

to replace any counters that may be lost or damaged in the course of playing the game.

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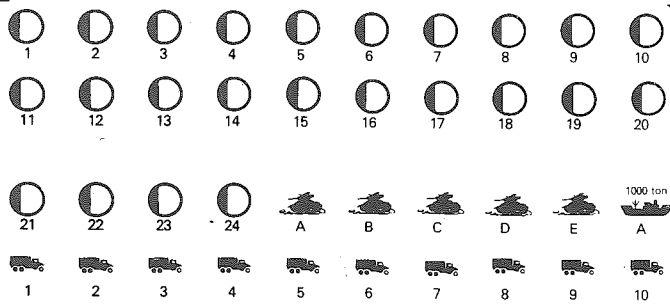
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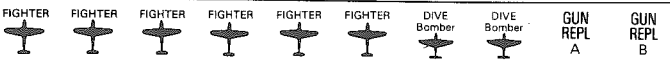
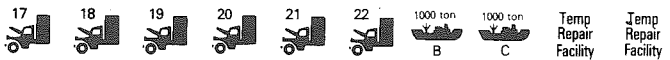
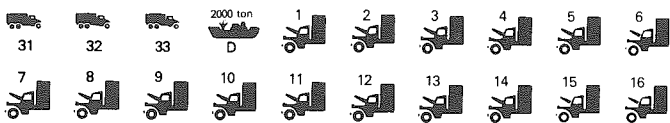
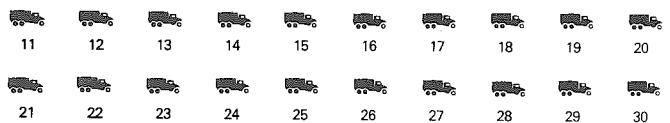
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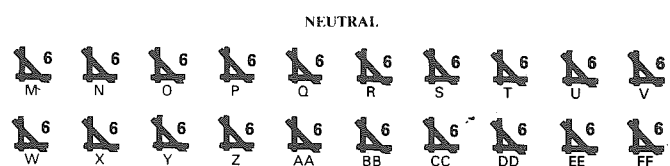
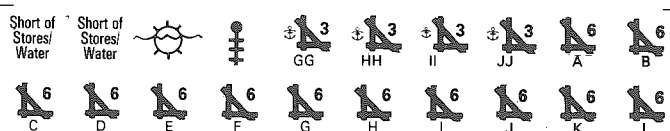
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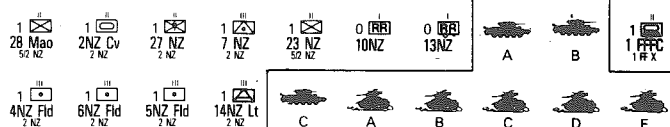
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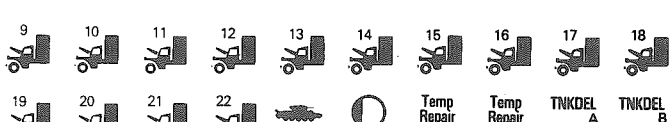
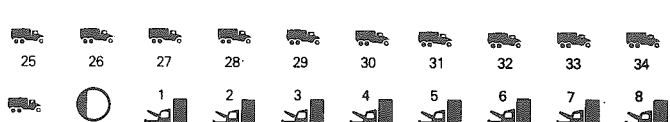
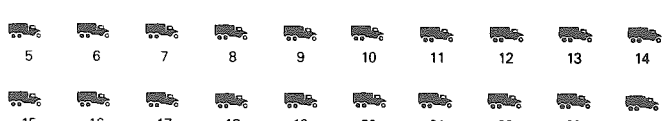
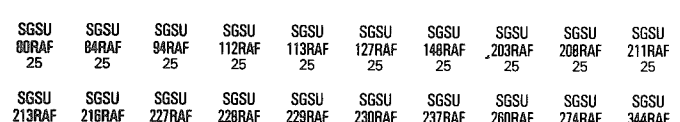
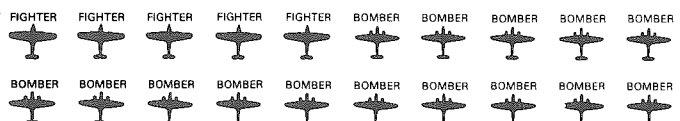
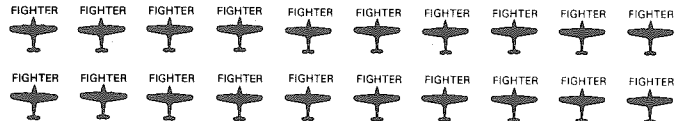


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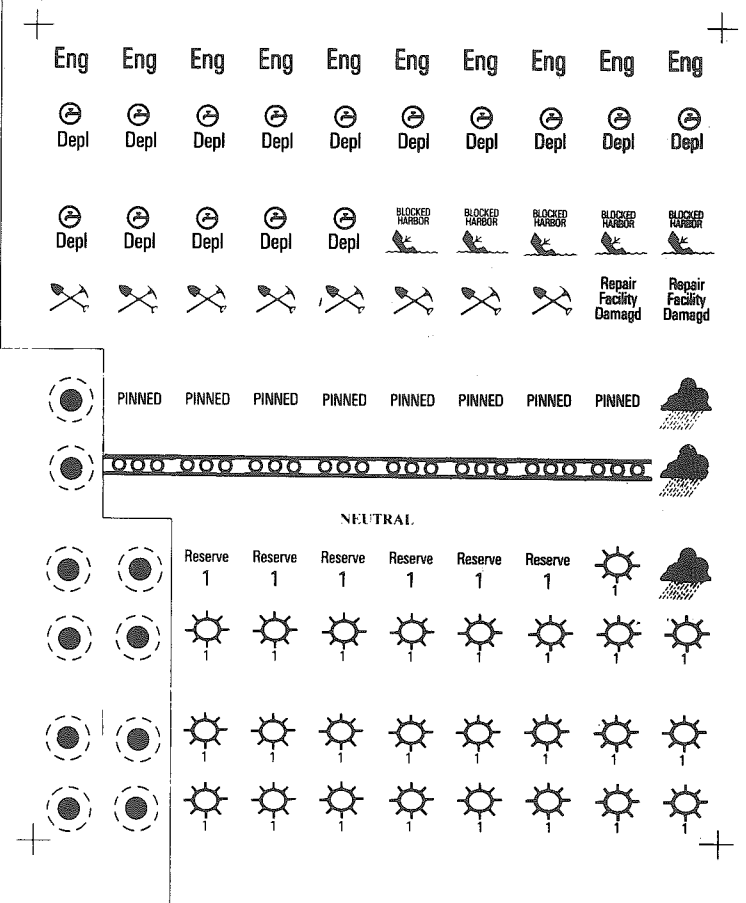
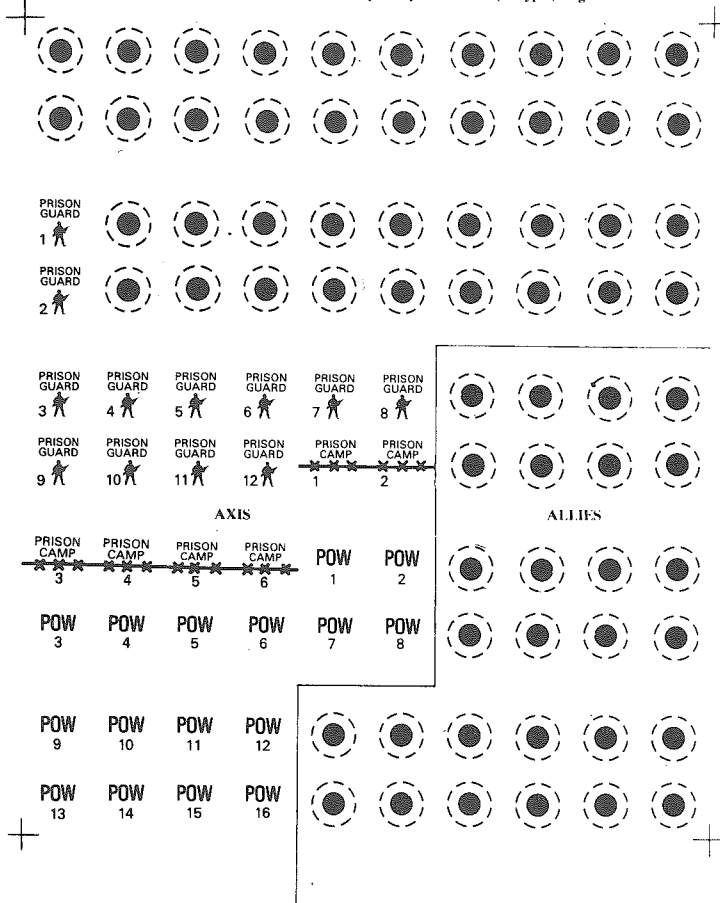
SOUTH AFRICAN

INDEX

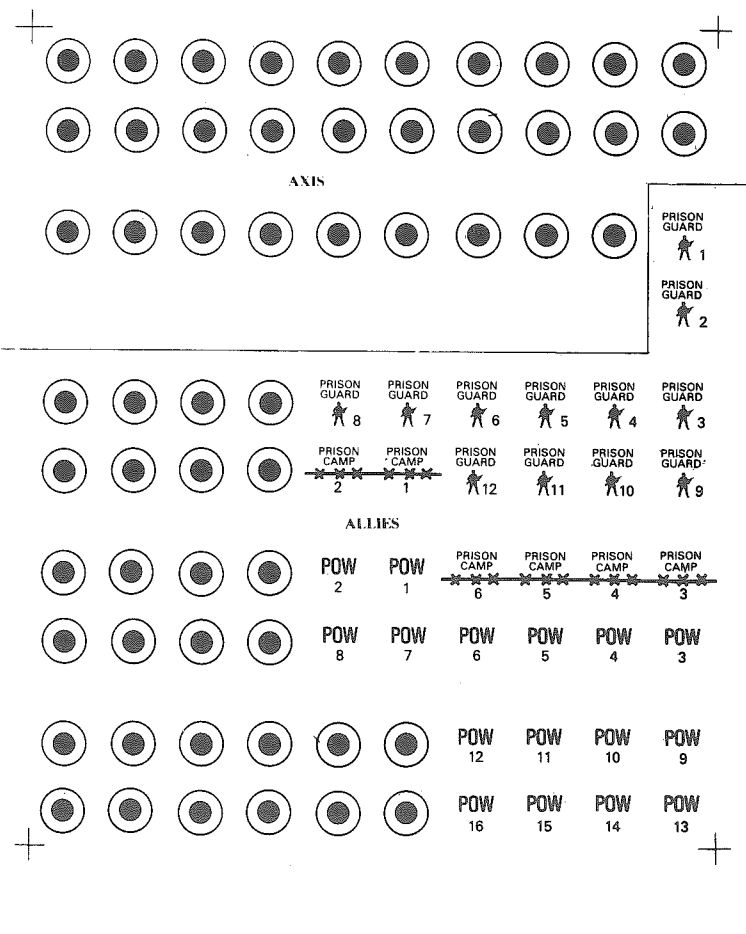
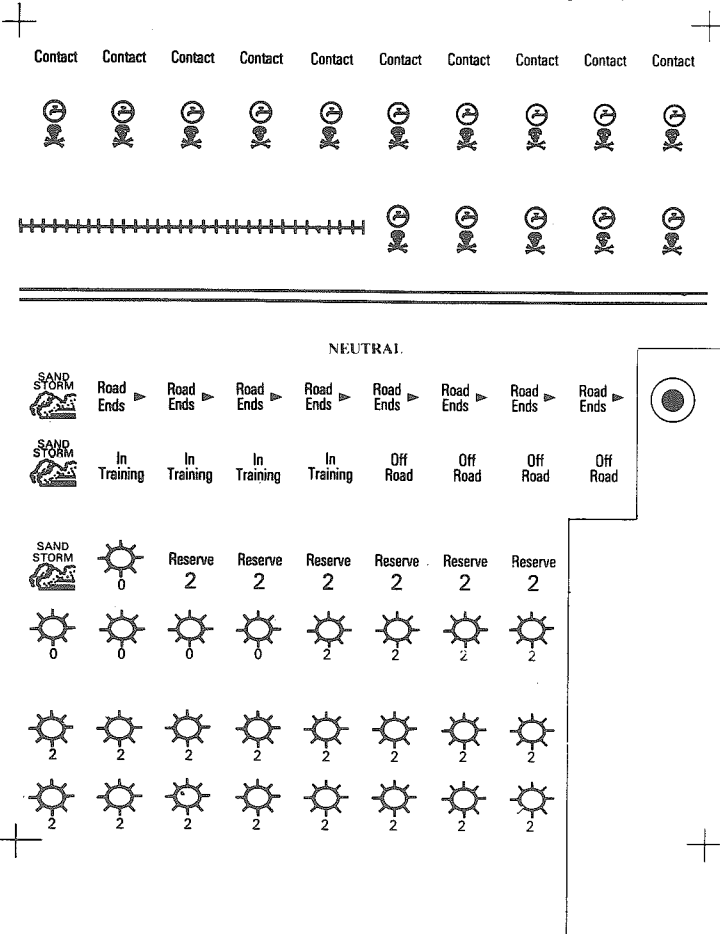
COMMONWEALTH

Campaign for North Africa Counter Section Nr. 9 (200 pieces): Front
 Quantity of Sections of this identical type: 1. Total quantity of Sections (all types) in game: 9.

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Campaign for North Africa Counter Section Nr. 9 (200 pieces): Back



THE CAMPAIGN FOR NORTH AFRICA

CHARTS & TABLES COMMON TO BOTH PLAYERS

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[6.3] CAPABILITY POINT EXPENDITURE SUMMARY

Action	Capability Point Expenditure
Detach a unit (Parent Formation and detaching unit)	1
Attach an assigned unit (Parent Formation and attaching unit)	1
Attach an unassigned unit (Parent Formation and attaching unit)	2
Absorb 2 TOE Replacement Strength Points (PF + absorbing unit)	1
Enter a non-minefield hex	TEC
Enter a friendly minefield hex:	
Any unit with an Eng unit	0 + TEC
Non-motorized unit, no Eng unit	1 + TEC
Motorized unit, no Eng unit	4 + TEC
Enter an enemy minefield hex:	
Non-motorized unit with an Eng unit	2 + TEC
Motorized unit with an Eng unit	4 + TEC
Non-motorized unit, no Eng unit	4 + TEC
Motorized unit, no Eng unit	CPA + TEC
Break Contact	2
Disengage	4
Be placed in Reserve	0
Phasing unit:	
Barrage and/or an Assault other than a Probe	5
Undergo a Barrage	3
Probe	2
Non-Phasing unit:	
Barrage and/or undergo a Barrage and/or defend against an Assault other than a Probe	3*
Defend against a Probe	2*
Patrol	0
Desert Raider Raid	5
Attempt to Poison a water source	1
Attempt to sweeten a poisoned water source	5
Attempt to blow a supply dump	1/3 CPA
Draw water, load/unload troops and/or supplies on trucks during an Organization Phase	0
Draw Water other than during an Organization Phase	1
Load/Unload troops and/or supplies on trucks other than during an Organization Phase	2
Construct a Real or Dummy/Non-Dump Supply Dump	3/2
Construction/Demolition of items other than Supply Dumps	0
Railroad or inter port transport of troops	0
Transport of troops by air	0
Paradrop paratroops	0
Paradrop commandos	5
Commando amphibious landing	5 or 10 + TEC
Ready airplanes	10

TEC = Expenditure listed on the Terrain Effects Chart for the hex's terrain type. CPA = The unit's Capability Point Allowance. *If the final adjusted differential is -4 or worse, the defending units 'reacquire' two of the capability points they originally expended in order to defend if the units were not barraged nor engaged in barrage. Note that certain actions, while not requiring the expenditure of capability points, restrict a unit's ability to expend Capability Points.

[15.53] ORGANIZATION SIZE CLOSE ASSAULT MODIFICATIONS CHART

Largest Unit On:		
Larger Side Is	Smaller Side Is	Adjustment
5 SP's	3 SP's	1
5 SP's	2 SP's	2
5 SP's	1 SP	4
5 SP's	0 SP	8
3 SP's	2 SP's	0
2 or 3 SP's	1 SP	2
2 or 3 SP's	0 SP	4
1 SP	0 SP	2

Largest Unit On = Each Player's largest unit, in size-equivalents, actually taking part in the combat in the hex. Note that only those TOE Strength Points actually partaking of combat (although not necessarily Close Assault) in *that* hex contribute toward determining size-equivalency. If, for example, a battalion did not contribute at least 50% of its maximum TOE Strength Point Limit (due to being understrength, engaging in a Probe, contributing to attacks in other hexes, whatever), that battalion is considered a shell, and may therefore affect the size-equivalent of any brigade or division it is currently attached to. **Larger Side Is/Smaller Side Is** = Either Player may possess the largest unit in that combat. Note that the numbers of the various size-equivalent units engaging in this Close Assault do not matter. Only the size of the largest unit. **5,3,2,1,0 SP's** = The largest unit's size in Stacking Points. **Adjustment** = The number of columns to be shifted in the Larger Size Player's favor in that Close Assault.

[16.6] PATROL SURVIVAL TABLE

DIE *	TOE Strength Point of Patrolling Force Killed / Captured
0...3	0 / 0
4	0 / 1
5	1 / 0
6	1 / 1

*If all Patrolling TOE Strength Points are from Recce units, subtract one from the die roll.

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[7.2] INITIATIVE RATINGS CHART

Player	Situation	Initiative Rating
Commonwealth	Game-Turns 1 thru 42 incl.	3
Commonwealth	Game-Turns 43 thru 90 incl.	4
Commonwealth	Game-Turns 91 thru 111 incl.	5
Axis	Rommel counter on the Game-Maps*	6
Axis	German land combat units but no Rommel counter on the Game-Maps*	3
Axis	No German combat units or Rommel counter on the Game-Maps*	1

*The Holding Boxes on Game-Map A for Tripoli/Tunisia region are not part of the Game-Maps for this purpose.

[9.4] STACKING POINT VALUES

Unit Equivalent

Full	Shell	Stacking Point Value
Division	-	5
Super* Brigade	Division	3
(Standard) Brigade, Battle Group	-	2
Battalion	Any Brigade Battle-Group	1
Company	Battalion, HQ any attached units	0
Truck Points in Convoy	-	½ for ea. 5 or fraction thereof†
Replacement Points	-	1 for ea. 5 or fraction thereof

*Term coined in order to distinguish those brigades which were in effect, infantry-light divisions from 'standard' brigades. †Only comes into play when determining Road/Track CP movement cost applicability.

[11.4] LAND COMBAT CALCULATIONS SUMMARY

Barrage:

Raw Barrage Points = Sum of the various Ratings × the respective assigned TOE Strength Points

Actual Barrage Points = Raw Points ÷ 10

Ability to combine to attack a single target is affected by whether the barraging unit is Forward or Back and/or Italian. Terrain may benefit the target.

Anti-armor:

Raw Anti-armor Points = Sum of the various Ratings × the respective assigned TOE Strength Points

Actual Anti-armor Point = Raw Points ÷ 10

Terrain may benefit either Player.

Minefields benefit the non-Phasing Player.

Non-phasing Player is benefitted by the Phasing Player's forces having to enter the Non-Phasing Player's hex.

Close Assault

Raw Close Assault Points = Sum of the Various respective Offensive or Defensive Ratings × the respective assigned TOE Strength Points

Phasing Player may be restricted because he is performing a Probe

Either Player's Ratings may be reduced because their tank TOE Strength Points are not involved in a Combined Arms attack.

Actual Close Assault Points = Raw Points ÷ 10

Terrain may benefit either Player.

Morale may benefit either Player.

Organizational size of the units involved may benefit either Player.

2:1 superiority in Raw Close Assault Points may benefit either Player.

Minefields may benefit the non-Phasing Player.

Engineer Units may benefit the Phasing Player.

[12.6] BARRAGE AGAINST LAND UNITS COMBAT RESULTS TABLE

Barrage Points									Target Class	Result
1,2	3,4	5,6	7,8	9,10	11,12	13,14	15,16	17+		
11...53	11...44	11...41	11...34	11...24	11...16	-	-	-	Infantry	No effect
54...66	45...66	42...65	35...64	25...61	21...44	11...35	11...32	11...31	Infantry	Pinned
-	-	66	65...66	62...66	45...66	36...64	33...56	32...55		1
-	-	-	-	-	-	65...66	61...66	56...66		2
11...62	11...54	11...45	11...36	11...31	11...22	11...14	-	-	Armor	No effect
63...66	55...66	46...66	41...66	32...66	23...66	15...63	11...62	11...54	Armor	Pinned
-	-	-	-	-	-	64...66	63...66	55...66		1
11...61	11...54	11...46	11...36	11...26	11...22	11...12	-	-	Gun	No effect
62...66	55...66	51...66	41...66	31...66	23...64	13...56	11...54	11...36	Gun	1
-	-	-	-	-	65...66	61...66	55...66	41...66		2
11...66	11...64	11...62	11...56	11...55	11...53	11...46	11...42	11...32	Truck	No effect
-	65...66	63...66	61...66	56...63	54...63	51...61	43...61	33...61	Truck	1
-	-	-	-	64...66	64...66	62...66	62...66	62...66		2

Pinned = The unit may not Retreat Before Assault, may not participate in Assault. 1 = Infantry* or Armor: Lose one TOE Strength Point and Pinned. Guns or Trucks: Lose one TOE Strength Point. 2 = As above, except lose two points. *If the Infantry is in Trucks, eliminate one Truck Point.

[14.6] ANTI-ARMOR FIRE COMBAT RESULTS TABLE

DICE	Anti-Armor Points																
	0*	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16+
11,12	-	-	-	1	2	3	4	6	8	10	11	12	14	16	18	20	22
13,14	-	-	-	1	3	4	5	6	8	10	11	13	15	17	19	20	22
15,16	-	-	1	1	3	4	5	7	9	10	12	13	15	17	19	21	23
21,22	-	-	1	2	4	5	6	8	9	11	12	14	16	18	20	21	23
23,24	-	-	1	2	4	5	6	8	10	11	13	14	16	18	20	22	24
25,26	-	-	2	3	4	6	7	9	10	12	13	15	17	19	20	22	24
31,32	-	1	2	3	5	6	8	9	11	12	14	15	17	19	21	23	25
33,34	-	1	2	3	5	7	8	10	11	13	14	16	18	20	21	23	25
35,36	-	1	3	4	5	7	9	10	12	13	15	16	18	20	22	24	26
41,42	-	2	3	4	6	7	9	11	12	14	15	17	19	21	22	24	26
43,44	-	2	3	5	6	8	10	11	13	14	16	17	19	21	23	25	27
45,46	-	2	4	5	7	8	10	12	13	15	16	18	20	22	23	25	27
51,52	1	3	4	6	7	9	11	12	14	15	17	19	20	22	24	26	28
53,54	1	3	4	6	8	9	11	13	14	16	17	20	21	23	25	27	29
55,56	1	3	5	7	8	10	12	13	15	16	19	21	22	24	26	28	30
61,62	1	4	5	7	8	10	12	14	15	17	19	22	23	25	27	29	31
63,64	1	4	6	8	9	11	13	14	16	18	21	22	24	26	28	29	32
65,66	2	4	6	8	9	11	13	15	17	19	21	23	24	26	28	30	32

*Only if 1...4 Raw Anti-Armor Points (before any terrain effects). Nr = Number of Armor Protection Points that must be lost.

Modifiers:

Phasing Player decreases his dice roll by one row (an 11 or 12 is unaffected).

Terrain: Shift *left one column* if the non-Phasing Player's units are in Level 1 Fortification, Rough or Heavy Vegetation. Shift *left two columns* if the non-Phasing Player's units are in a Level two or three Fortification or Mountain hex. In addition, shift *left one column* if all of the attacking forces are attacking through a slope hexside, or Shift *left two columns* if all of the attacking forces are attacking through a ridge hexside, or the attacker only shifts two columns left if attacking down an escarpment hexside (his forces may not be harmed). A shift to less than zero is treated as zero Anti-Armor Points.

[15.89] PRISONERS CAPTURED RESULTS TABLE

DIE	1	2	3	4	5	6
% Prisoners	10	25	33	50	50	75

% Prisoners = Percentage of Casualties of Infantry, Armor, and Gun losses that are taken as Prisoners by the opposing Player.

[16.7] PATROL RECONNAISSANCE TABLE

DIE	Net Patrol Points		
	1	2	3
1	0	0	1
2	0	1	1
3	1	1	2
4	1	2	3
5	2	3	4
6	3	4	All

= Number of Battalion size-equivalent units in that hex about which the Patrolling Player receives Reconnaissance Information.

[16.8] OBJECTIVE LOSS TABLE

DIE	Objective Losses in TOE Strength Points
	Killed / Captured
1...4	0 / 0
5	0 / 1*
6	1 / 0

*Results are losses suffered by units in the hex under Reconnaissance by the Patrolling Force. If the Patrolling Force was completely eliminated, treat as 1 TOE Strength Point Killed.

[27.93] SPECIAL AIR SERVICE BRIGADE RAID TABLE

DIE	1	2	3	4	5	6
% Planes:	0	10	25	25	33	50

% Planes = Percentage of the planes on the ground in that hex that are destroyed.

Final Close Assault Differential

Overrun									
+11 +14 +12 +15 +13					+17 ET Seq Losses				
50%					50%				
40%	-	-	-	-	11-13	14-25	21-26	40%	
30%	-	-	-	-	14-22	26-33	31-36	30%	
25%	-	-	-	-	23-33	34-43	41-46	25%	
20%	-	-	-	-	34-46	44-55	51-56	20%	
15%	-	-	-	-	51-62	56-63	61-63	15%	
10%	11	11	11-13	12-15	14-21	14-23	21-26	10%	
5%	12-16	12-23	14-26	16-31	22-41	24-44	31-46	5%	
0	21-66	24-66	31-66	32-66	42-66	45-66	51-66	0	
Capt	-	-	-	-	-	-	-	Capt	
Retreat:	1Hex	-	-	10	9	8	5-6	4-6	3-7
	2Hexes	-	-	-	-	-	-	11	11
	3Hexes	-	-	-	-	-	-	-	-

If the dice sum does not lie within the range of numbers listed the result is In Contact. **Note:** The attacking forces are never required to perform an Involuntary Retreat.

[17.4] MORALE MODIFIER TABLE

Disorganization	Morale Modifier									
Level	+4	+3	+2	+1	NO	-1	-2	-3	-4	SURR
+8+	11...26	31...46	51...65	66	-	-	-	-	-	-
+7	11...16	21...36	41...56	61...66	-	-	-	-	-	-
+6	11...12	13...25	26...36	41...66	-	-	-	-	-	-
+5	11	12...21	22...33	34...65	66	-	-	-	-	-
+4	-	11...14	15...31	32...56	61...66	-	-	-	-	-
+3	-	11...13	14...21	22...44	45...66	-	-	-	-	-
+2	-	-	11...12	13...36	41...66	-	-	-	-	-
+1	-	-	-	11...26	31...66	-	-	-	-	-
0	-	-	-	11	12...65	66	-	-	-	-
-1	-	-	-	-	11...54	55...65	66	-	-	-
-2	-	-	-	-	11...46	51...56	61...66	-	-	-
-3	-	-	-	-	11...42	43...46	51...66	-	-	-
-4	-	-	-	-	11...33	34...41	42...55	61...66	-	-
-5	-	-	-	-	11...23	24...32	33...46	51...66	-	-
-6	-	-	-	-	11...12	13...25	26...36	41...63	64...66	-
-7	-	-	-	-	11	12...16	21...33	34...51	52...63	64...66
-8	-	-	-	-	-	11...13	14...31	32...44	45...61	62...66
-9	-	-	-	-	-	11	12...26	31...41	42...55	56...66
-10	-	-	-	-	-	-	11...21	22...34	35...53	54...66
-11	-	-	-	-	-	-	11...14	15...33	34...46	51...66
-12	-	-	-	-	-	-	11	12...26	31...44	45...66
-13	-	-	-	-	-	-	11	12...21	22...36	41...66
-14	-	-	-	-	-	-	-	11...16	21...33	34...66
-15	-	-	-	-	-	-	-	11	12...26	31...66
-16	-	-	-	-	-	-	-	-	11...21	22...66
-17-	-	-	-	-	-	-	-	-	-	11...66

NO=No change to the unit's Basic Morale. **SURR**=All of the units Surrender. (exception: See case 17.26). **Number**=Add (or subtract) that number to the unit's Basic Morale for that combat. Note limitations on maximum and minimum adjusted Basic Morale.

[17.6] TRAINING CHART

Type of Replacement Point	Number of OpStages
Infantry (except Commando)	3
Tank or Recce	6
Gun	1
Commando	12
Commonwealth Unit	6

= Number of Operations Stages needed to train that type of Replacement Point or raise any type of an untrained Commonwealth unit's Basic Morale one point. The Commando Replacement Point must be trained by its assigned unit. Note that the time required for training guns is more of an uncrating cost than training necessity.

[52.8] POISONING AND SWEETENING WELLS TABLE

Attempt: Poison Well

Die 1: Well poisoned.

Die 2...6: No effect.

Attempt: Sweeten (Unpoison) Well

Die 1...3: Well sweetened, water points may be drawn as normal.

Die 4...6: No effect (well still poisoned).

[40.8] STRAFING TABLE

TacAir Points										
1...5	6...10	11...15	16...20	21...25	26...30	31...35	36...40	41...45	46+	Result
11...63	11...56	11...53	11...46	11...42	11...34	11...25	11...16	-	-	0
64...66	61...66	54...65	51...64	43...63	35...62	26...61	21...56	11...55	11...46	1
-	-	66	65...66	64...66	63...66	62...66	61...66	56...65	51...64	2
-	-	-	-	-	-	-	-	66	65...66	3

Shift one column to the right if Strafing Trucks in Convoy. Shift one column to the right if Strafing a Supply Dump. **Infantry:** Shift one column left if in a Fortification Level zero or one hex, two columns left if in a Fortification Level two hex. **Armor:** Shift one column left if in a Fortification Level two hex.

RESULTS: **Trucks**= That number of Truck Points and their cargo eliminated. **Infantry**= That number of TOE Strength Points, and their transporting Truck Points, if any, that are eliminated. **Supply Dump**= Multiply by 10% to determine the percentage of all types of Supplies in that Dump that are eliminated. **Pipeline**= A result of '2' or '3' destroys the pipeline. **Grounded Airplanes**= The number of dice to be rolled. The sum of the numbers of the dice roll is the number of planes destroyed. **Armor**= The number of dice to be rolled. The sum of the numbers of the dice roll is the minimum number of Armor Protection Points eliminated. **Notes:** Strafing of Ports is considered, in determining results only, to be part of any Bombardment Mission attacking that Port.

6 [19.5] MAXIMUM ATTACHMENT CHART

Allied Unit Type	Maximum Attachment(s)
Armor Division (GT-1-67)	2 units. 1 Infantry; no Tank.
Armor Division (GT 68+)	3 units. 1 Infantry and/or 1 Tank.
Infantry Division	2 units. 1 Infantry; no Tank.
Tank Brigade (GT 1-67)	2 units. 1 Infantry; no Tank.
Tank Brigade (GT 68+)	1 unit. No Tank.
other Brigade	1 unit.
any Battalion	1 Company.
Matruh Garrison (GT 1-12)	6 units no tank, 1 gun class
Selby Force (GT 13+)	5 units no tank, no recce, 3 infantry
German Unit Type	Maximum Attachment(s)
Armor Division	1 Brigade and 1 unit <i>or</i> 4 units. No Tank.
Infantry Division	1 Brigade <i>or</i> 3 units. No Tank.
Infantry or Armor Regiment	1 unit.
Battle Group	4 units.
Artillery Brigade HQ	1 Artillery unit.
any Battalion	1 Company.
Italian Unit Type	Maximum Attachment(s)
Armor Division/Tank Group	2 units.
Infantry Division	2 units. 1 Infantry <i>or</i> 1 Tank.
Brigade/Regiment	1 unit.
Battle Group	3 units.
any Battalion	No units.

Unit=Refers to a counter which is at full strength, a Battalion—or Company-Equivalent. **Tank**=Specifically Bn-Eq units composed of tank TOE Strength Points. **Infantry**=All Infantry-type units with the exception of Machinegun and Heavy Weapons units (whether or not they are motorized).

Modifications:

Any nation's Division- and Brigade-equivalent units may attach two non-Shell company-Equivalent units at no cost to the maximum attachments. Note that three company-eq are equal to one Bn-eq unit and that the company-equivalents must comply with any restrictions as to the type of units which may be attached to that Division or Brigade.

[39.5] AIRCRAFT MISSION SUMMARY

STRATEGIC MISSIONS:

Commonwealth

Reconnaissance of Axis Naval Convoys (Case 42.5)
Bombing Axis Naval Convoys (Case 41.6)
Defensive CAP over Axis Naval Convoys (Case 41.63)
Flak Suppression of Axis Naval Convoys (Case 41.63)
CAP over Malta

Axis

Bombing Maltese air facilities (Case 44.2)
CAP over Malta
Flak suppression of Maltese air facilities (Case 44.2)
Offensive CAP over Axis Naval Convoys (Case 41.63)

LAND SUPPORT MISSIONS:

Fighter

Scramble* (Case 40.3)
Combat Air Patrol* (CAP), Offensive/Defensive (Case 40.2)
Strafing (Cases 40.5 and 40.6)

1st Line (attached) Trucks

Flak Suppression

Grounded Aircraft

Infantry-type units

Ports†

Tanks‡

Trucks in Convoy (2nd-3rd Line)

Water Pipeline

Bombing (Section 41.0)

Air facilities

Fortifications/Major Cities*

Flak Destruction

Mining Harbors

Personnel

Ports*

Railroad/Road*

Commonwealth Ships

Supply Dumps

Non-Combat

Airdrop (Case 42.4)

Reconnaissance of land units (Case 42.2)

Transfer (Case 42.1)

Transport (Case 42.3)

*May be performed at Night. †Performed on the Air Bombardment CRT. ‡Only Commonwealth Player owned (i.e., non-captured) Hurricane IID's. Note that certain planes may be assigned two Land Support Missions in the same OpStage (See Combined Missions, Case 39.2 and the Aircraft Characteristics Charts 4.6)

[29.61] WEATHER TABLE

Season	Game-Turn			Weather			
				Normal	Hot	Sand storm*	Rain storm*
Spring	1...12	49...60	97...108	11...42	43...55	56...64	65...66
Summer	13...24	61...72	109...110	11...23	24...55	56...66	-
Fall	25...36	73...84	-	11...35	36...54	55...61	62...66
Winter	37...48	85...96	-	11...52	-	-	53...66

*Proceed to Table 29.62, Foul Weather Location Table

[20.3] REPLACEMENT POINT CONVERSION CHART

One Replacement TOE Strength Point of	Requires	Replacement Points
Any Headquarters Unit		2 Inf
Commando ^a		3 Inf
Paratroop Infantry ^b		2 Inf
Italian Bersaglieri Infantry		2 Inf
Machinegun ^c		2 Inf
Heavy Weapons ^d		1 Inf & any 1 Gun
Any other Infantry		1 Inf
Engineer Battalion ^e		2 Inf
Engineer Company		1 Inf
Road Construction or Railroad Construction ^f		none
Motorcycle Reconnaissance, Camel Cavalry or Reconnaissance		2 Inf
Armored Reconnaissance		1 ArmR or 1 Lt Tank
Armored Car		2 ArmR or 1 Lt Tank
Tank		1 Tank
Artillery		1 Artillery
Airdropable Artillery		1 Italian
		ParaArt or 1 German
		7.5cm Lt Gun
Anti-tank		1 Anti-tank
Airdropable Anti-tank ^g		1 Lt AT
Anti-air		1 Anti-air
Squadron Ground Support Unit		1 Inf & Supplies

Key:
a = Specifically the Layforce or Special Air Service Brigade. **b** = Specifically units of the Italian 185th Folgore or German Ramcke Brigade. **c** = Whether or not it bears a motorized symbol. **d** = All units except those of the Ramcke Brigade. **e** = Including the Australian Pioneer unit. **f** = Specifically the Commonwealth Player's one South African Road Construction Battalion and two New Zealand Railroad Construction Battalions. If eliminated, the unit is returned to play as a reinforcement during the sixth Operations Stage after the Operations Stage in which the unit was eliminated. **g** = Only applies if assigning anti-tank TOE Strength Points to headquarters of the 185th Folgore Division and/or the headquarters of the Ramcke Brigade. **Inf** = Infantry. **Gun** = Artillery, anti-tank or anti-air, whatever its ratings. **ArmR** = Armored Reconnaissance/Armored Car. Italian ArmR replacement points are always treated as Autobelinda 41's (ID Code 'vv'), even if not able to upgrade the unit under the listed Upgrading Armored Car Effectiveness rules (Case 20.5). **Lt Tank** = Light Tank of the type specified in Case 20.5. This only applies if upgrading the unit. **Lt At** = Only Italian 47/32 or German 2.8cm or 28/20 Pak anti-tank guns.

[21.38] BREAKDOWN TABLE

Accumulated Breakdown Points

	0...3	4...10	11...20	21...30	31...40	41...50	51...60	61...70	71 +	% Breakdown
11...66	11...66	11...42	11...32	11...26	11...23	11...16	11...14	-	-	0
-	43...64	33...62	31...55	24...53	21...46	15...42	11...33	11...25	10	10
-	65	63...64	56...62	54...61	51...56	43...54	34...52	26...43	25	25
-	66	65	63...65	62...64	61...63	55...63	53...62	44...55	33	33
-	-	66	66	65...66	64...65	63...64	56...63	50	50	50
-	-	-	-	-	66	65...66	64...66	64...66	75	75

%Breakdown = The percentage of that type of vehicle that breaks down. BAR (Breakdown Adjustment Rating) of a particular type of vehicle is found by indicating the number of columns shifted (right or left) before rolling to determine Breakdown. **Hot Weather**: shift one column to the right. **Sandstorms**: If at least 50% of the accumulated Breakdown Points were acquired in map sections containing Sandstorms, shift the Breakdown Points one column to the right. BAR for vehicles and Weather are cumulative. Note that certain vehicles (e.g., motorcycles, artillery with inherent transport) never Breakdown.

[22.15] VEHICLE REPAIR SUPPLY COSTS CHART

Location	State	Type of Vehicle	Supplies Expended Before Attempt
Field	Bd	Truck, AC, Recce	None
Field	Bd	Tank, SPA, TD	1 Fuel
Facility	Bd	All	1 Fuel and 1 Stores
Field	De	Tank*	2 Stores and 2 Fuel
Facility	De	Tank	2 Stores and 2 Fuel

Supplies = Amount of that number of points of that type of supply. **Facility** = Major or Minor Repair Facility. **Bd** = Broken down vehicle. **De** = Destroyed tank (no other destroyed vehicle points may be repaired). **AC** = Armored Car. **SPA** = Self-propelled artillery. **TD** = Tank Destroyer. *Only destroyed Commonwealth tanks by a Tank Delivery Squadron. Only German tanks by the Mobile Tank Repair Squad. **Note**: If not indicated on the chart, the repair may not be performed.

[22.44] DESTROYED TANKS REPAIR TABLE

DIE	Field	Axis Repair Facility			CW Repair Facility
		German	Italian	Commonwealth	All Nationalities
1	R	R	R	R	R
2	-	R	R	R	R
3	-	R	-	-	-
4	-	-	-	-	-
5	-	-	J	-	-
6,7	-	J	J	J	J

Add one to the die roll if attempting repair through use of a Minor Repair Facility. **Field:** Repair in the Field. May only be attempted by a Tank Delivery Squadron on Commonwealth tanks and the Mobile Tank Recovery Squadron on German tanks. **R** = The tank TOE Strength Point is repaired. **J** = The tank TOE Strength Point is Junked (lost). **-** = No effect (may attempt repair again on another Op. Stage.).

[22.8] BROKEN DOWN VEHICLE REPAIR TABLE

DIE	Truck	Field		Repair Facility	
		AC/R	Tank/SPA/TD	Temporary	Major
0,1	2	1	25%	50%	75%
2	1	0	10%*	33%	50%
3	0	0	10%*	25%	50%
4	0	0	10%*	25%	50%
5	0	0	0%	10%*	33%
6	0	0	0%	10%*	33%
7	na	na	na	10%*	25%
8	na	na	na	10%*	10%

Field, Repair Facility = The place in which the repair attempt is being made; **AC/R** = Armored Car/Recce; **SPA** = Self-propelled Artillery; **TD** = Tank Destroyer. **1** = One Light or Medium Truck Point is repaired; **2** = One Heavy Truck Point (or any combination of Medium and Light Truck Points equal to two) is repaired.

AC/R: 1 TOE Strength Point of either armored cars or recce is repaired. **%:** That percentage of that type of vehicle is repaired, rounding fractions up. **10%*** = If only attempting to repair a single TOE Strength Point, treat as a 0% (zero) result. **Die Roll Modifications:** Subtract one from the die roll if attempting to repair a *broken down tank* in the same hex as Tank Delivery Squadron or Mobile Tank Repair Squad. Add one to the die roll if the Facility is in a Major City hex whose Fortification Level is at present two. Add two to the die roll if attempting repairs at a repair facility in a Major City hex whose Fortification Level is at present one or zero.

[29.7] FOUL WEATHER LOCATION TABLE

DIE	1	2	3	4	5	6
Map Sections	A,B	C,D	D,E*	B,C	B,D	B,C,D

Map Sections = The map sections on which the Sandstorm or Rainstorm appear. Normal Weather occurs on the unaffected map sections. ***** = A Sandstorm occurring on a (Nile) Delta hex is treated as Normal Weather. **Note:** Sandstorms do not extend north of the coastal hexes.

[27.91] DESERT RAIDER RAIDS TABLE

Target: Water Pipeline

Die 1...4: Target destroyed

Die 5,6: No effect

Notes: Each Desert Raider is limited to one attempt per OpStage.

Target: Airfield

Die 1,2: Reduce one level of effectiveness

Die 3...6: No effect

Notes: May not be raided if any combat units are present. An airfield may not be raided more than once per Game-Turn.*

Target: Airplanes on the ground

Die 1,2: 10% of the planes are destroyed, the planes being determined by the raiding Player

Die 3...5: No effect

Die 6: Planes unaffected, desert raider eliminated.

Notes: May not be raided if combat units are present. Flying boats are never affected by raid results.*

Target: Rommel

Notes: Only by Commonwealth Long Range Desert Groups. See Raid on Rommel Table.

Target: Supply dump with combat units in hex

Notes: Roll two dice. If the dice sum is equal to or greater than the total Raw Close Assault Defense Points (ignore parentheses when calculating strength) of the combat units in the hex the desert raider survives and attacks as below. If the dice sum is less than the total Raw Defensive Close Assault Points, the desert raider is eliminated.

Target: Supply dump with no combat units in hex or one in which the desert raider survived

Die 1,2: 10% of supplies in the dump destroyed

Die 3...5: No effect

Die 6: Supplies unaffected. If there are combat units present, the desert raider is eliminated.

Notes: If the desert raider completes the raid without being eliminated, the opposing Player must inform the raiding Player if the supply dump is real, a non-dump, or a dummy.

Target: Trucks in convoy

Die 1,2: 1 Truck Point eliminated. The raided Player determines the type of Truck Point eliminated, the raiding Player then determines the cargo carried eliminated, with Replacement Points named only if unavoidable.

Die 3...5: No effect

Die 6: Trucks unaffected. If any of the trucks in that convoy are transporting infantry Replacement Points, roll one die again. If a 5 or 6 is rolled, the desert raider is eliminated.

Notes: If all of the trucks in the convoy are transporting infantry Replacement Points, the desert raider is eliminated without the raiding Player determining if the trucks are affected.

GENERAL NOTES: All raids require the expenditure of 5 CP's by the desert raider. No target in a Major City hex may be raided. *The raiding Player may combine attacks on an airfield and the planes on the ground in that hex into a single raid. The desert raider must expend 10 CP's. The results of "both" raids are determined separately with the airfield attacked first. No desert raider may perform the same raid in the same hex or attack the same truck convoy more than once per OpStage.

[24.17] CONSTRUCTION CHART

Item	Situation	Unit(s) Required	Supplies Required*	Nr. Op Stages	Restrictions†
1 Level of Fortification	Build/Rebuild‡	AnyE + Inf Bn with 3 + TOE	30 Stores	3	May not be built in a Salt Barsh, Delta or Major City hex. May be rebuilt in a Major City hex in an enemy ZOC. No construction other than construction/demolition of ports and/or flying boat facilities may simultaneously occur in same hex.
Real Minefield	Build	EBn, ECoy or CHQ ^E	15 Ammo + 15 Stores	1	May only be built in Clear, Sand/Gravel and Salt Marsh hexes.
Fake Minefield	Build	EBn, ECoy or CHQ ^F	3 Stores	1	As above.
Railroad	Build	NZRR	1 Stores	1	Building limited to head of track.
Railroad	Rebuild	AnyE	1 Stores/hex	1	One hex and any two adjacent hexes may be rebuilt simultaneously.
Road	Build/Rebuild	a) Inf Bn with 3 + TOE or ECoy, HQ ^F b) EBn or ECoy, HQ ^E + Inf Bn with 3 + TOE	2 Stores/hex	1	May only build in "unfinished road" hexes. a) Only the hex occupied. b) The hex occupied and any two adjacent hexes. May combine building and rebuilding except as restricted above.
Temporary Repair Facility	Build	AnyE	50 Fuel + 250 Stores	1	May only be built in a Major City or Village/Town hex.
Repair Facility	Rebuild 1 Level‡	AnyE	10 Fuel + 50 Stores	1	May be rebuilt in an enemy ZOC.
Water Pipeline	Build/Rebuild	EBn or CHQ ^E	10 Stores	1	May only be built from a Major City hex. Building limited to head of pipe.
Airfield	Build	EBn or CSGSU	50 Fuel + 100 Stores	3	May only be built in Clear, Major City, Desert and Sand/Gravel hexes.
Air Field or Air Landing Strip	Rebuild 1 Level‡ Build	EBn, ECoy or SGSU	10 Fuel + 20 Stores	1	As Airfield above.
Flying Boat Basin	Build	EBn or CSGSU	25 Fuel + 50 Stores	3	In any Coastal hex (only) regardless of terrain.
Flying Boat Basin or Flying Boat Alighting Area	Rebuild 1 Level‡ Build	EBn, ECoy or SGSU	10 Fuel + 10 Stores	1	As Flying Boat Basin above.
Port	Block 1 Level‡	EBn or CHQ ^E a) Tobruk b) Other	50 Ammo + 25 Stores 25 Ammo + 10 Stores	1	The ports of Tripoli, Bizerta (Tunis), Alexandria (E3613), Aboukir (E3815) and Rosetta (E4019) may not be blocked.
Real Supply Dump	Build	Any unit with 1 + TOE	10 Stores	3 CP's	None.
Fake Supply Dump	Build	Any unit with 1 + TOE	none	2 CP's	May not be constructed if the percentage of Fake Supply Dumps in play is, or would exceed, 50% of the Real Supply Dumps in play.

Key: Nr. Op Stages = Number of Operations Stages of constructions required. AnyE = Any engineer battalion, engineer company or headquarters unit with engineering capability. EBn = Engineer battalion. ECoy = Engineer company. HQ = Any headquarters with engineering capability. CHQ = Allied headquarters with engineering capability. SGSU = Any Squadron Ground Support Unit. CSGSU = Allied Squadron Ground Support Unit. Inf Bn with 3 + TOE = Any infantry-type battalion possessing at least three TOE Strength Points. CP's = Capability Points.

Notes: † = Unless otherwise stated, an item may not be constructed in an enemy ZOC. * = If the hex is affected by Hot Weather, an additional 10 Water Points are required for each item, other than a Supply Dump, under construction. ‡ = Only one may be built, rebuilt or blocked at a time

[24.18] DEMOLITION CHART

Item	Situation	Unit Required	Nr. Op Stages	Restrictions/Commentary
Fortification	Reduce 1 Level	Not allowed		May be reduced by air bombardment or barrage.
Fake Minefield	Clear	Any unit	1	Cleared at the end of the Movement Segment in which it is entered.
Real Minefield	Clear	Any E or tank bn with 6+ TOE of Scorpions	1	
Railroad	Destroy	Any E or inf bn with 3+ TOE	1	May also be destroyed by air bombardment or barrage.
Road	Destroy	Not allowed	1	May only be destroyed by air bombardment or barrage.
Repair Facility	Dismantle	Any E	1	May also be reduced by barrage or air bombardment. Recover 25 Fuel and 120 Stores when dismantled.
Water Pipeline	Destroy	HQ E or any units with 1+ TOE	1	May also be destroyed by barrage, strafing, and desert raider raids.
Air Facility	Reduce 1 Level	Not allowed		Airfields may be reduced, but not destroyed, by desert raider raids and may be reduced by air bombardment.
Port of Tobruk	Unblock 1 Level	E bn or CHQ E	1	As "other" port, except unblocking 1 Level requires 25 Ammo + 10 Stores.
Port of Benghazi	Unblock 1 Level	Total 2 E bn and/or CHQ E	1	As "other" port, except unblocking 1 Level requires 100 Ammo + 50 Stores.
Port (other)	Unblock 1 Level	E bn or CHQ E	1	This applies whether the port was blocked by mines or ships. Air bombardment damage is automatic. Requires 50 Ammo + 25 Stores to unblock 1 Level.
Fake Supply Dump	Destroy	Any unit	upon entry	May also be eliminated by a desert raider.
Real Supply Dump	Blow	Any unit	% of CP's	May be damaged by a desert raider raid, air bombardment, or barrage.

Nr. OpStages = Number of Operations Stages required to demolish the item. **Any E** = Any engineer battalion, engineer company, or headquarters with engineering capability. **E bn** = Engineer battalion. **HQ E** = Head-

quarters with engineering capability. **CHQ E** = Allied headquarters with engineering capability. **Inf bn with 3+ TOE** = Any infantry-type unit with at least 3 TOE Strength Points.

[35.23] SQUADRON CAPACITY CHART

Nationality	Ready	Reserve	Total
Italian Squadriglia	9	3	12
German Staffel	12	4	16
Commonwealth Squadron (1940-41)	12	4	16
Commonwealth Squadron (1942-43)	18	6	24

[38.37] AIRCRAFT REFIT TABLE**A. By Plane:**

	Commonwealth	German	Italian
Dice Roll to Refit	2...8	2...7	2...6

#...# = Plane refitted. **Modifier:** Add two to the dice roll if the plane is not assigned to the Squadron Ground Support Unit attempting the refit.

B. By Squadron:

Dice Roll:	1	2	3	4	5	6,7	8,9
% of Planes Refitted:	100	80	70	60	50	40	33

Modifiers: German Squadron Ground Support Unit add 1. Italian Ground Support Unit add 2. Planes attempting refit not assigned to Squadron Ground Support Unit attempting refit add 1.

[45.6] PILOT AND PLANE RECOVERY TABLE

DIE	1	2	3...6
Strafed:	R	R	L
Flak/Air-Air	R&A	L&A	L&K

Strafed = Any plane lost thru a destroyed result on the Strafing CRT may be rolled for. **Flak/Air-Air** = Only fighter/fighter-bombers and their pilots lost in plane-plane combat or from Flak Fire may be rolled for. **R** = The plane is repairable. **L** = The plane is permanently eliminated. **A** = The pilot survives. **K** = The pilot is eliminated. Note that plane and pilot recovery may only be attempted, and 1 Fuel and 1 Stores Point expended for each repairable plane, during the first eligible Maintenance Segment/Phase after the plane was lost. A repaired plane is considered unfueled, unarmed and unfit and normal maintenance may not be performed on it until one Game-Turn has elapsed from this saving die roll. A surviving pilot may not be returned to his squadron until one Game-Turn after this saving die roll.

[40.4] SCRAMBLE TABLE

Distance	0...2	3,4	5,6	7,8	9,10	11...15
Scramble	6	5	4	3	2	1

Distance = Distance in hexes to the Target Hex. **Scramble** = A die roll equal to or less than the number listed successfully Scrambles that Squadron's planes.

[42.27] AIR RECONNAISSANCE OF LAND UNITS TABLE

DIE	1,2	3,4	5,6	7	8	9	10	11	12	13+
Reveal:	0	1	2	3	4	5	6	7	8	All

Add one to the die roll for every four planes completing the Reconnaissance Missions (ignore fractions). **Reveal** = Number of battalion-equivalents revealed. **All** = Reveal all units in the hex, regardless of size-equivalency.

[49.19] FUEL CONSUMPTION CHART

CP's Expended in Movement	Vehicle's Fuel Consumption Rate						
	1	2	3	4	5	6	7
1	0.2	0.4	0.6	0.8	1.0	1.2	1.4
2	0.4	0.8	1.2	1.6	2.0	2.4	2.8
3	0.6	1.2	1.8	2.4	3.0	3.6	4.2
4	0.8	1.6	2.4	3.2	4.0	4.8	5.6
5	1	2	3	4	5	6	7
10	2	4	6	8	10	12	14
15	3	6	9	12	15	18	21
20	4	8	12	16	20	24	28
25	5	10	15	20	25	30	35
30	6	12	18	24	30	36	42
35	7	14	21	28	35	42	49
40	8	16	24	32	40	48	56
45	9	18	27	36	45	54	63
50	10	20	30	40	50	60	70

= The number of Fuel Points consumed by each vehicle.

[41.39] MINING HARBOR BY PLANE TABLE

Bombload Capacity:	0...5	6...10	11...15	16...20	21...25	26...30	31+
Harbor Mined:	0	1	2	3	4	5	6

Bombload Capacity: The average Bombload of the six planes attempting to Mine the Harbor.
Harbor Mined: If the die roll is equal to or less than this number, the Harbor is Mined.

[45.4] AIR-AIR COMBAT MANUEVER ADJUSTMENT CHART

Manuever Differential:	0	1...3	4...7	8...14	15...20	21+
Adjustment:	0	1	2	3	4	5

Manuever Differential: The absolute difference between the Manuever Ratings of the two planes.
Adjustment: The amount added to (subtracted from) the TacAir Rating of the plane with the higher (lower) Manuever Rating in the Air-Air combat.

[46.4] FLAK ADJUSTMENT CHART

Nr. non-Fighter Planes	Columns Adjusted in Favor of Player Firing AA
1 to 23	0
24 to 47	1
48 to 71	2
72 or more	3

[45.5] TACAIR KILL

Adj TacAir Diff:

	-13 or less	-5...-12	-2...-4	-1	0	+1	+2	+3	+4	+5	+6	+7	+8	+9	+10	+11	+12 or more
Kill:	na	11	12	13	14	15	16	22	24	26	32	34	36	43	46	53	56

Adj TacAir Diff: The result of subtracting the defender's (modified) TacAir Rating from the total of the attacker's (modified) TacAir Rating plus the Manuever Adjustment (which may be a negative number) for

those two planes. **Kill:** A dice roll, read sequentially, equal to or less than that number results in the elimination of the defending plane. **na** = The defending plane may never be eliminated.

[41.5] AIR BOMBARDMENT AND SECONDARY BARRAGE TARGETS TABLE

Weapons Type

Torpedo Points^a 1...5 6...12 13...24 25...50 51...80 81...120 121...160 161...210 211...270 271...340 341 +

Barrage Points^b 1,2 3,4 5,6 7,8,9,10 11,12 13,14 15,16 17,18 19,20 1 +

Target Bomb Points 1...20 21...40 41...80 81...120 121...160 161...200 201...260 261...320 321...390 391...470 471 + Result Target

	11...62	11...55	11...51	11...42	11...36	11...31	11...15	11...12	-	-	0	
Airfields	63...66	56...66	52...66	43...66	41...62	32...56	16...55	13...46	11...43	11...36	11...25	1
Air Landing Strips	-	-	-	-	63...66	61...66	56...65	51...64	44...63	41...55	26...52	2
Ports	-	-	-	-	-	-	66	65...66	64...65	56...63	53...61	3
	-	-	-	-	-	-	-	-	66	64...66	62...66	4

0%

10%

20%

30%

40%

50%

75%

Supply Dump

Supply Dump

0%

10%

20%

30%

40%

50%

75%

Fortification

Fortification

No Effect

Reduced

Railroad

No Effect

Destroyed

No Effect

Destroyed

No Effect

Destroyed

Road

No Effect

Destroyed

Trucks, Flak Suppression, Combat Units, Commonwealth Fleet

Trucks, Flak Suppression, Combat Units Commonwealth Fleet

0

1

2

3

4

5

6

7

0%

5%

10%

20%

25%

30%

40%

50%

Axis Naval Convoys

Axis Naval Convoy

0%

5%

10%

20%

25%

30%

40%

50%

Key to Air Bombardment and Secondary Barrage Targets Table

Key: a = Use only when attacking ships of the Commonwealth Fleet. Attacks consisting of Bomb and Torpedo Points are performed as two attacks. Attacks by planes armed with Torpedos against Ports or Axis Naval Convoys are carried out using the Bomb Points row (see Case 41.7). b = Calculated in the same manner as for the Master Barrage Against Units CRT. **Results:** An Airlanding Strip receiving a result of 1 or greater is eliminated. **Airfield:** That number of Squadron GroundSupport Units may no longer use the Airfield's Ready Capacity. **Ports:** Reduce the Port by that number of Efficiency Levels. **Supply Dump:** Eliminate that percentage of all Supplies in that Dump. In addition, lose 1 Truck Point for each 10%, if possible, defender's choice. **Trucks:** That number of Truck Points and their

cargo destroyed. **Flak Suppression:** That number of TOE Strength Points eliminated. **Combat Units:** That number of Battalion-equivalents (and then Company-equivalents) Pinned (see Case 41.9). **Commonwealth Fleet:** That number of Damage Points. **Axis Naval Convoys:** That percentage of the Shipping that is eliminated. **Fortification:** Reduced one Level or not affected. **Railroad:** Destroyed or not affected. **Road:** Reduced to a Track or not affected. **Modifications:** If at least 50% of the *planes* attacking an Axis Naval Convoy are armed with Torpedos, increase the total Bomb Points by 25%. Subtract one from the results of Land Support Bombing against armor units.

[46.3] ANTI-AIRCRAFT COMBAT RESULTS TABLE

Target	Number of Flak Points											Planes on other missions										Planes on fighter missions									
	1...4	5...8	9...12	13...16	17...20	21...24	25...28	29...32	33...36	37+	Result	No. Planes Destroyed					No. Planes Aborted					Nr. Planes Destroyed									
11...55	11...45	11...36	11...32	11...21	11...12	-	-	-	-	-	0																				
56...66	46...65	41...62	33...56	22...55	13...53	11...45	11...36	11...32	11...24	1																					
-	66	63...66	61...64	56...64	54...63	46...61	41...54	33...51	25...43	2																					
-	-	-	65...66	65...66	64...66	62...65	55...64	52...62	44...56	3																					
-	-	-	-	-	-	66	65...66	63...65	61...64	4																					
-	-	-	-	-	-	-	-	-	66	65...66	5																				
11...53	11...42	11...31	11...22	11...13	-	-	-	-	-	-	0																				
54...66	43...64	32...62	23...55	14...52	11...44	11...31	11...21	11...12	-	1																					
-	65...66	63...66	56...64	53...62	45...55	32...52	22...43	13...34	11...26	2																					
-	-	-	65...66	63...65	56...63	53...62	44...55	35...51	31...43	3																					
-	-	-	-	66	64...66	63...66	56...65	52...63	44...54	4																					
-	-	-	-	-	-	-	66	64...65	55...62	5																					
-	-	-	-	-	-	-	-	66	63...65	6																					
-	-	-	-	-	-	-	-	-	66	7																					
11...64	11...56	11...52	11...42	11...31	11...16	-	-	-	-	0																					
65...66	61...66	53...66	43...66	32...65	21...62	11...56	11...46	11...34	11...16	1																					
-	-	-	-	66	63...66	61...66	51...64	35...62	21...56	2																					
-	-	-	-	-	-	-	65...66	63...66	61...66	3																					

Notes: Each Target Group is attacked separately. Attacks against a Target Group comprised of Planes on Other Missions containing a multiple of twelve Bomber-Class or Transport-Class aircraft or any combination shift the Nr. of Flak Points one column to the right for each twelve aircraft exceeding twelve aircraft (i.e., 24 aircraft shift one column to the right).

[58.5] ABSTRACT TRUCK LOSS CHART

Year	Month											
	Jan	Feb	Mar	Apr	May	June	July	Aug	Sept	Oct	Nov	Dec
1940	-	-	-	-	-	-	-	-	-	6/2	5/3	4/4
1941	2/3	2/3	10/1	9/2	7/3	5/4	4/3	3/3	3/3	3/3	3/5	3/4
1942	2/4	3/3	3/3	3/3	3/3	3/4	3/4	3/4	3/4	3/4	4/6	3/5
1943	2/4	-	-	-	-	-	-	-	-	-	-	-

Nr/ = Percentage of Commonwealth Motorization Points in N. Africa that are destroyed.

/Nr = Percentage of Axis Motorization Points in N. Africa (Tripoli/ Tunis/etc are part of N. Africa) that are destroyed.

[50.2] AMMUNITION CONSUMPTION RATES TABLE

Logistics Game Played

Action	Ammo
Barrage	4
Anti-armor	3
Close Assault: Armor-class, Gun-class, MG-inf, HvyWpn-inf	2
Anti-air at single target group	2
Rearm any portion of a squadron's planes' tacair rating	1
Rearm one plane with bombs, torpedoes, or mines	1
Engage in air-to-air combat and/or strafe	TacAir Bombload

Logistics Game Abstracted

Action	Ammo
Barrage: Phasing Battalion-Eq	4
Barrage: non-Phasing Battalion-Eq	2
Assault: Phasing Battalion-Eq	1
Assault: non-Phasing Battalion-Eq	1
Anti-air Fire: Gun-class unit at a single target group	1

Rearm any portion of a non-fighter squadron's

tacair ratings and bombload
Flight by a non-fighter plane

1

Bombload
Ammo = If the logistics game is being played, the number of Ammo Points that must be expended in order for one TOE Strength Point to perform the action. If the Logistics Game is being abstracted, the number of Ammo Points that must be expended for the unit to perform the action.

[52.7] WATER AVAILABILITY TABLE

DIE	1	2	3	4	5	6
Town:	100	150	200	300	350*	500*
Bir:	50	100	150	200*	300*	400*

Number = Number of Water Points drawn.

*Roll one die again, secretly. If the die roll is a '1', the Town or Bir in that hex is Depleted. Note that a Player may draw an unlimited number of Water Points from an Oasis or Major City, which never deplete.

[54.12] SUPPLY DUMP CAPACITY CHART

Location	Ammo	Fuel	Stores	Water
Tunis/Tripoli	U	U	U	U
Major City	U	U	U	U
Village	2,500	8,000	3,000	1,000
Other Terrain	1,500	5,000	1,000	1,000
Non-Dump	50	0	50	0

Tunis/Tripoli = Each of the four Axis Off-map boxes for the Tripoli through Tunis region. **Location** = The hex in which the dump is located or the type of dump. **Nr** = The number of points of that type of supply that may be stored in a dump in that location. **U** = An unlimited number of points may be stored. Note that water may be drawn as well as stored in a village, major city hex or Tunis/Tripoli box. Note that the Tunis/Tripoli/In Transit boxes may not be used as supply dumps.

[54.17] SUPPLY DUMP DEMOLITION TABLE

DIE :	-2	-1	0	1	2	3	4	5	6	7	8 or more
% Supplies	0	33	0	10	20	33	50	75	100	33	100

Die = The die roll, modified as listed below. **% Supplies** = Percentage of all supplies in that dump that are destroyed.

Modifications (cumulative):

- + 1...for each additional 1/3 or the unit's Basic CPA expended *and*
- + 1...if the attempting unit is a full (non-shell) division, *or*
- 1...if the attempting unit(s) total one Stacking Point or less, *and*
- 2...if the attempt is being made in a Major City hex. If not, *then*
- + 1...if the total of the dump's supplies is 500 or less points, *or*
- 1...if the total of the dump's supplies is 4000 or more points, *and*
- + 1...if the attempting unit(s) "just" captured the dump, *or*
- 1...if the dump has not just been captured and the nearest enemy unit is at least 20 CP's (via a medium truck) away.

[54.2] TRUCK CHARACTERISTICS CHART

MAXIMUM TRANSPORT CAPACITY

Truck Type	CPA			TOE Strength Points				Supply Points			Fuel Capacity	Fuel Consumption Factor	BAR
	Inf	Guns	Supplies*	Inf	Arty	AA	Ammo	Fuel	Stores	Water			
Light	25	na	40	½†	na	1	2	50	6	40	8	1	2L‡
Medium	20	15	30	1	1	2	4	120	15	100	6	1	2L
Heavy	20	15	30	2	1	4	8	250	30	200	6	1	2L

Inf = Infantry. **Arty** = Artillery. **AA** = Anti-aircraft. **BAR** = Breakdown Adjustment Rating, in column shifts to the left. **CPA** = The Capability Point Allowance of the Truck Point when transporting (motorizing) the item listed. **na** = Not allowed. *Applies only to Trucks in Convoy carrying Supplies. †Requires two Light Truck Points to motorize one Infantry TOE Strength Point. ‡Light Trucks, if not being moved along a Road, garner one break down point in addition to those acquired by entering the hex or crossing the hexside. e.g., moving along a track one hex across a wadi into a rough hex accumulates 18 BP's (½ × 8 for the wadi + ½ × 8 for the rough + 1 each for the wadi hexside and the rough hex).

[54.5] EQUIVALENT WEIGHTS CHART

In Tonnage

Supplies by Naval Convoy, Interport Transfer, Railroad or Air Transport:

One Point Of	Equivalent Tonnage
Ammo	4
Fuel	1/8
Stores	1
Water	1/6
One Replacement Point By	Equivalent Tonnage
Axis Naval Convoy	Varies, see Axis Replacement Pool
Interport or Railroad	na (See below)
Air	2*
Truck/Motorization Point by	Equivalent Tonnage
Axis Naval Convoy	Varies, see Axis Replacement Pool
Railroad	n/a (see below)
Interport	50†
Air	P

In Stacking Points

Railroad/Interport	Stacking Point Equivalent
Truck/Motorization Point‡	1/10
Replacement Point	1/5
Squadron Ground Support Unit	1/2
Unit of Stacking Points	1/2

Key: n/a = Not applicable. P = Prohibited. * = Note that only infantry-class replacement points may be transported by air. † = Only applies to unattached points or those attached points in excess of the sum total of the TOE Strength Points comprising the Parent unit to which they are attached. ‡ = Only applies to points being transported by rail that are unattached or those attached points in excess of the sum total of TOE Strength Points comprising the Parent unit to which they are attached.

[55.3] PORT CAPACITY AND EFFICIENCY LEVEL CHART

Stacking Points

Port	Efficiency Level	In	Out	Maximum Tonnage
Tripoli	10	10	15	15,000
Bizerta*	10	10	15	3,333
Alexandria	10	5	10	15,000
Tobruk†	5	1	3	1,700
Benghazi	3	2	5	2,500
Mersa Matruh	1	1	2	250
Bardia	1	0‡	1	400
Sollum	1	0‡	1	250
Derna	1	na	1	300
All others	1	na	na	100

Stacking Points: The number of stacking points of items that may be shipped in *and* may be shipped out in one Operations Stage. **Maximum Tonnage:** The total tonnage of supplies that may be shipped in and/or out in one Operations Stage. **Efficiency Level:** A rating expressing a port's ability to absorb damage without being affected. A loss of one level of efficiency decreases the port's capacity by a fraction equal to one over the listed efficiency level. Note that reinforcements are never affected by a port's capacity or current efficiency; that the stacking points handled and tonnage handled by a port have no effect upon each other; and that replacement points shipped by an Axis Naval Convoy are off-loaded by the Stacking Point schedule, not the maximum tonnage schedule. *May not be used by the Axis Player before Game-Turn 35, at which point he may roll for activation. †Begins the campaign with an efficiency below the listed five due to the *San Giregio* partially blocking the harbor. ‡Any one unit of zero Stacking Points may be shipped in.

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